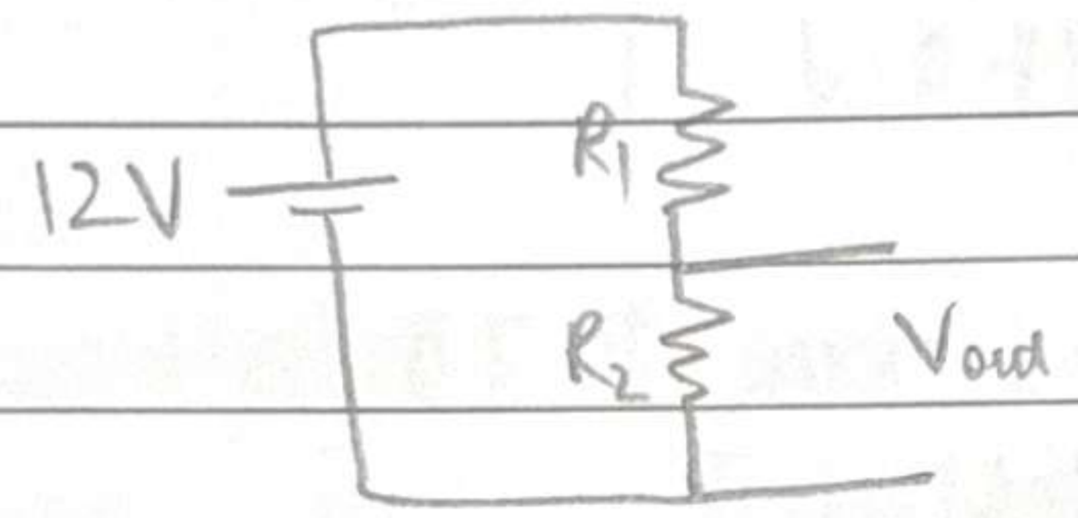
PoL DC/DC Converters

12V $\longrightarrow$ 1V
(500A)

How to convert 12V DC to 1V DC ?

Option 1) Resistance Divider

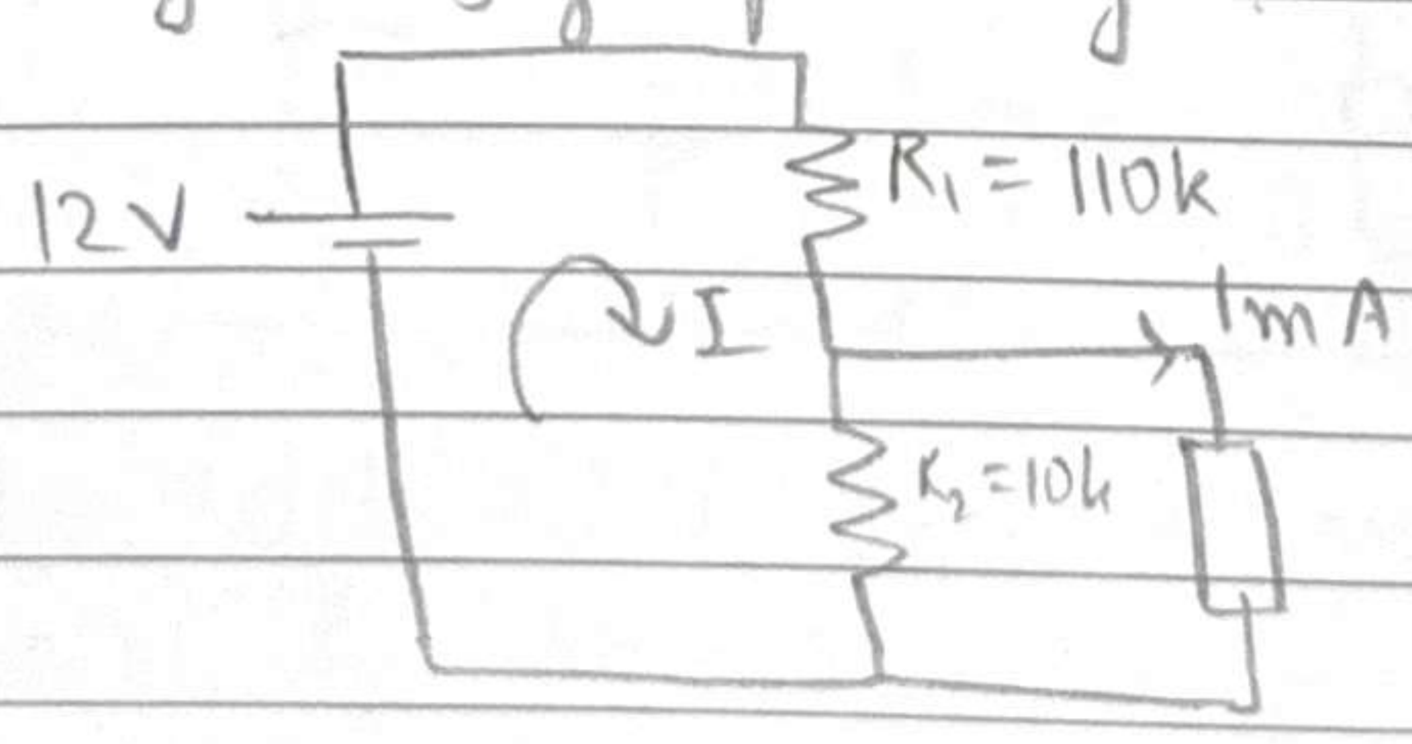


$$\frac{R_2}{R_1+R_2} = \frac{V_0}{12}$$

To minimize power loss, let us choose large R_1 and R_2
Let us choose $R_2 = 10k$
$R_1 = 110k$
tolerance 0.1% $\rightarrow$ gives us accurate output

No load loss $= \frac{(12)^2}{120k} \approx 1mW \rightarrow$ good enough!

Problem ? As soon as we connect load, even when drawing very small load current, output voltage changes significantly



$\cancel{12 - 110k\,I - 10k\,I =}$

$12 - 110k\,I - 10k\,(I - Im) = 0$

$12 - 110k\,I - 10k\,I + 10 = 20$

$22 = 120k\,I$

$I = \frac{22}{120k}$

$V_0 = \frac{22}{12} = 1.9\,V$!

Very bad, even at loads as low as 1mA
Bad voltage regulation!

To improve voltage regulation, reduce R_1 & R_2
$\Rightarrow$ High losses!

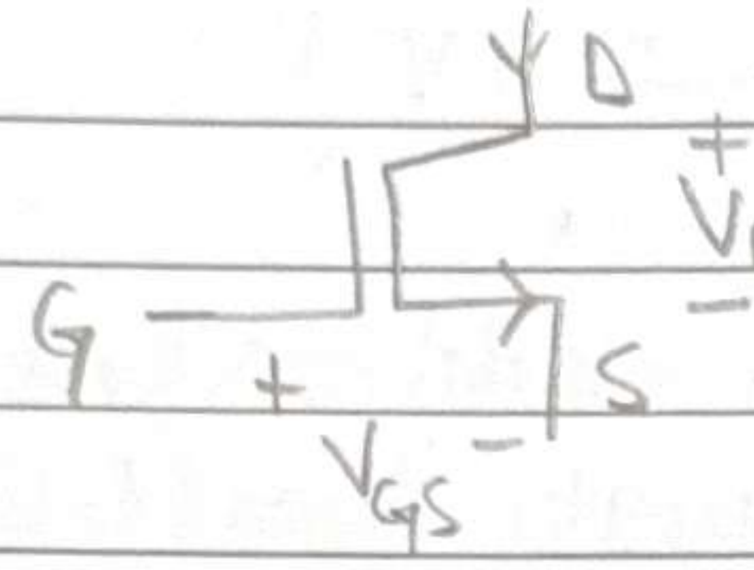
Limitations —
i) Efficiency (i.e. Power loss)
ii) Voltage regulation

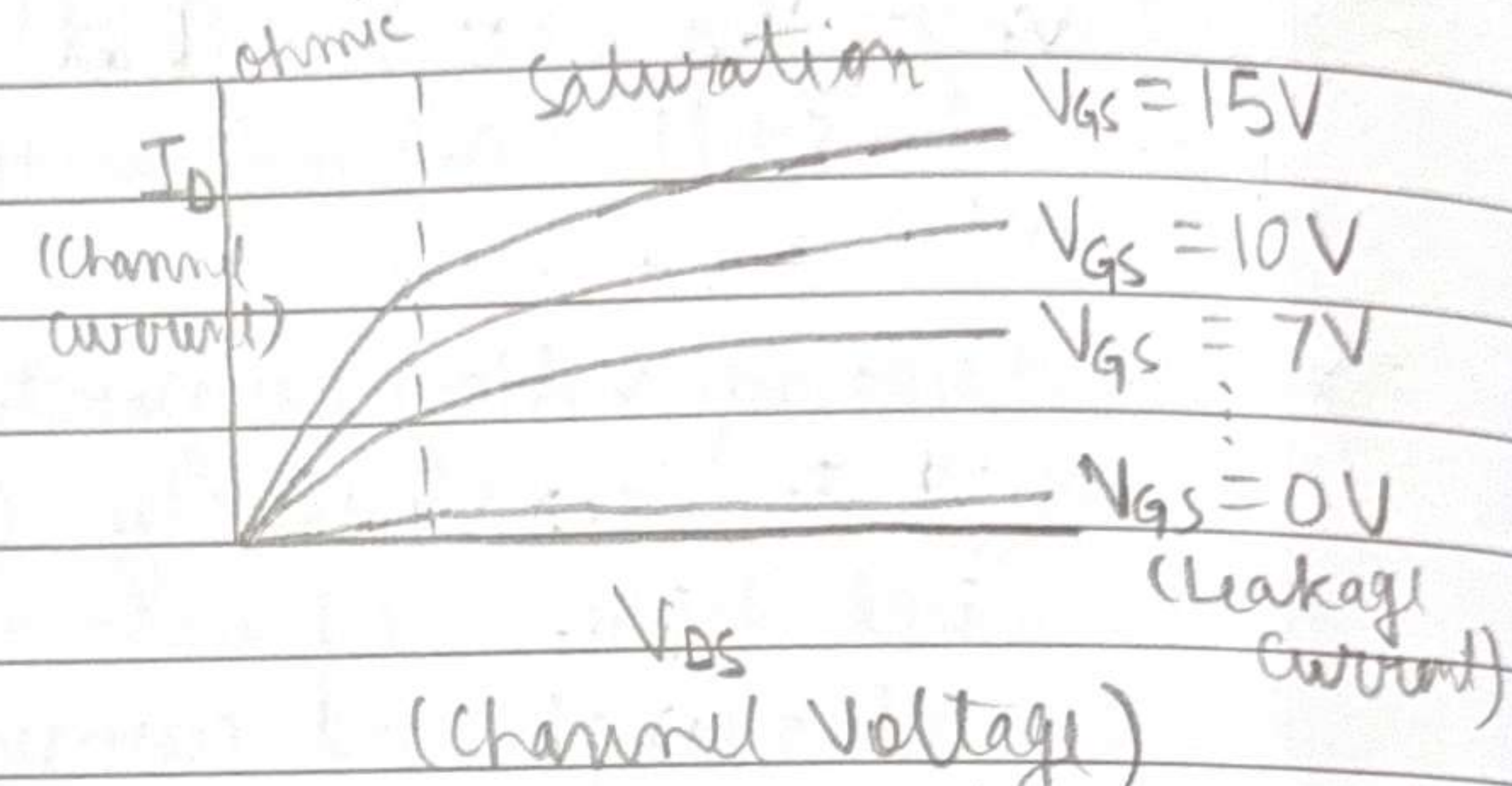
Conclusion — Not suitable for power

Only good for very load currents (e.g. to power opamps)

Option - 2) Using semiconductor

MOSFET

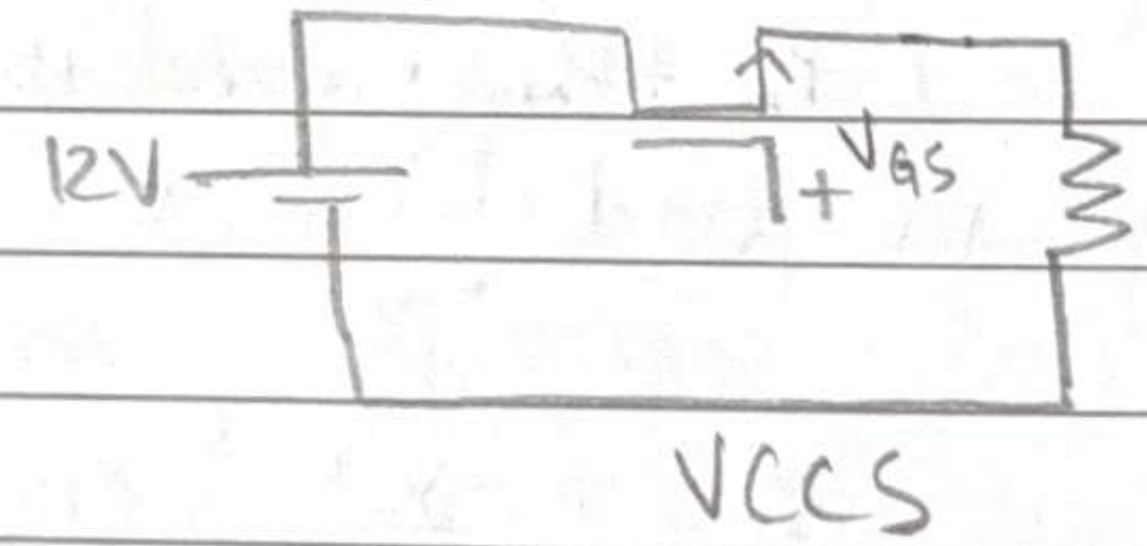




MOSFET characteristics

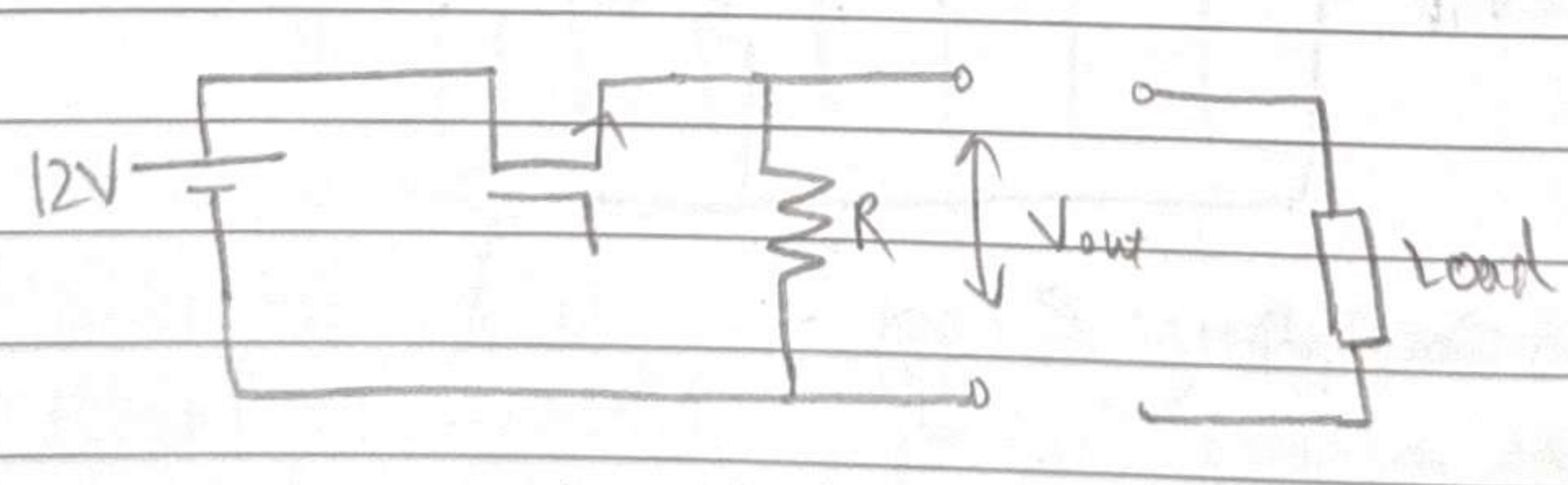
Can use MOSFET as current source by operating in saturation region to force (nearly) constant current
Can change V_{GS} to change value of constant current $\longrightarrow$ Voltage-controlled Current Source !
(VCCS)

Analog



VCCS

Use this to convert 12V to 1V



Choose V_{GS} such that $I_D = 0.1mA$
Choose $R = 10k\Omega$
$V_{out} = 1V$ (at no load)

Now consider load current of 0.1 mA

⇒ Change V_{GS} such that $I_D = 0.2\,mA$!
Still $0.1\,mA \times 10k = 1V$!

⇒ Problem of voltage regulation addressed
Need to continuously adjust V_{GS} to keep V_{out} at 1V
by checking Load current, ⇒ Feedback Control

What about efficiency ?

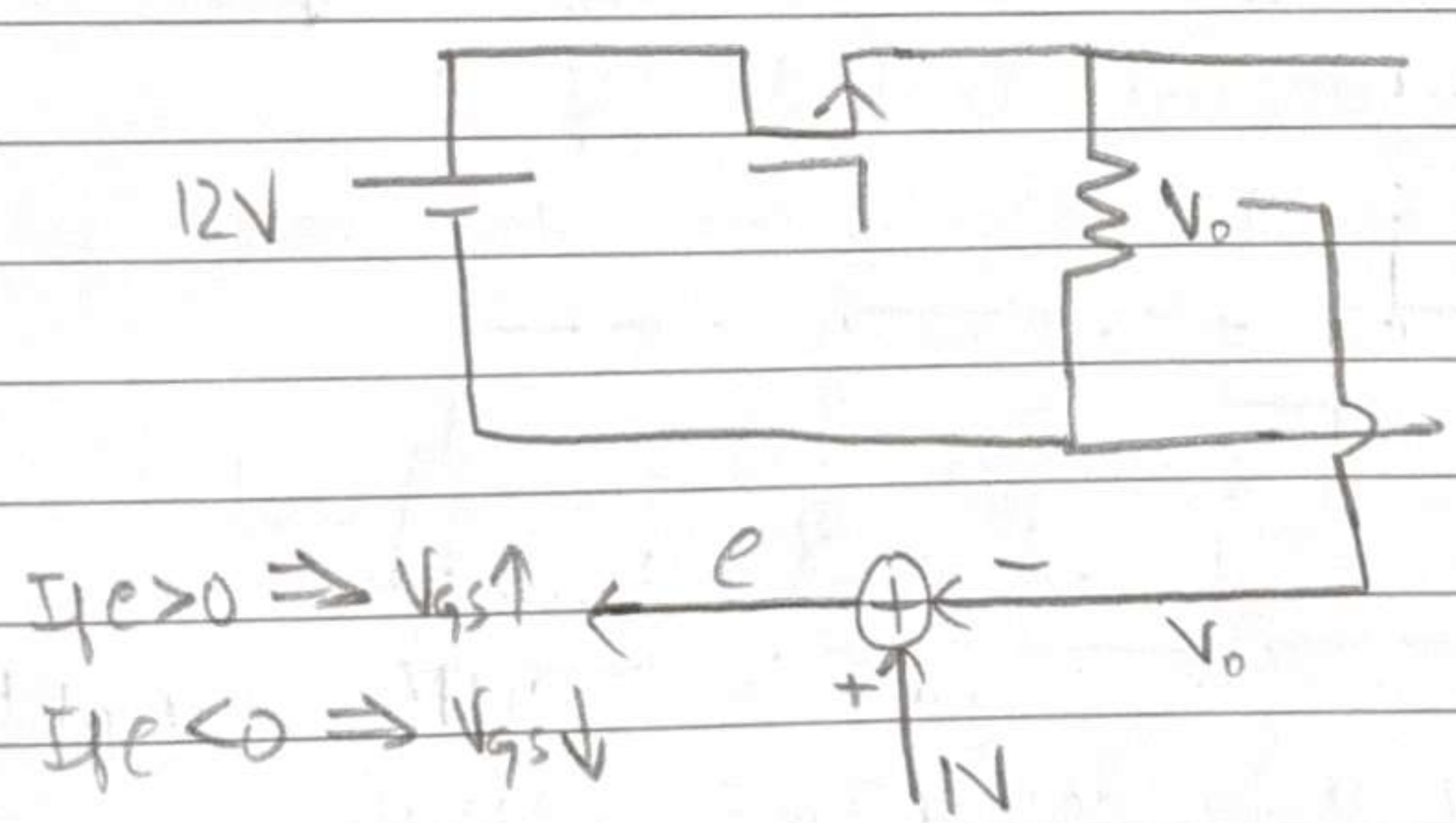
$P_{out} = 100A \times 1V = 100\,W$

$P_{R_2} = \frac{1^2}{10k} = 0.1\,mW$

$P_{MOS} = V_{DS} \times I_{DS} \approx 11V \times 100A \approx 1kW$

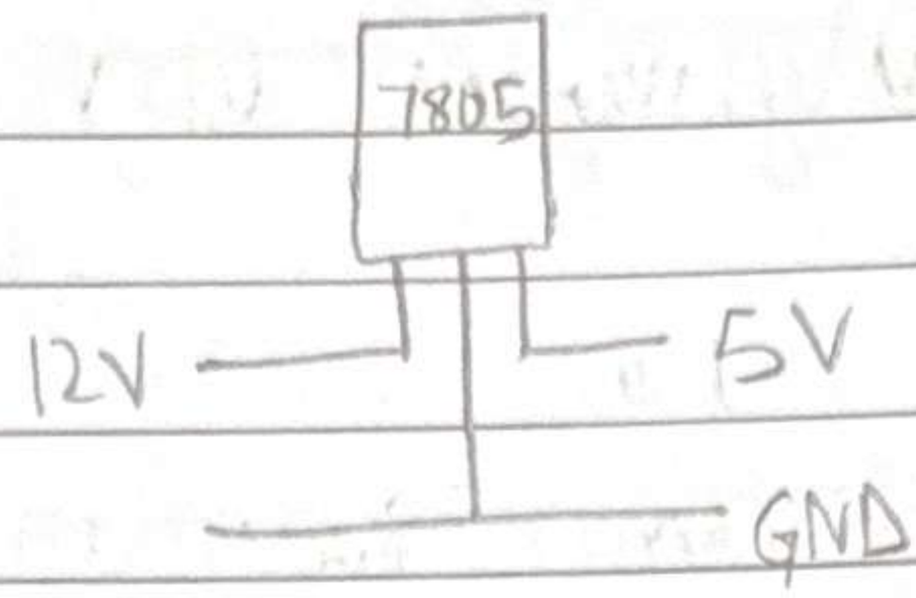
$\eta = \frac{P_{out}}{P_{R_2} + P_{MOS}} = \frac{100}{0.1m + 1k} \approx 10\%$ ⇒ Bad !!

η is bad !!



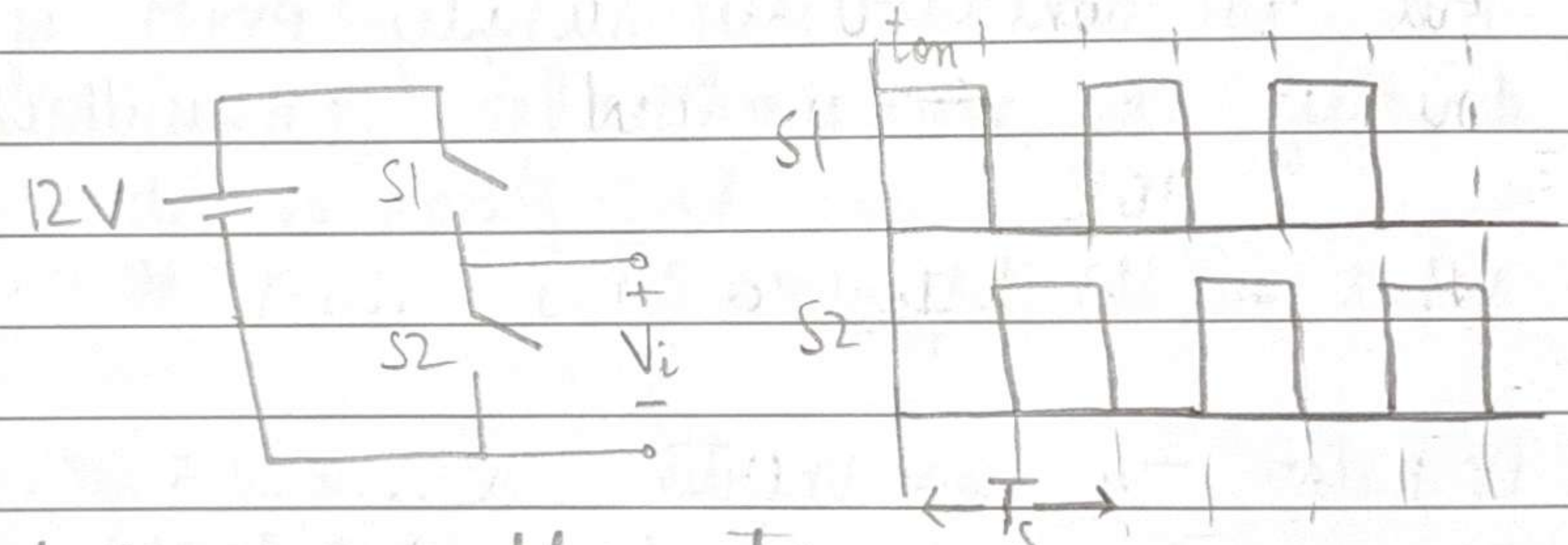
This type of ckt is called Linear Regulators
aka LDO (Linear Dropout Regulators)

E.g 7805 IC (12V to 5V)



Can't use to power microprocessors (which draw lots of load current)
Can be used to power opamps (which draw very small load current)

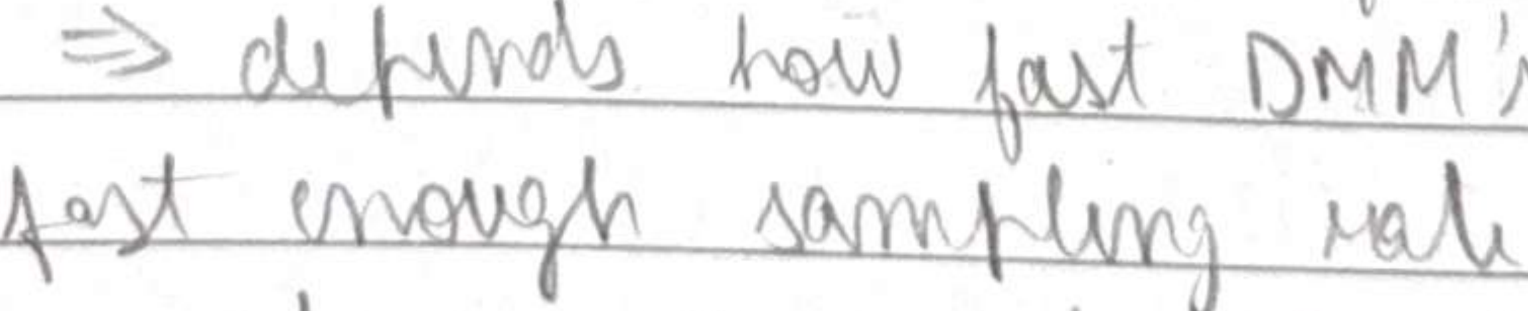
Option 3) DC - DC Buck converter



$F_s = \frac{1}{T_s}$

S1 and S2 are complementary

Whenever $(S1, S2) = (1, 0)$, $V_i = 12V$
$(S1, S2) = (0, 1)$, $V_i = 0V$



On DSO ⇒ We'll see this waveform
on DMM ⇒ depends how fast DMM's sampling is
For fast enough sampling rate, suppose DMM shows average (or RMS) value

$$\langle V_i \rangle_{T_s} = \frac{t_{on}}{T_s} \times 12V$$

$\frac{t_{on}}{T}$ is called Duty Cycle (D)

HW: Simulate Resistance Divider
Find other ICs (LDO) ~~on~~, add and simulat them

PWM! (Pulse Width Modulation)

We can supply PWM with $D = 1/12$ (say), and the motor will run at speed corresponding to (roughly) $1/12$ Vin.
But we can't directly supply PWM of $\frac{1}{12} \times 12V$ directly to microcontroller (it will blow)

What is the difference?

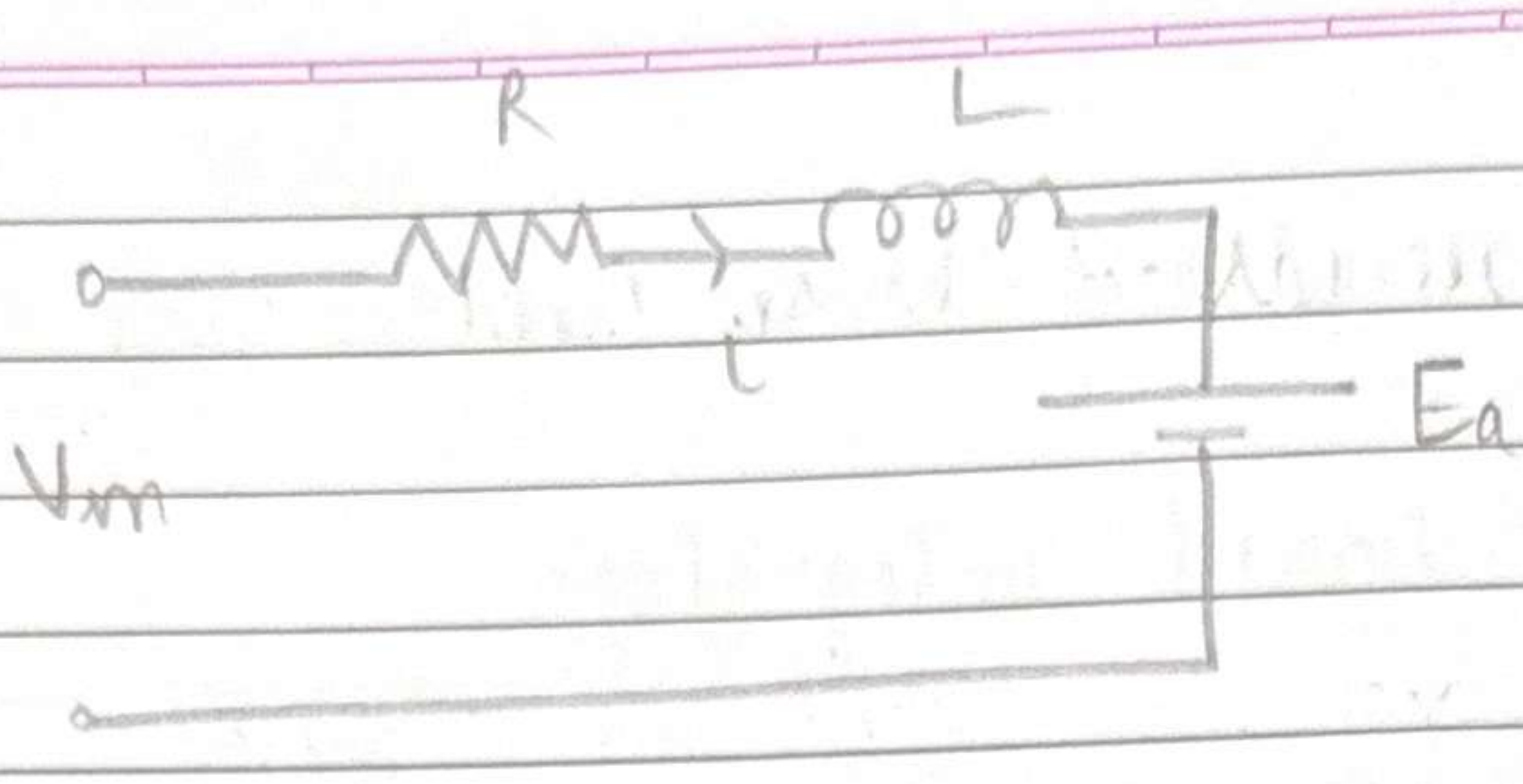
DC Motor - has inertia
has inductance → acts like LPF
↓
averages PWM

$$12 - i_L R - L\frac{di_L}{dt} = 5$$

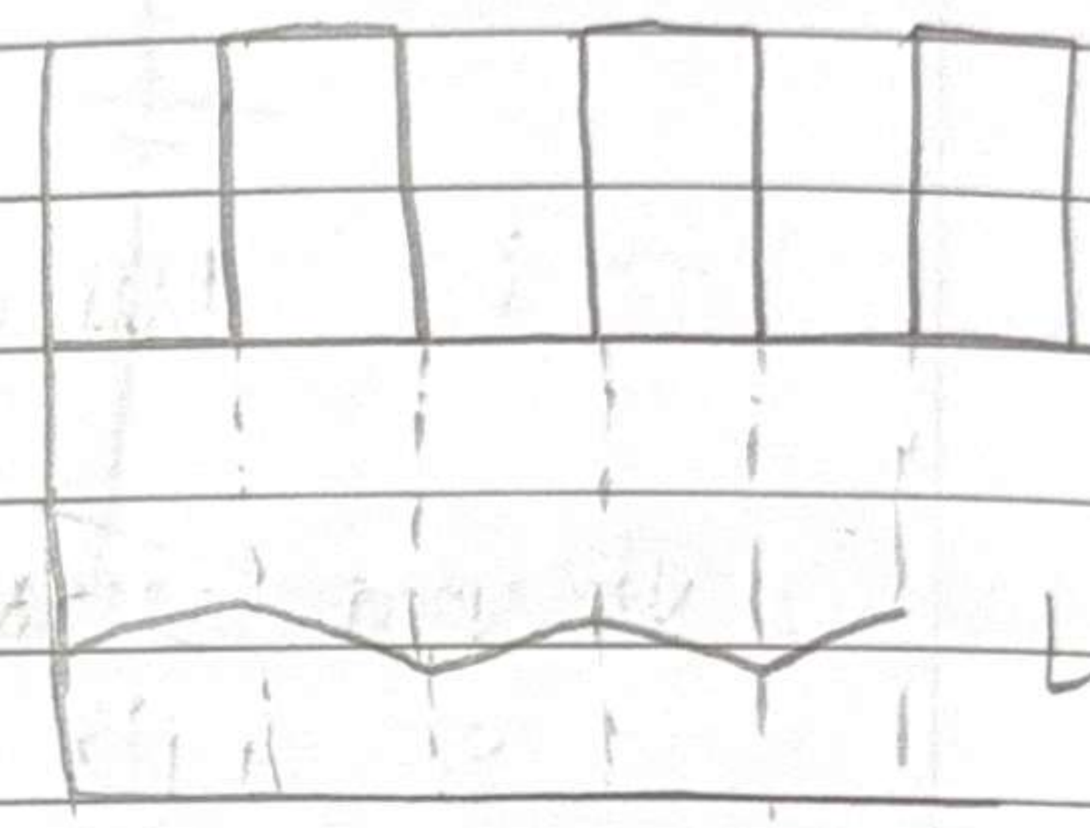
$$-L\frac{di_L}{dt} = i_L R - 7$$

$$-\frac{di_L}{dt} = \frac{i_L R - 7}{L}$$

DC Motor



Assume, L large
R small



When $V_{in} = 0$, $E_a = 5V$

$$V_{in} - iR - L\frac{di}{dt} = E_a$$

$$\frac{di}{dt} = \frac{V_{in} - E_a - iR}{L}$$

When $V_{in} = 0$,
$\frac{di}{dt}$ is small and $-ve$

When $V_{in} = 12V$,
$\frac{di}{dt} = \frac{5 - iR}{L}$ is small and $+ve$

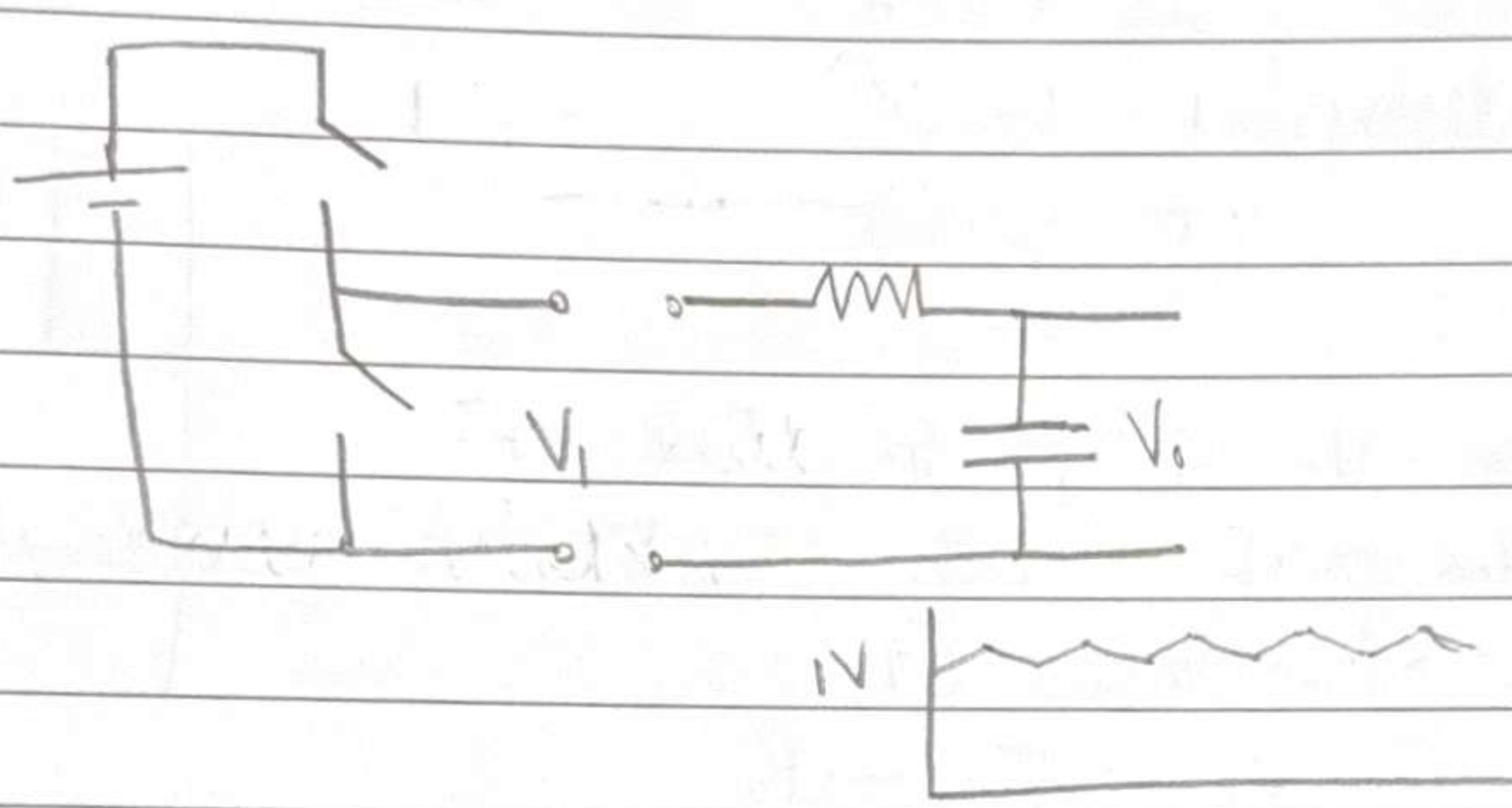
LPF attenuates high frequency components' amplitudes, and retains low freq components (including DC component)

$\therefore$ Effectively, DC Motor has 2 LPFs
i) Inductance
ii) Inertia

LPF circuits —

i) RC

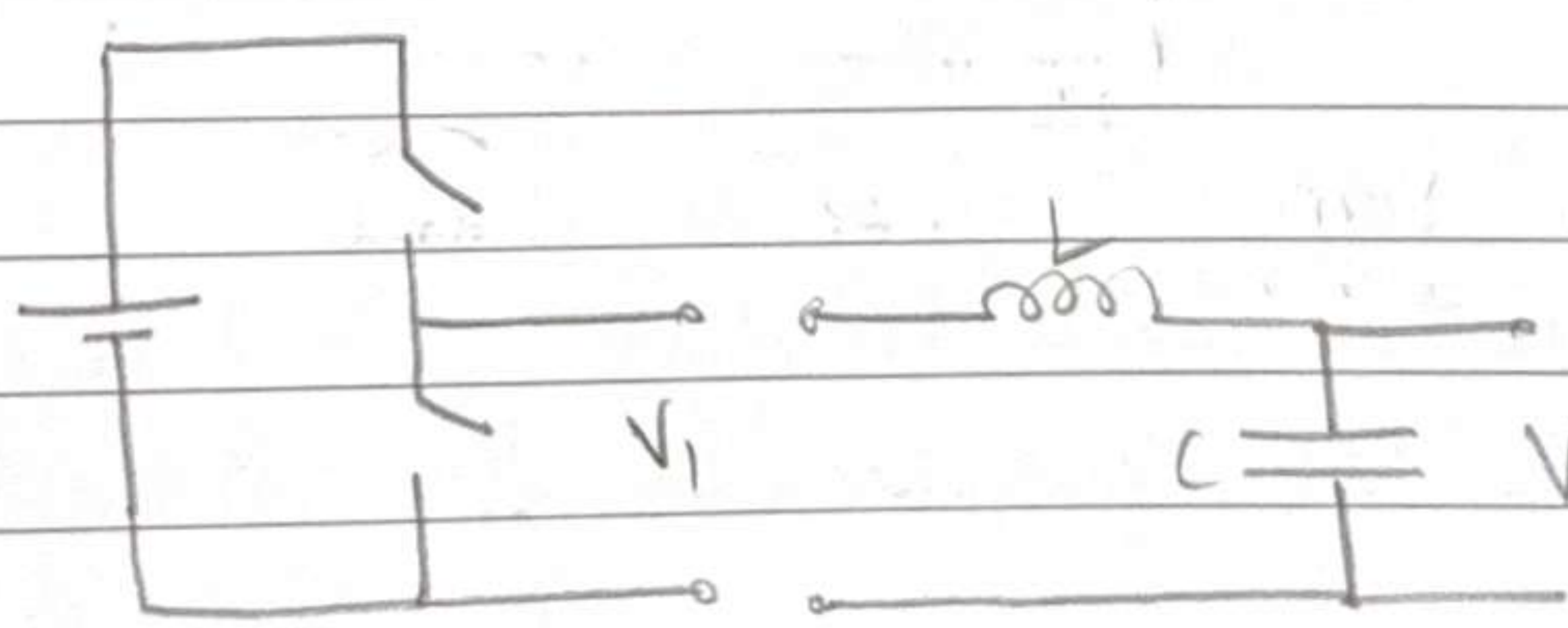
ii) Opamp-based integrator



To make a better RC LPF, increase capacitance to increase time constant.

RC is lossy because of R

Better option — LC

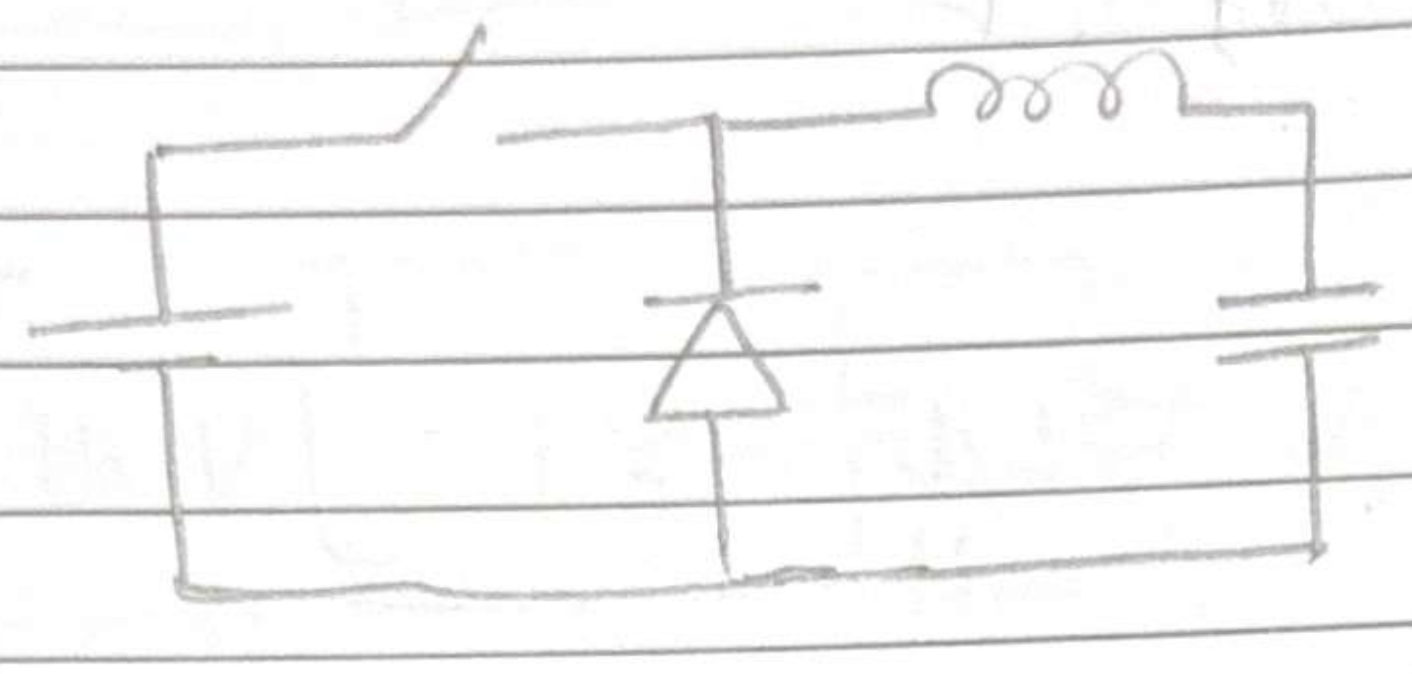


RC is a 1st-order filter

LC is a 2nd-order filter

DC/DC

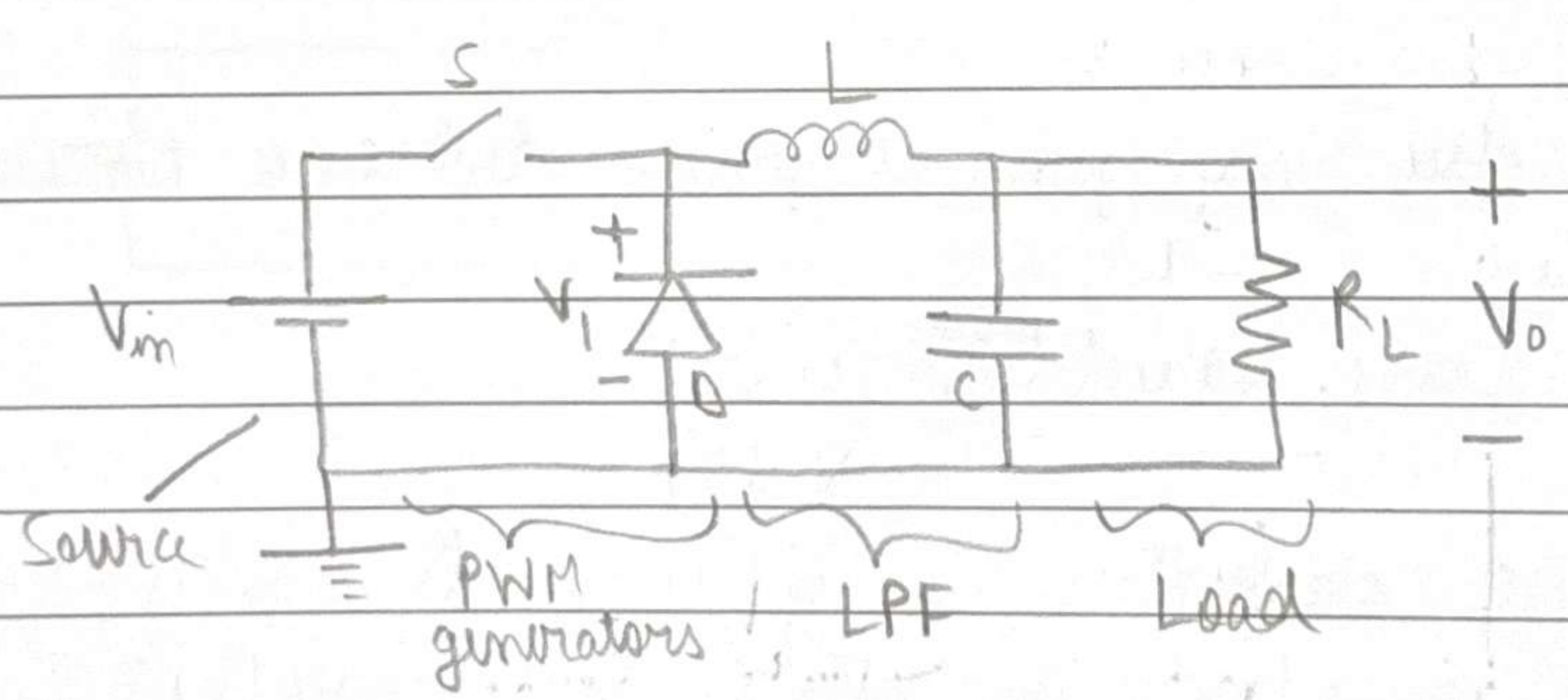
Another ckt of buck-converter



This is a more realistic ckt
The switch is also realized using transistors / diodes etc. in practice.

Now let us perform analysis of this ckt

Time Domain / Freq Domain



(Microcontroller is not R_L-type load
For R_L when $V_0 \uparrow$, $I \uparrow$
For microcontroller, current ~~at~~ input depends on instructions input to it and operations performed)

Assume ideal ckt,
$\Rightarrow$ All elements are ideal

Assume load is resistive

Special Class

Visiting Faculty from Spain

Important to feel confident, can solve any problem using only

10 commandments –

physical eqns
- KVL
- KCL
- Ohm's Law
- Inductor $\Rightarrow V_L = L\frac{di}{dt} \Rightarrow i = \int \frac{V_L}{L} dt + I_0$
- Capacitor $\Rightarrow i_C = C\frac{dV_C}{dt} \Rightarrow V_C = \int \frac{i}{C} dt + V_0$
- $p(t) = v(t) \cdot i(t)$

Math eqns
- $\langle x(t) \rangle_T = \frac{1}{T}\int x(t)\, dt$
- RMS value
- $\langle V_L \rangle_T = 0$ in steady state
- $\langle i_C \rangle_T = 0$ in steady state

How to solve any general power electronic circuit?

Steps –

$\Rightarrow$ Find the state variables $i_L(t)$, $V_C(t)$

In DC/DC converter, we approximate / assume that they are constant in steady state, $i_L(t) \approx I_L$, $V_C(t) \approx V_C$

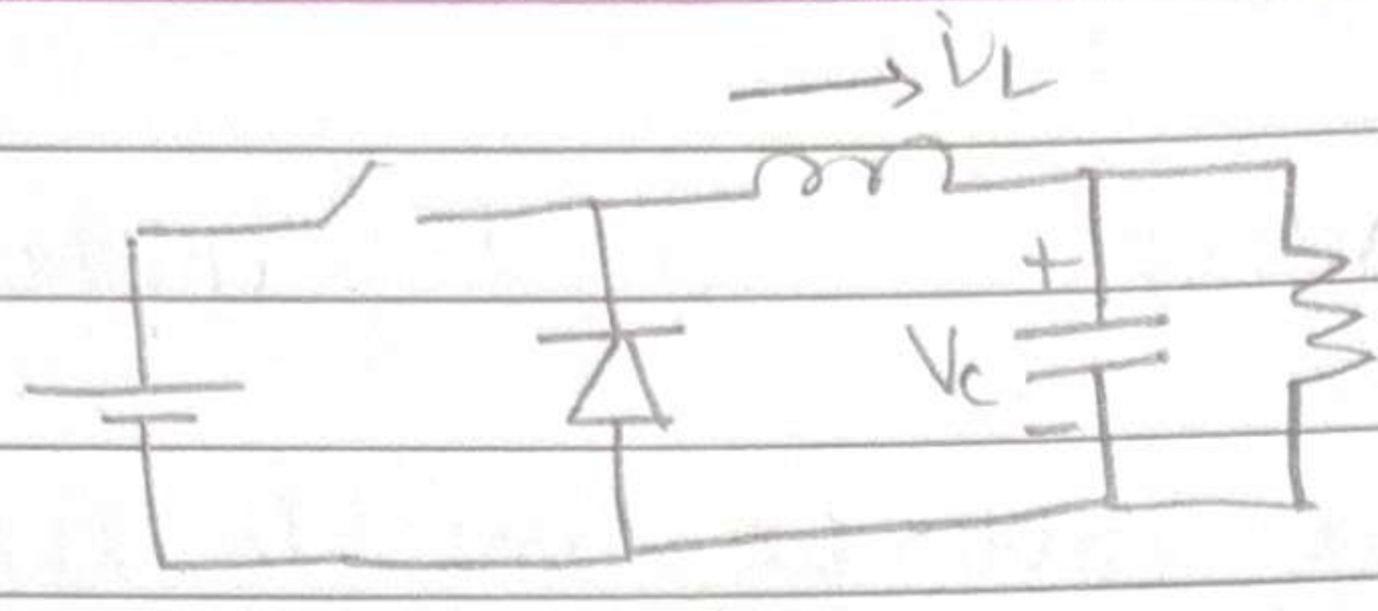
$\Rightarrow$ Find all possible circuits

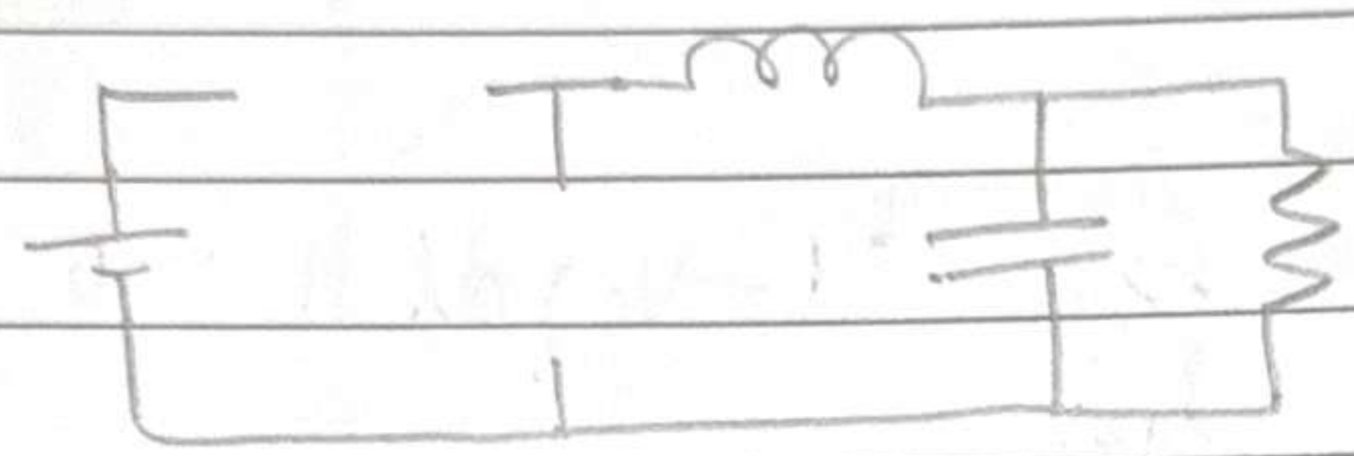
(DC/DC converters are nonlinear due to presence of switches. Make all possible ~~ct~~ switch configurations)

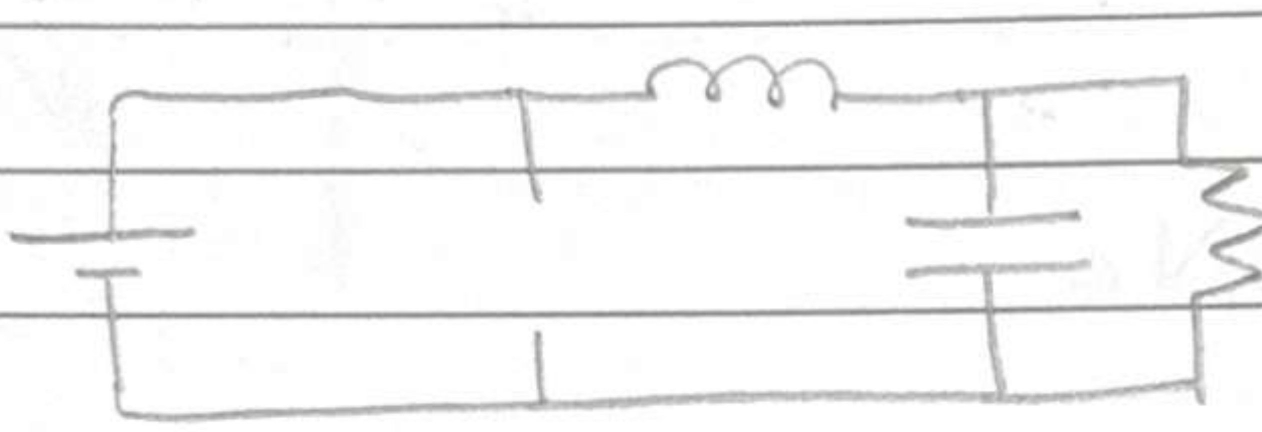
For n switches, 2^n possible circuits

$\Rightarrow$ Discard impossible circuits

In power electronics, we use switches, inductors and capacitors, Why?
Lossless!

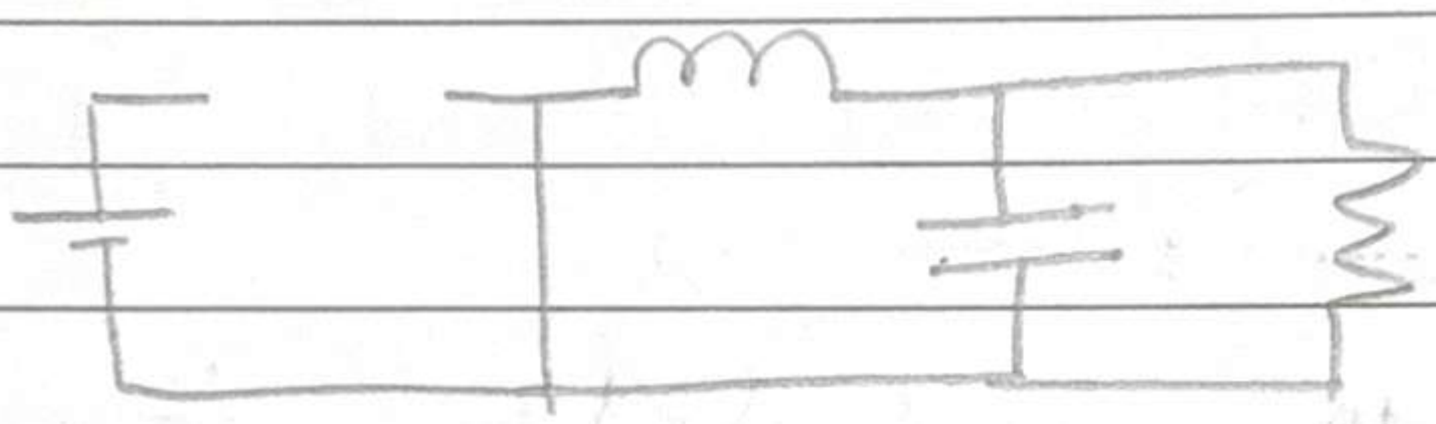


1)  X

2) 

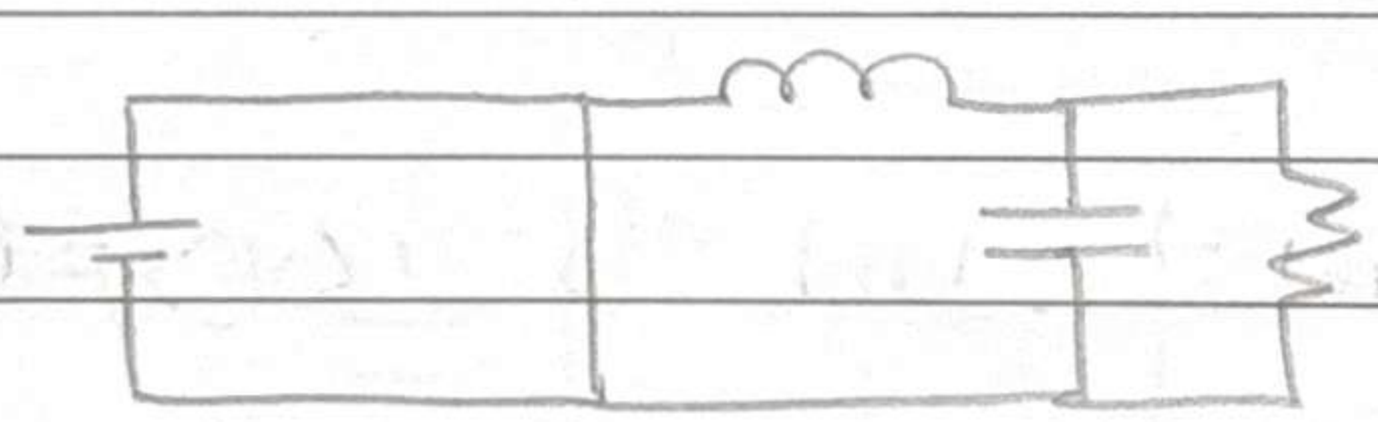
$$i_c = I_L - \frac{V_c}{R}$$

$$V_L = V_{DC} - V_c$$

3) 

$$i_c = I_L - \frac{V_c}{R}$$

$$V_L = -V_c$$

4)  X

⇒ Find V_L and i_c for the remaining circuits using parameters (V_{DC}, R, L, C, i_L, v_c)

(Note that we are approximating constant value even though actually there is ripple associated with it)

⇒ Solve the equations using $\langle V_L \rangle_T = 0$

$\langle i_c \rangle_T = 0$

PTO

$$\frac{1}{T}\int_0^T V_L(t)\,dt = 0$$

There are two times corresponding to the two circuits

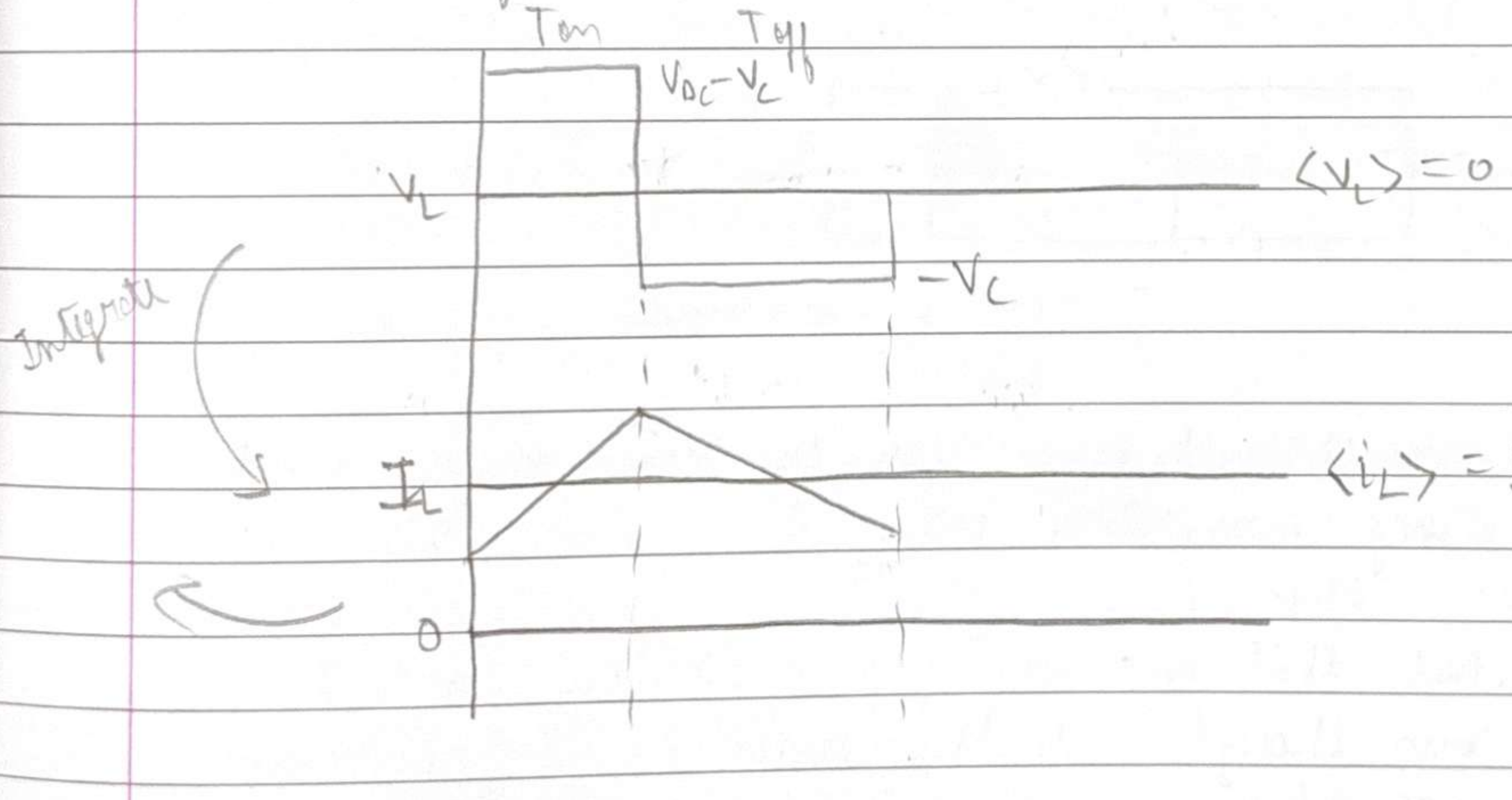
$$\frac{1}{T}\left[\int_0^{T_{on}} (V_{DC}-V_C)\,dt + \int_{T_{on}}^{T_{on}+T_{off}} (-V_C)\,dt\right] = 0$$

Solving we get,

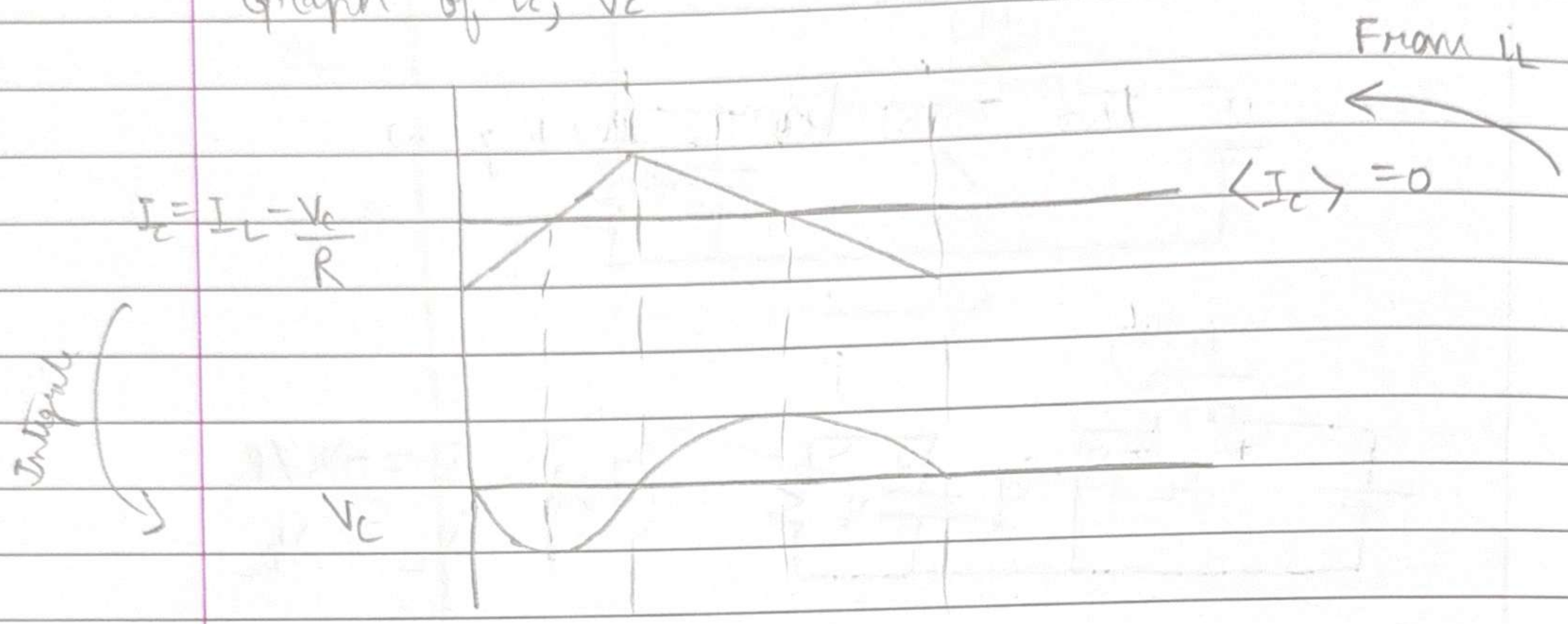
$$V_C = \frac{T_{on}}{T} V_{DC} = D V_{DC}$$

$$I_L = \frac{V_{DC}}{R}$$

Graph of V_L, i_L



Graph of i_c, V_c



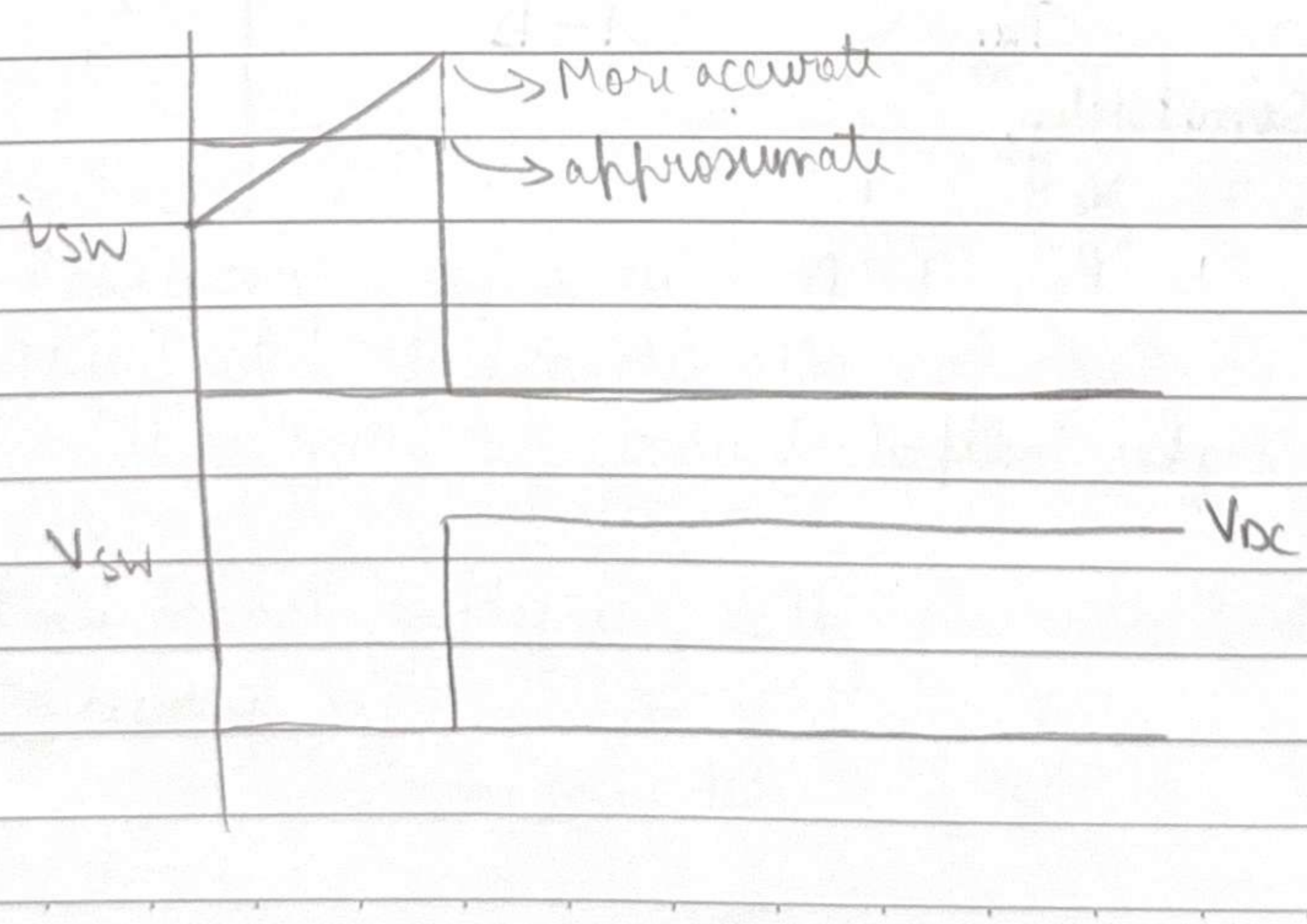
Now, we want to find the maximum peak-to-peak, which is the ripple

$$\Delta i_L = \frac{1}{T}\int_{t_1}^{t_2} V_L \, dt$$

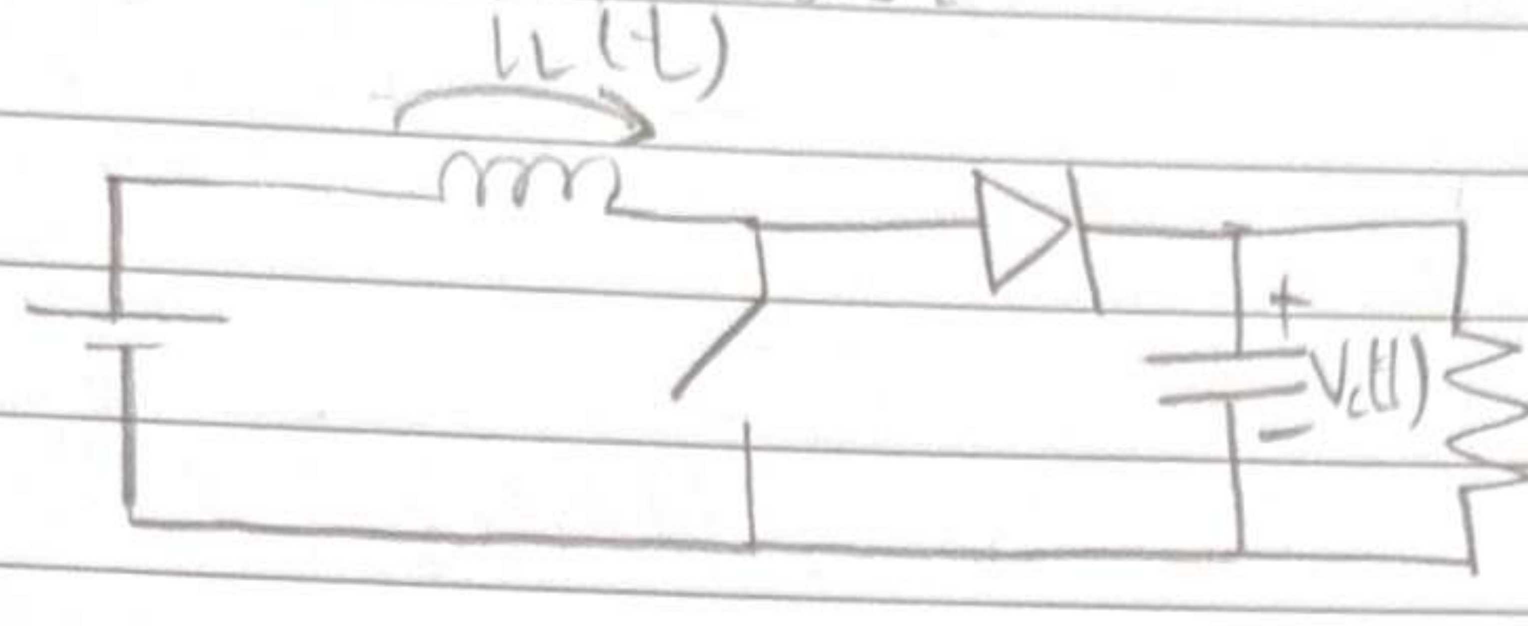
$\to$ Max

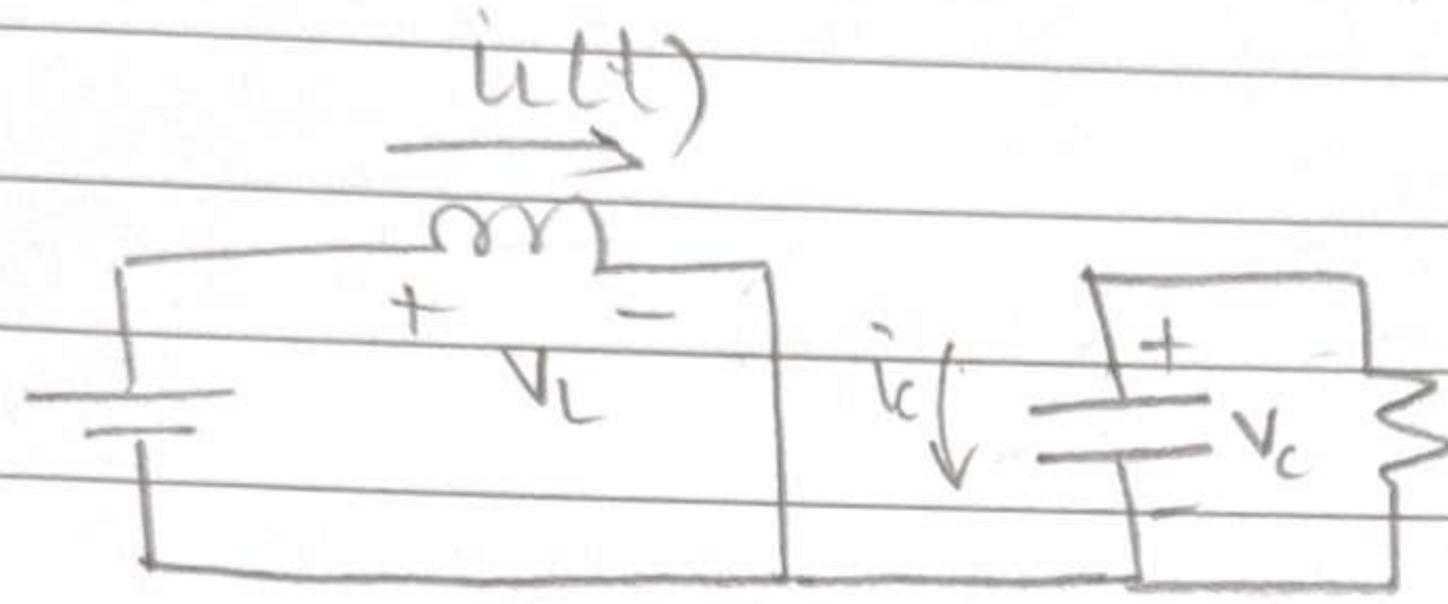
$$\Delta i_L = \frac{1}{T}\int_0^{T_{on}} (V_{DC} - V_{OCO}) \, dt$$

Let us plot i and v for the switch



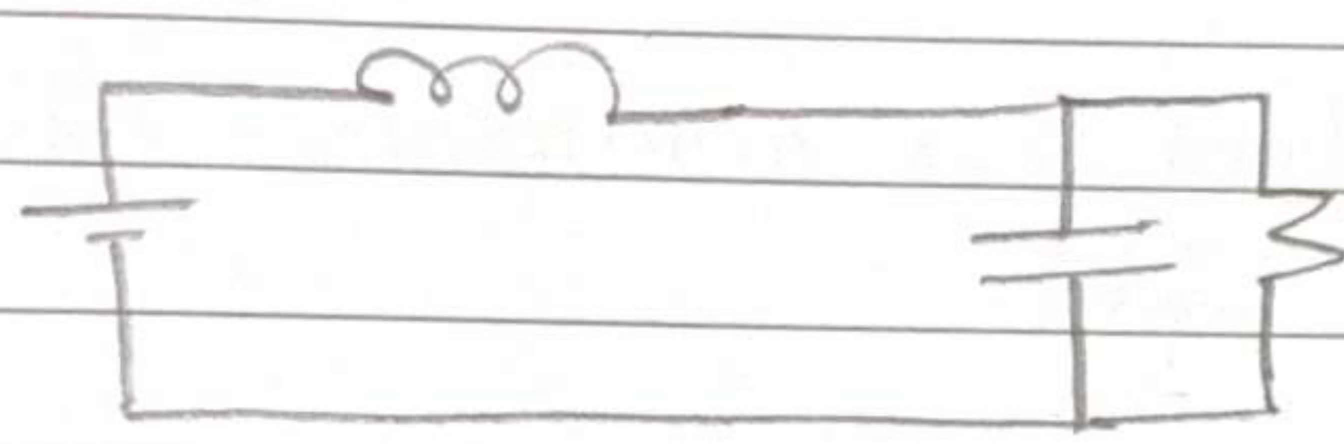
DC/DC Boost Converter





(T_{on}) $\quad i_c = -V_c/R$

$V_L = V_{DC}$



(T_{off}) $\quad i_c = i_L - V_c/R$

$V_L = V_{DC} - V_c$

$$\frac{1}{T}\left(V_{DC} T_{on} + (V_{DC} - V_c)(T - T_{on}) \right) = 0$$

$$V_{DC} T_{on} = (V_c - V_{DC})(T - T_{on})$$

$$\cancel{V_{DC} T_{on}} = V_c T - V_c T_{on} - V_{DC} T + \cancel{V_{DC} T_{on}}$$

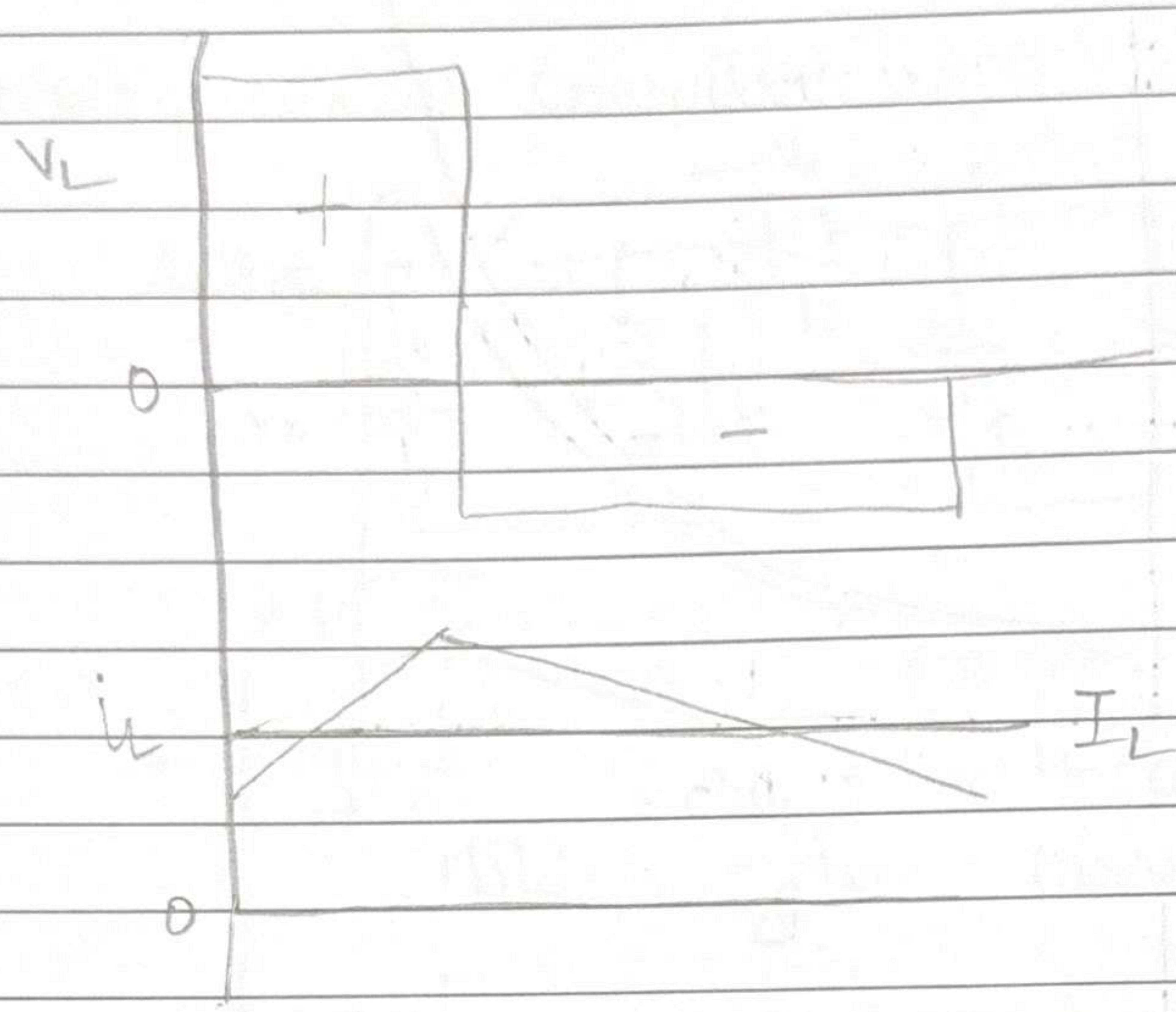
$$V_c T_{off} = V_{DC} T$$

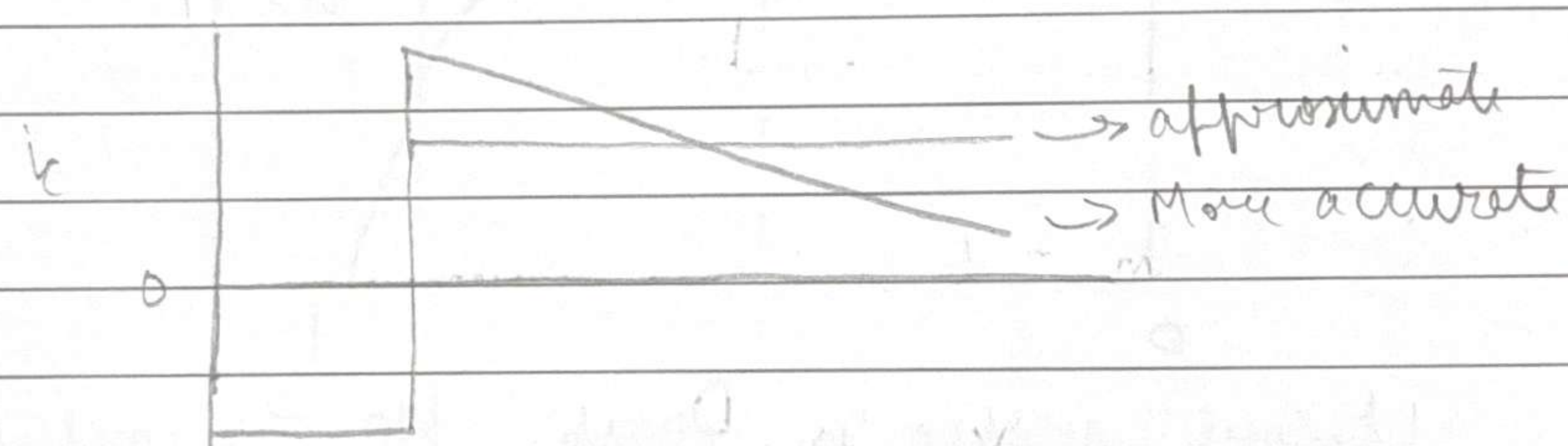
$$V_c = \frac{T}{T_{off}} V_{DC} = \frac{V_{DC}}{1-D}$$

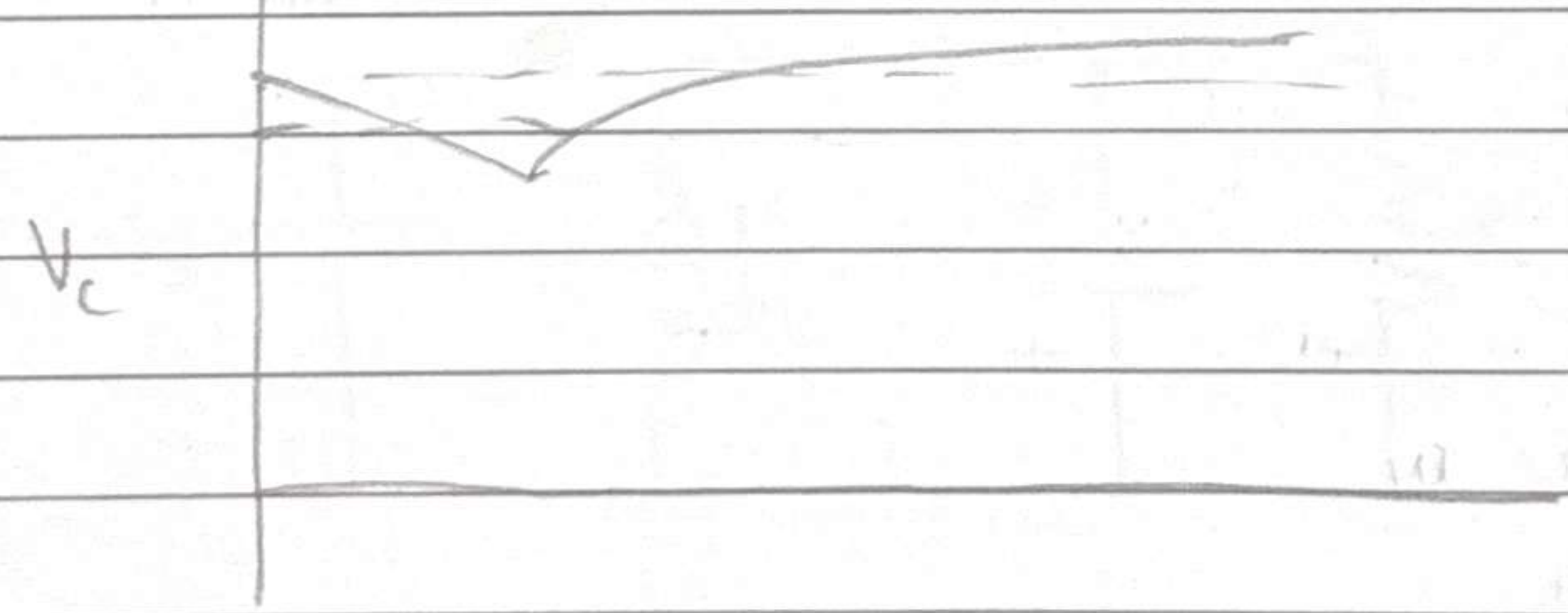
similarly,

$$I_L = \frac{V_c}{R} \cdot \frac{1}{1-D}$$

$\Rightarrow$ Provides higher output!



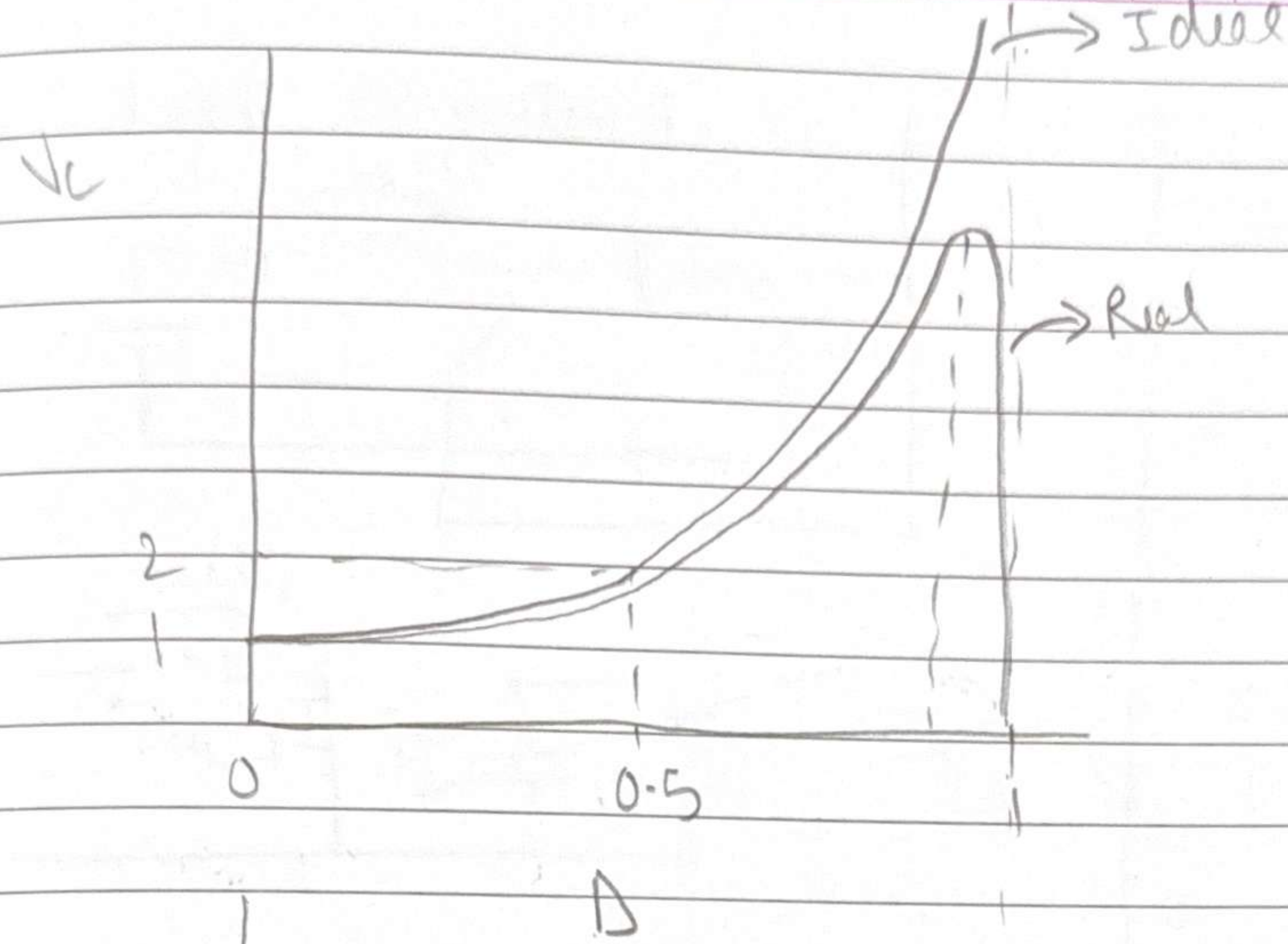




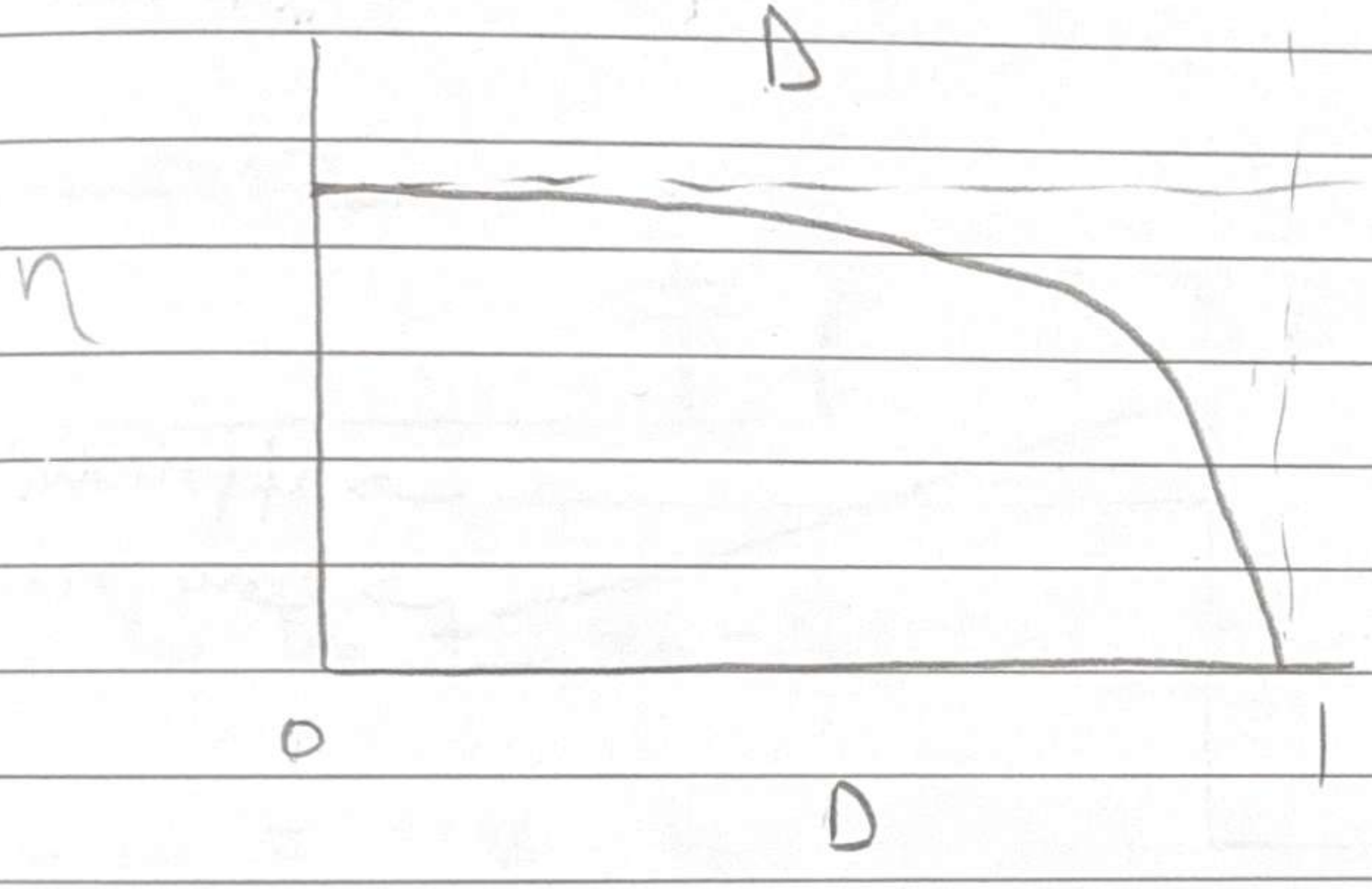
Right now, efficiency = 100%
Now what if consider the real case —
Resistance in inductor coil!

Same exact analysis with the modified ckt

$$\text{Efficiency, } \eta = \frac{P_{out}}{P_{out} + P_{loss}} = \frac{\frac{V_c^2}{R}}{\frac{V_c^2}{R} + R_{esr} I_L^2} = \frac{1}{1 + \frac{R_{esr}}{R} \cdot \frac{1}{(1-D)^2}}$$

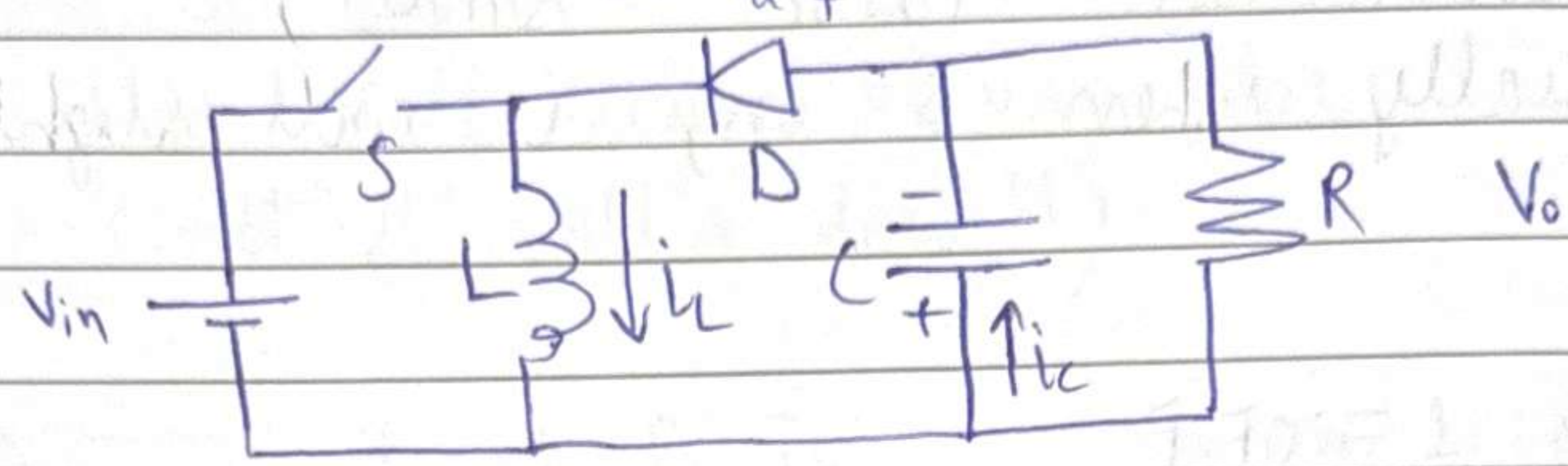


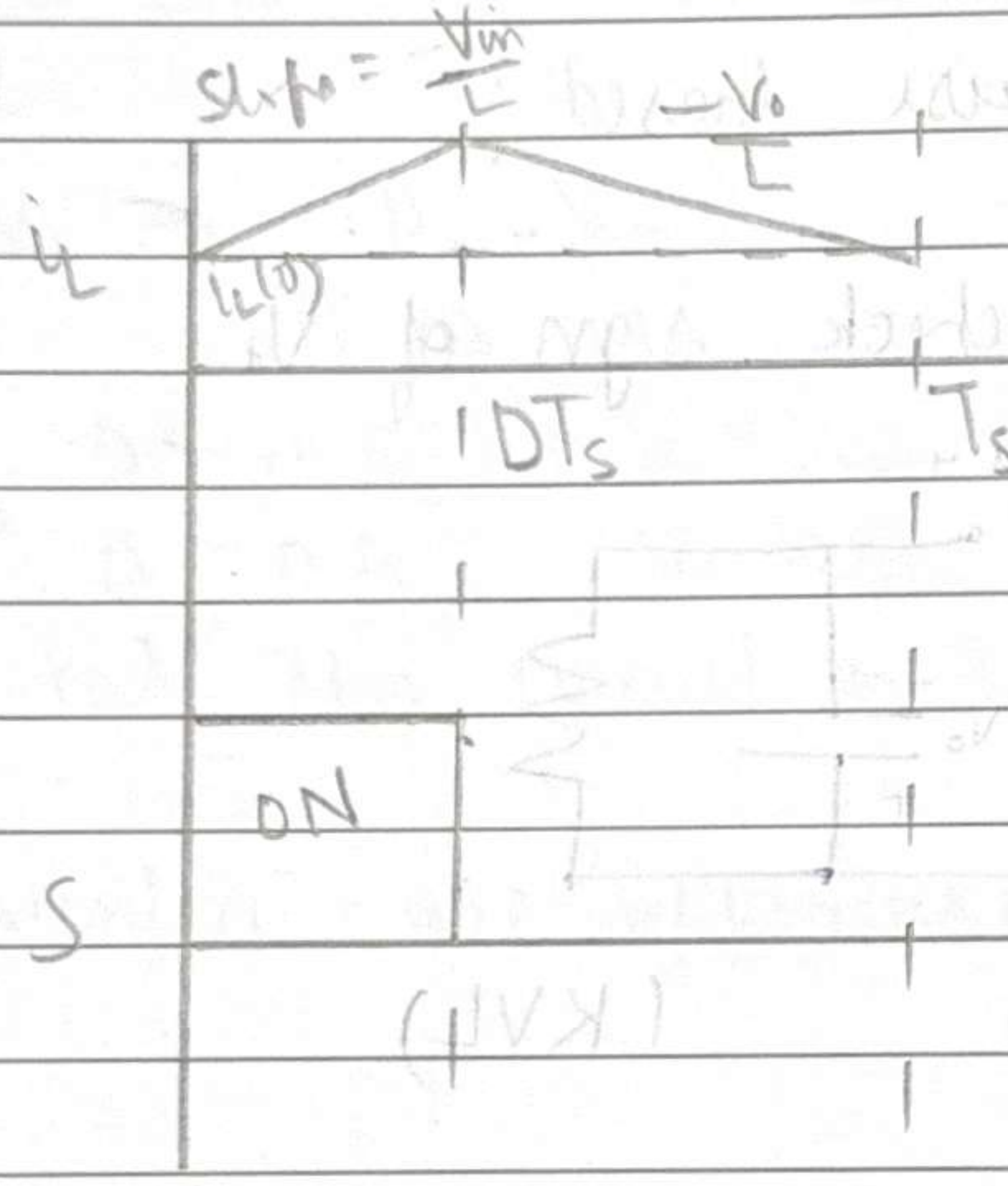
Ideal
Vc
Real
2
1
0
0.5
1
D



η
0
1
D

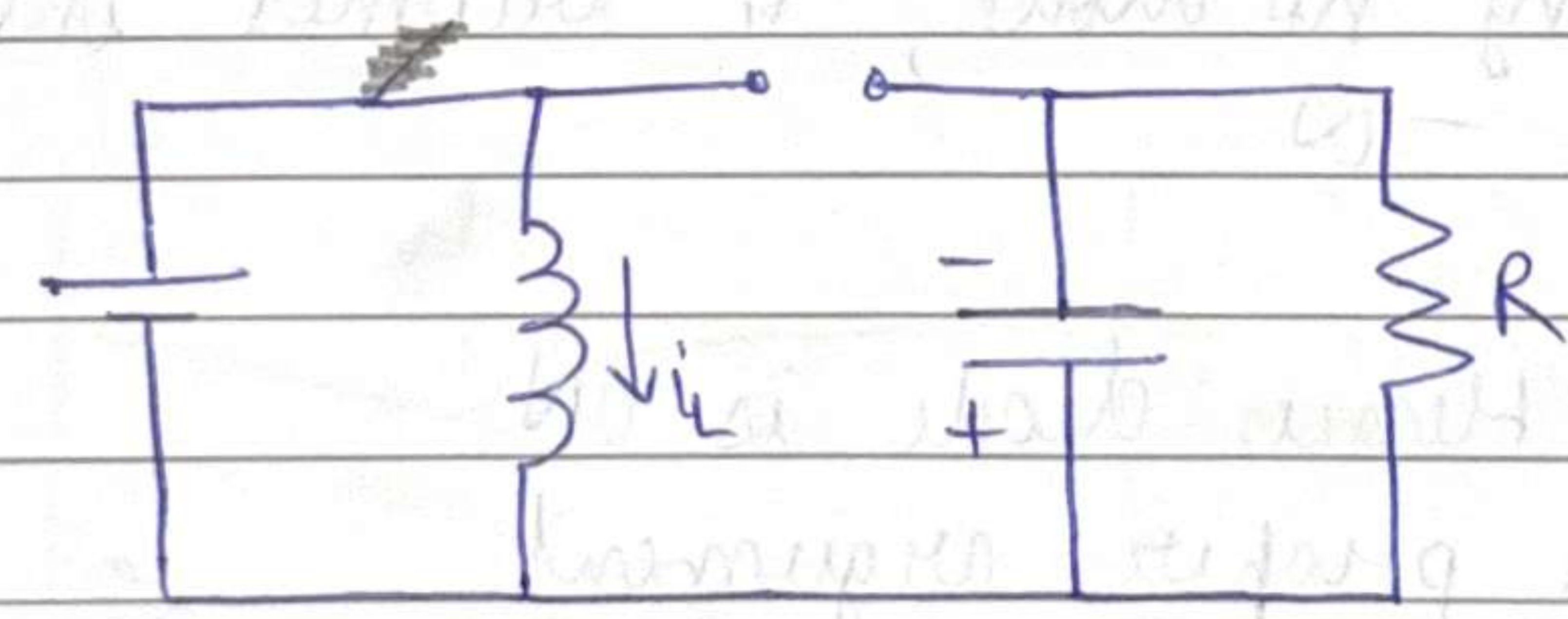
Solving another converter



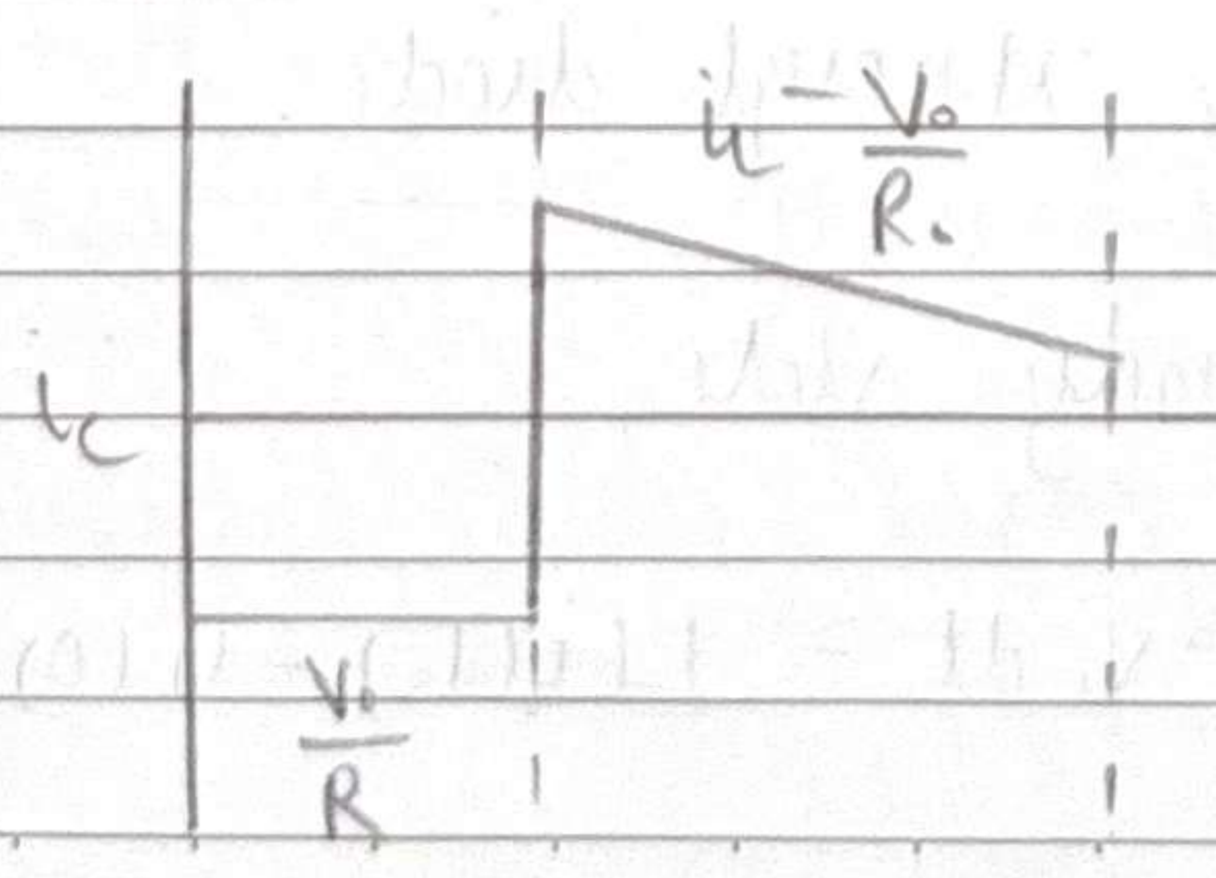


Periodic steady-state
$i_L(0+t) = i_L(T_s+t)$
(assumed reached)

When S = ON, diode is reverse biased



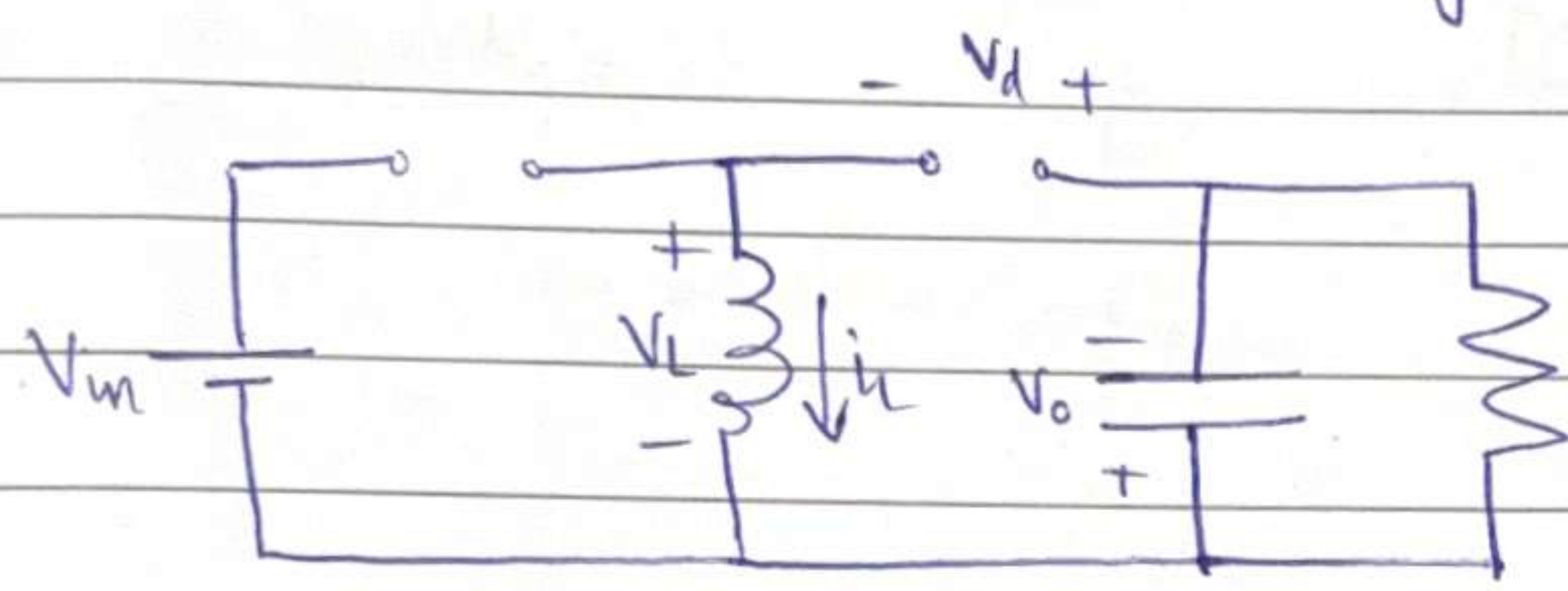
$V_{in} = V_L = L\frac{di_L}{dt}$, $i_c = \frac{V_o}{R}$



For the capacitor, we make the small ripple approximation ($\Delta V_c \rightarrow$ small)
Actually when S = on, i_c will slightly decrease

Now S = OFF
Is diode fwd / reverse biased?

Remove diode and check sign of V_d



$$0 + V_L + V_d + V_o = 0 \qquad \text{(KVL)}$$

$$V_d = -V_L - V_o$$

$$V_L = L\frac{di_L}{dt}$$

Assuming no diode, i_L becomes instantaneously 0

$$\therefore \frac{di_L}{dt} = -\infty$$

$\therefore$ $V_d > 0$ Hence diode is ON
This is a proper argument.

A quick intuitive argument is - as soon as S is turned off, inductor current needs a path to flow, so it flows through diode.

Due to periodic steady state,

$$i_L(0) = i_L(T_s)$$

$$V_L = L\frac{di_L}{dt} \Rightarrow \int_0^{T_s} V_L\,dt = L(i_L(T_s) - i_L(0)) = 0$$

Hence, in periodic steady state, average voltage across inductor is 0.
This is called Volt-second (V_s) law
(not 2nd volt's law !!)

$$\frac{V_{in}}{L} DT_s = \frac{V_o}{L}(1-D)T_s \qquad (\text{from } i_L(0) = i_L(T_s))$$

$$V_o = \frac{D}{1-D} V_{in}$$

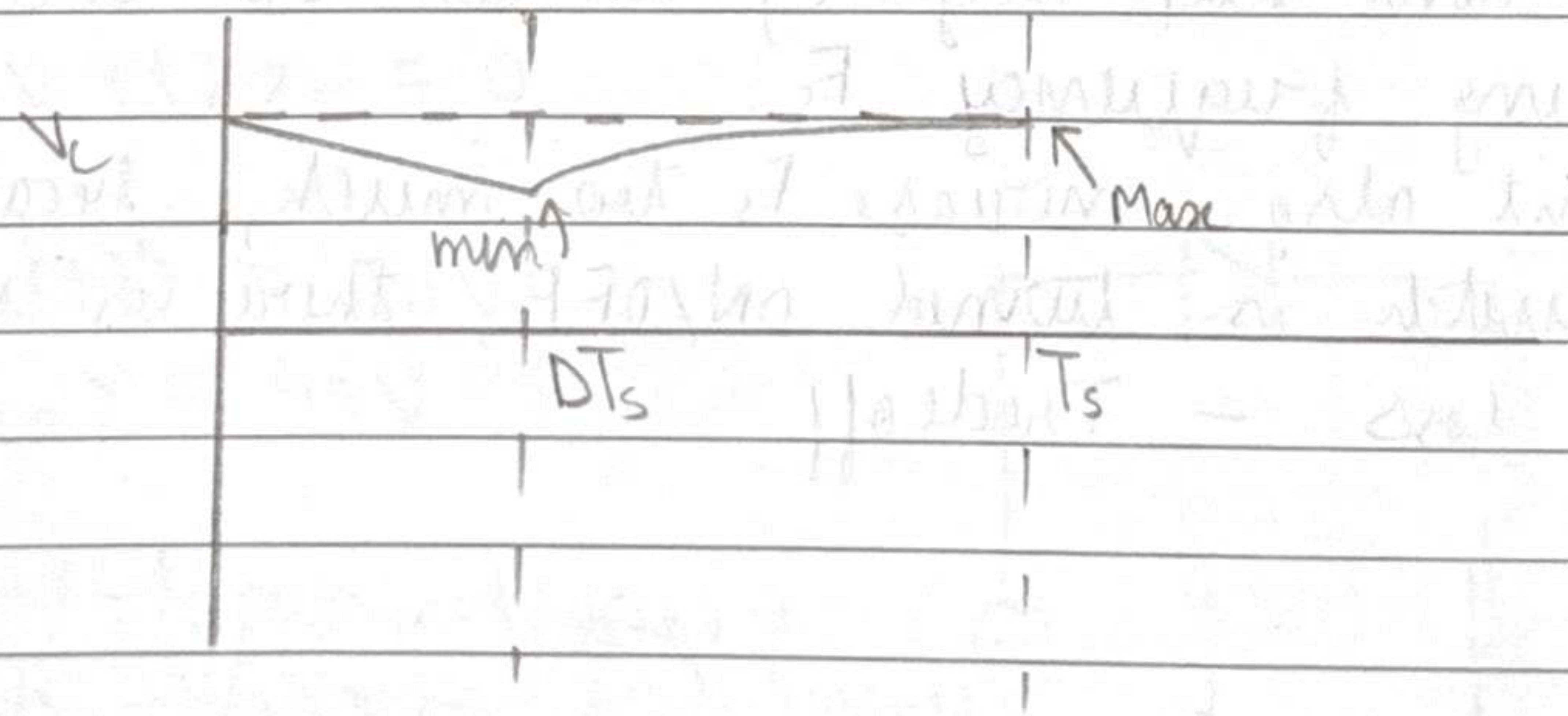
If $D < 0.5$, $V_o < V_{in}$
$D > 0.5$, $V_o > V_{in}$

We call this circuit a Buck-Boost Converter

(convention - use uppercase letters for average value)

When S = OFF,

$$i_c = -\frac{V_o}{R_o} + i_L$$



We can set the D according to what V_o we want.

To reduce ripple factor $\Rightarrow$ increase C

HW : (i) Find Δi_L

(ii) Find the condition for which * happens

$(V_c = V_0)$

Consider ripple factor ΔV_c

Need a relation b/w ΔV_c and C

$$\int_0^{DT_s} i_c \, dt = \int_0^{DT_s} C \frac{dv_c}{dt} \, dt$$

$$\Rightarrow \quad -\frac{V_0}{R_0} DT_s = C\left(V_c^{min} - V_c^{max}\right) \longrightarrow \text{Wrong}$$

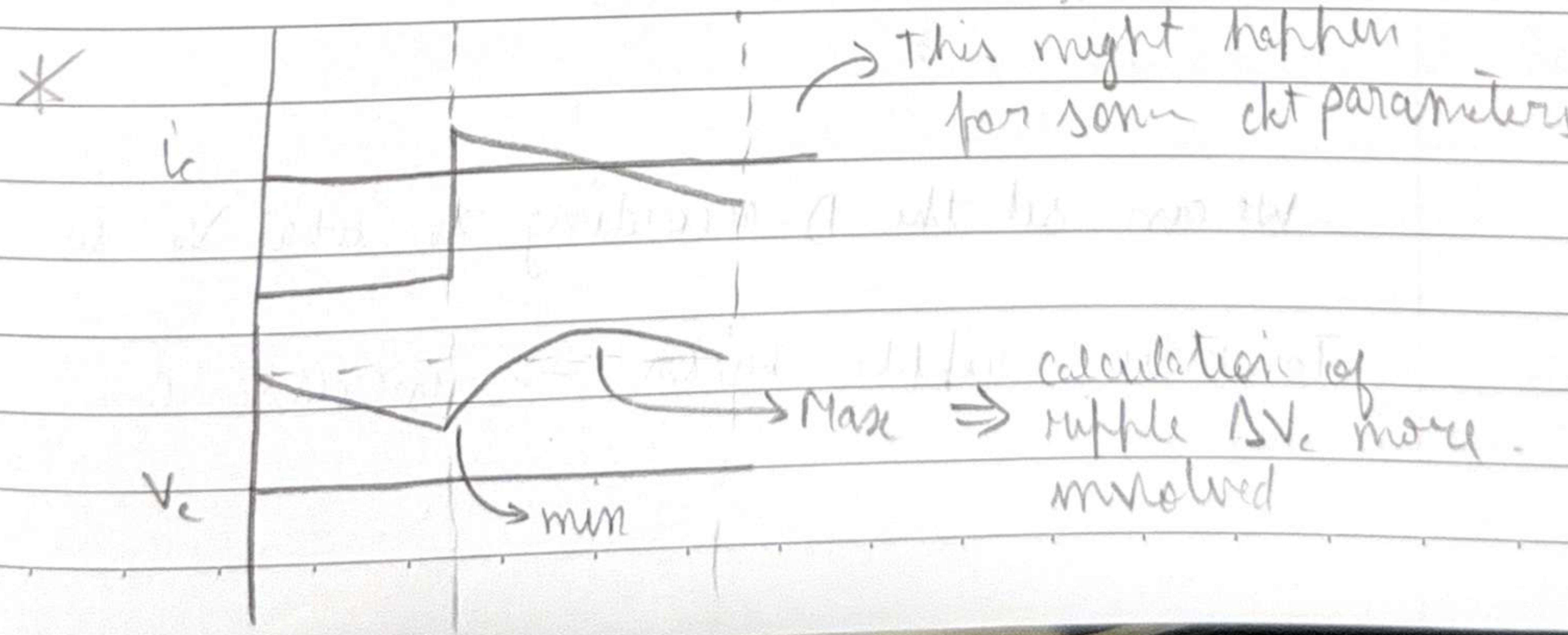
$$\therefore \quad \Delta V_c = \frac{V_0}{R_0} \frac{DT_s}{C}$$

Large load means large load current
means small R_L

$R \downarrow \Rightarrow \Delta V_c \uparrow$
$C \uparrow \Rightarrow \Delta V_c \downarrow$
$T_s \downarrow \Rightarrow F_s \uparrow \Rightarrow \Delta V_c \downarrow$

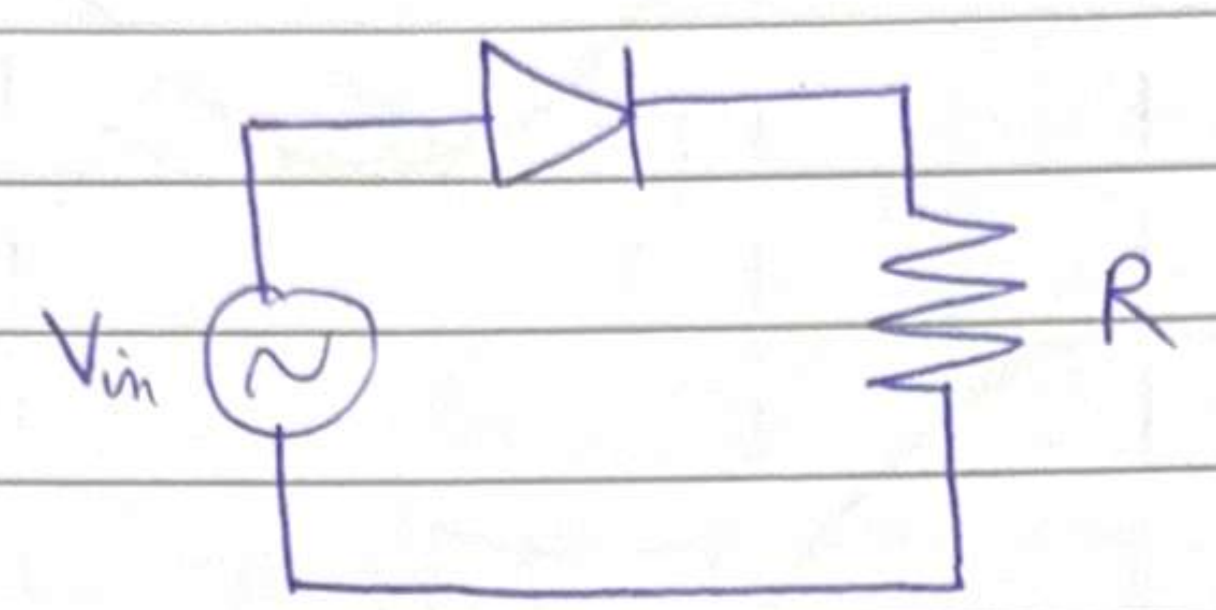
If we can't keep large C, we can also increase switching frequency F_s.
We can't also increase F_s too much, because every time switch is turned ON/OFF, there is some energy loss – Tradeoff

*



** 11.

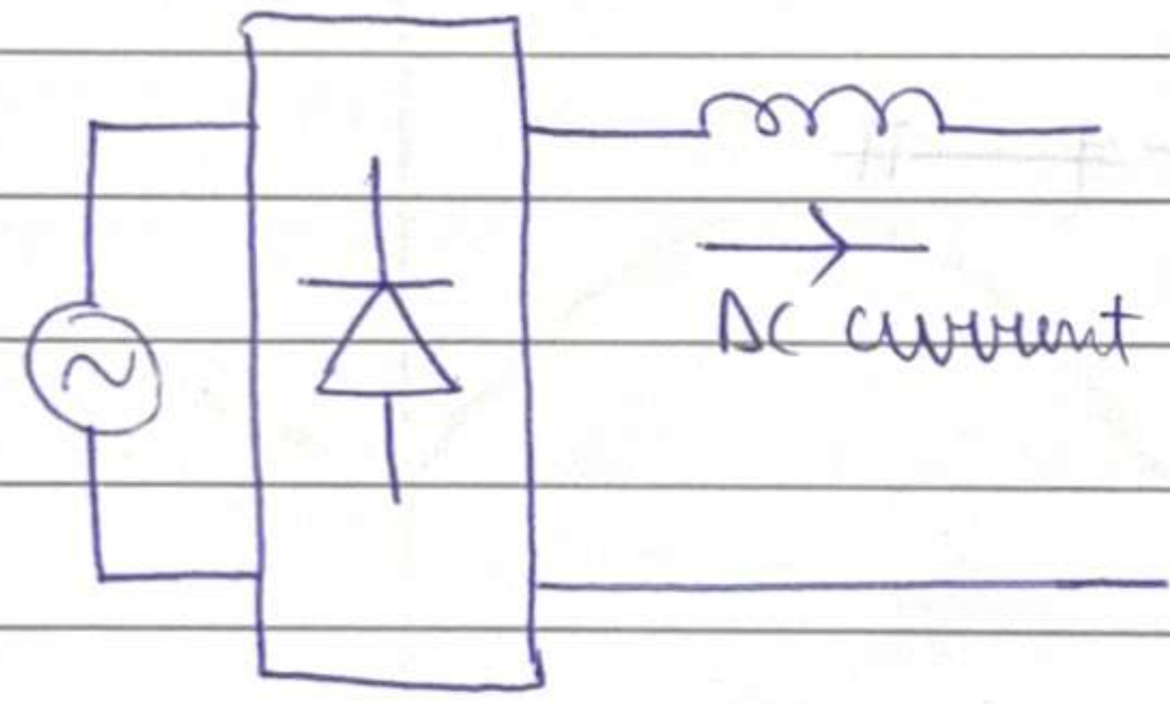
Diode Rectifiers

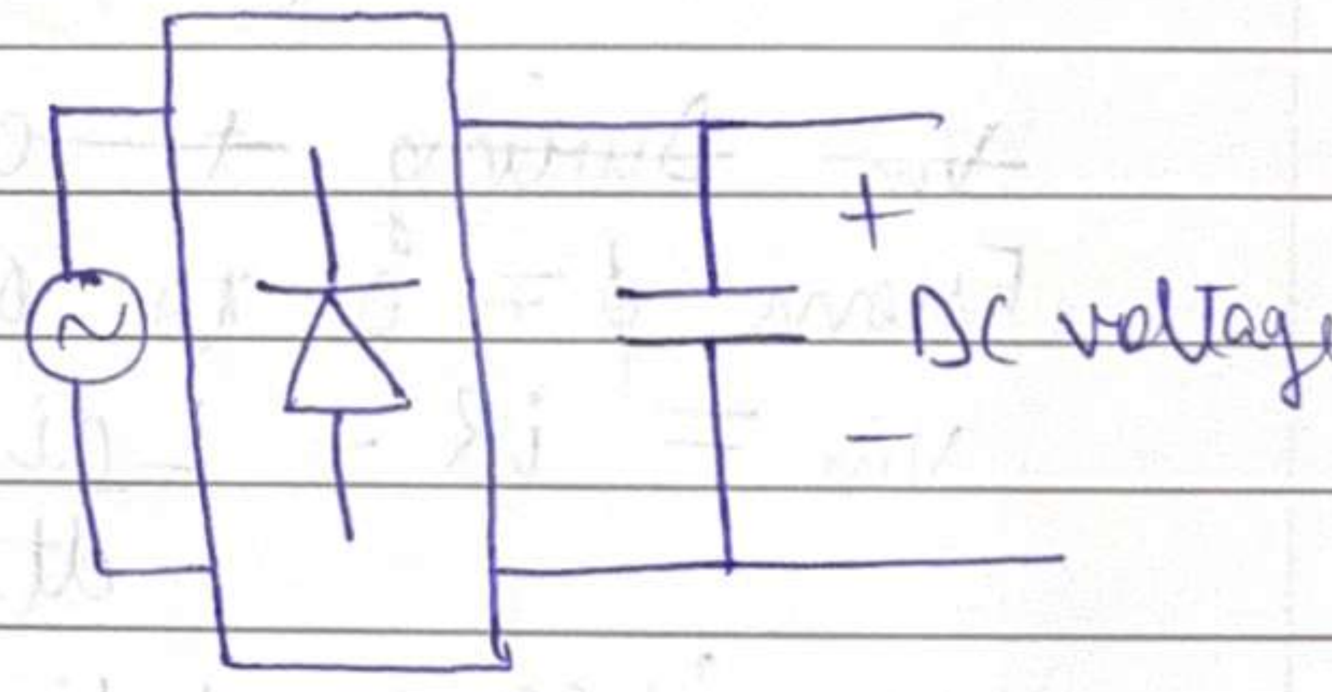


In power applications, there are broadly two types of loads -

- Capacitive output, e.g; phone charger
- Inductive output, e.g; DC Motor

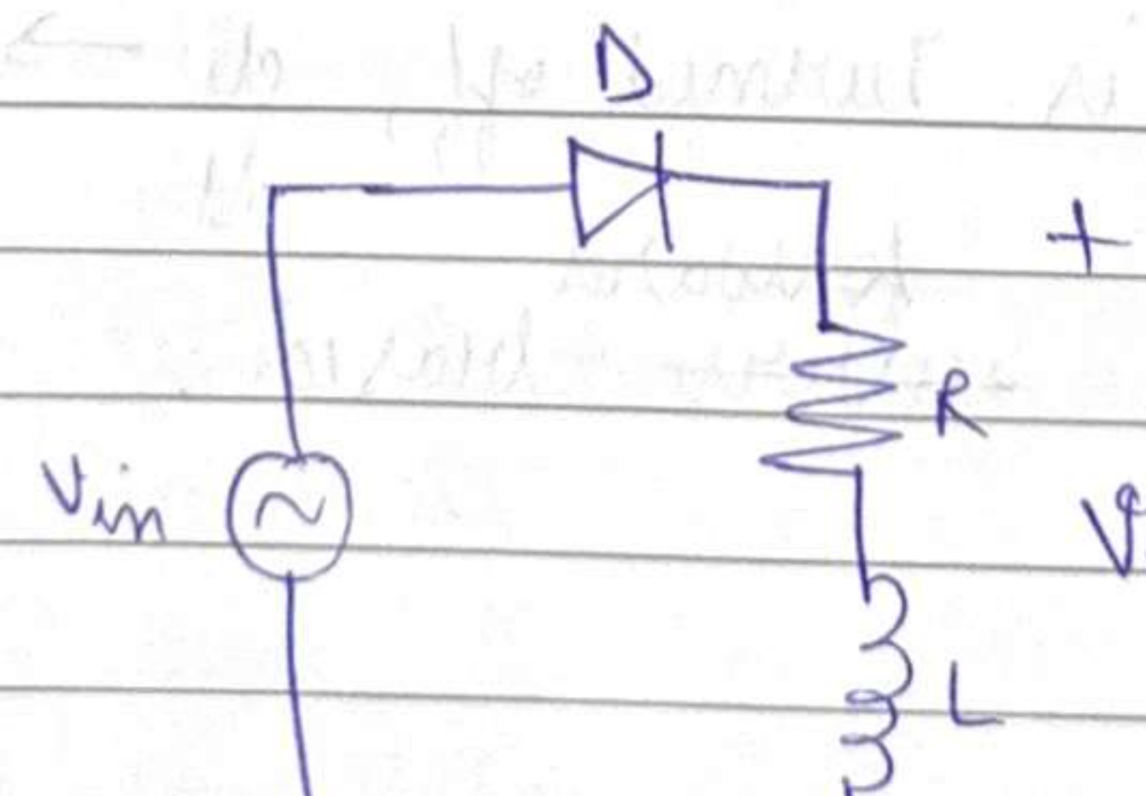
Rectifiers - convert AC to DC





This ckt had practical applications in earlier times	This ckt has practical applications today

Now let us consider the circuit -



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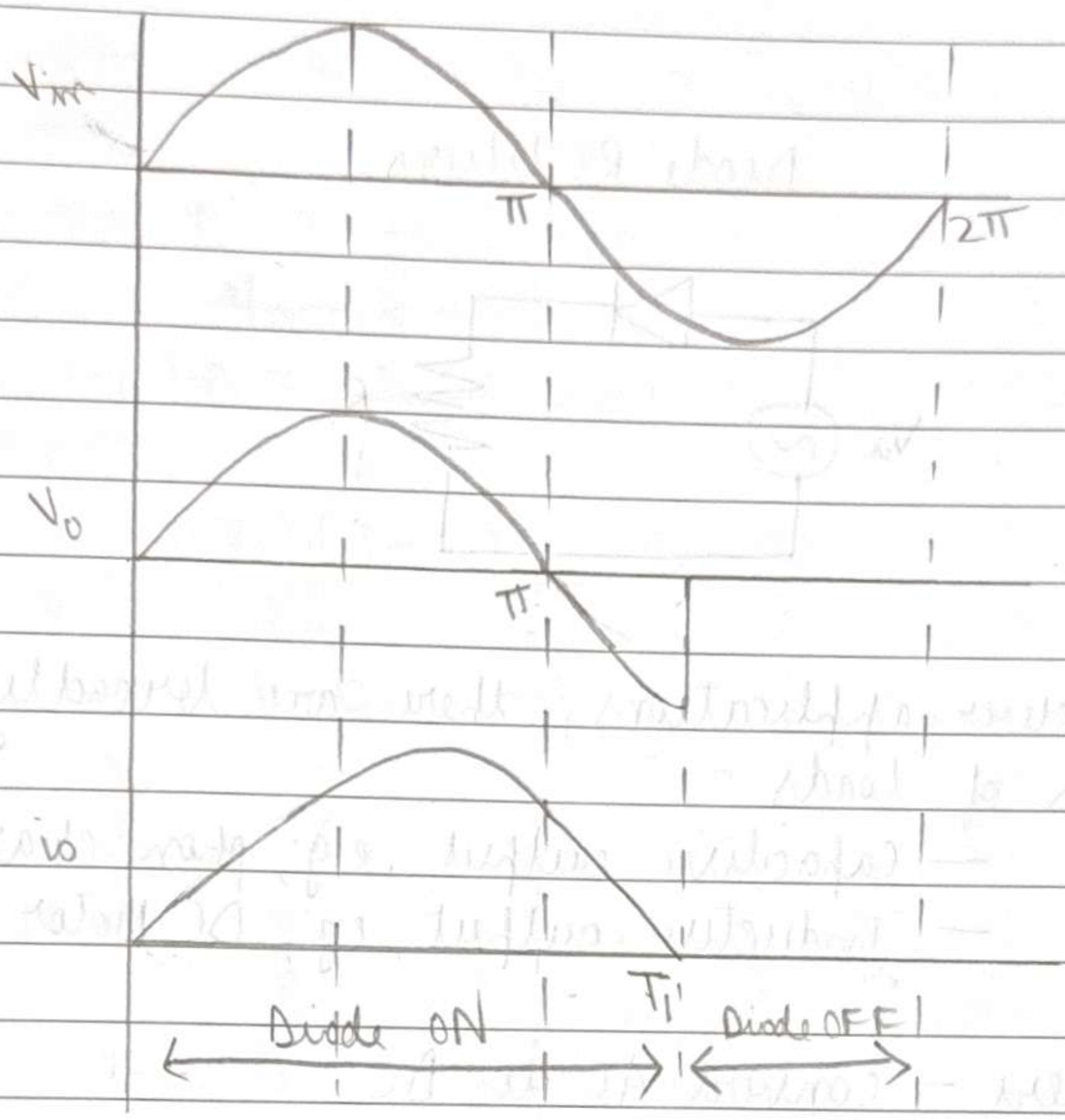


... (Periodic steady state)

~~V_{in} During $t = 0$ to $t = \pi$~~

From $\phi = 0$ to $\phi = \pi$

$$V_{in} = iR + L\frac{di}{dt}$$

$$(V_{in} - iR) = L\frac{di}{dt}$$

At $\phi = \pi^+$,

$$V_d = V_{in} - Ri - L\frac{di}{dt}$$

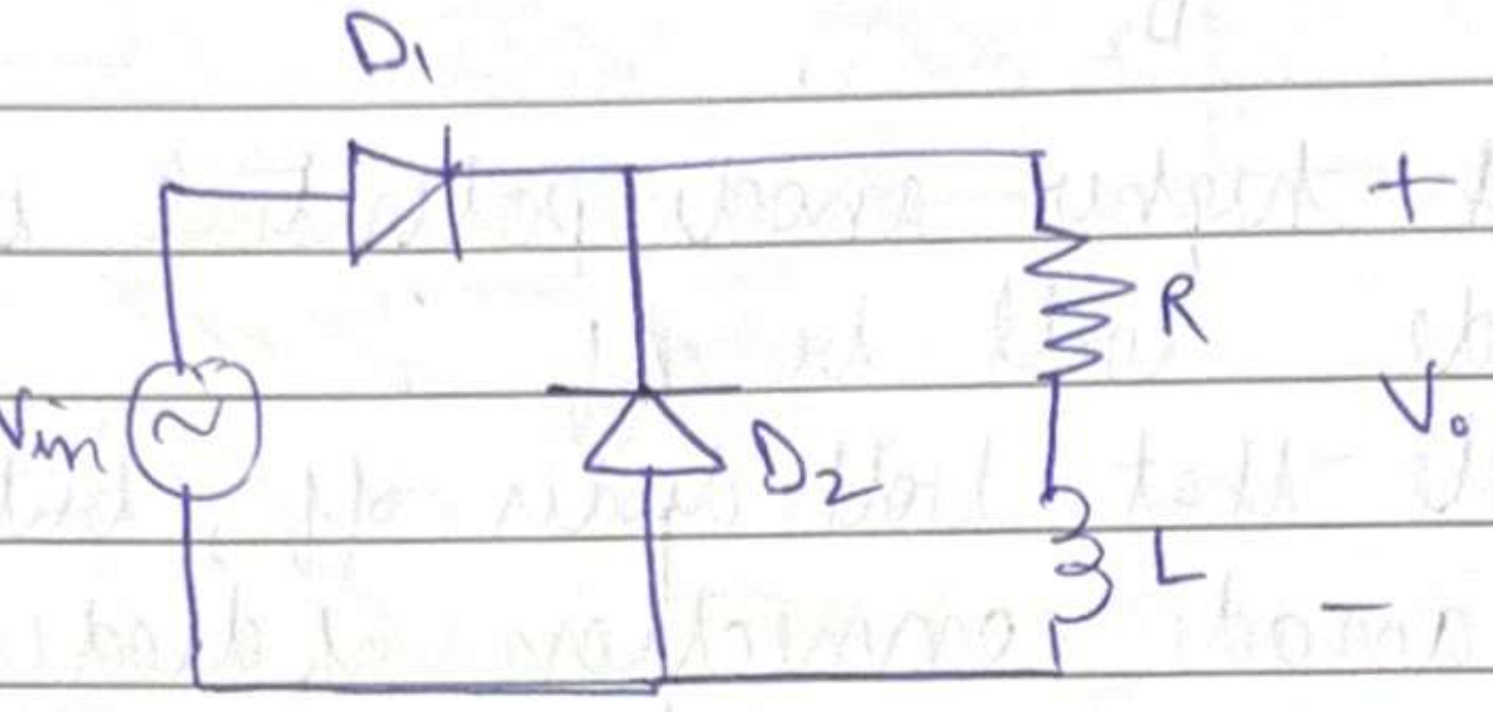
If the diode is turned off, $\frac{di}{dt} \to -\infty$, hence

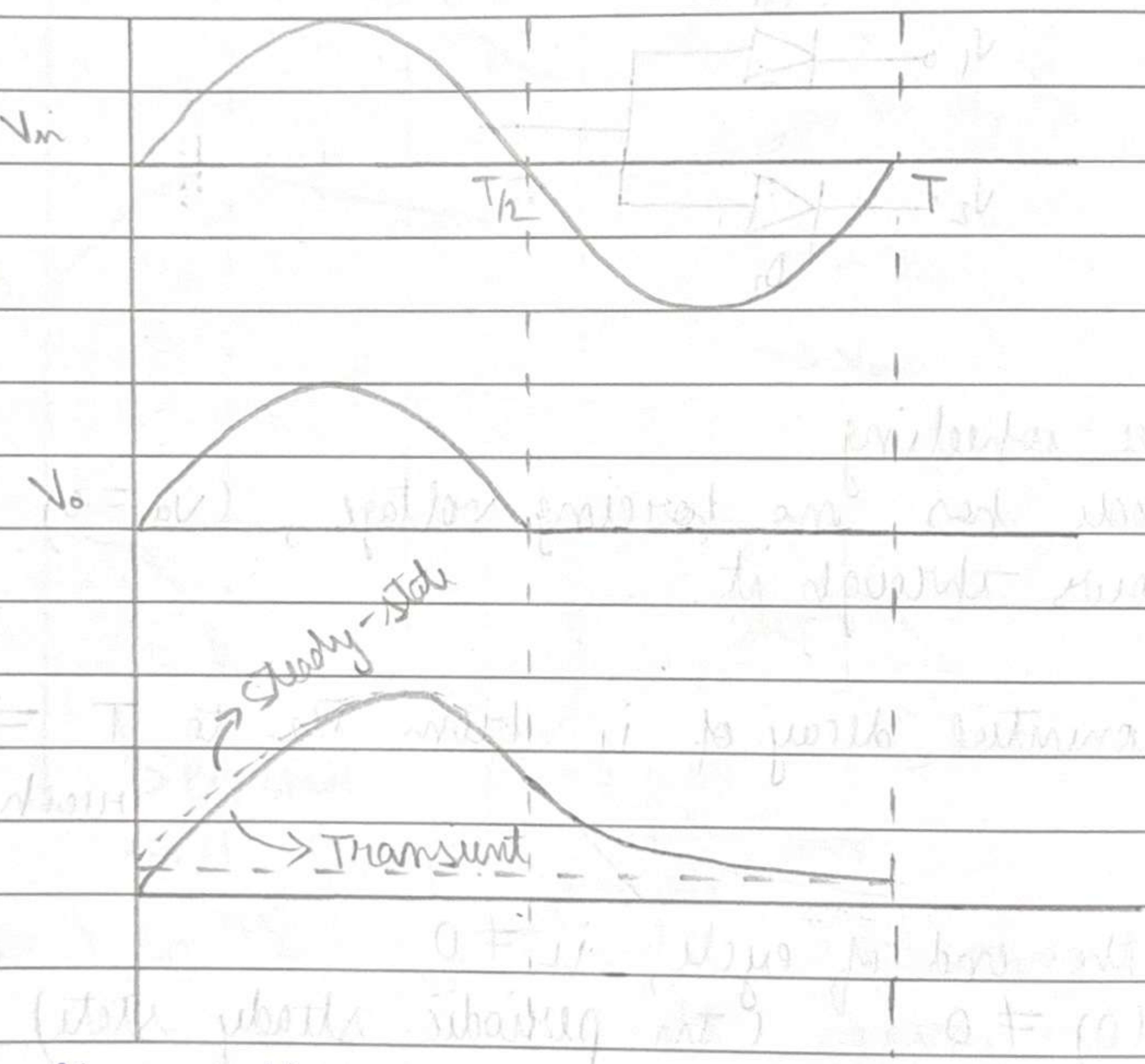
diode becomes ~~reverse~~ forward -biased.

~~$\pi \le T_1 \le 2\pi$~~ $\quad T/2 \le T_1 \le T$

~~If $R=0$, $T_1 = 2\pi$~~ $\quad$ If $R = 0$, $T_1 = T$

Note that $\langle v_L \rangle_T = 0$





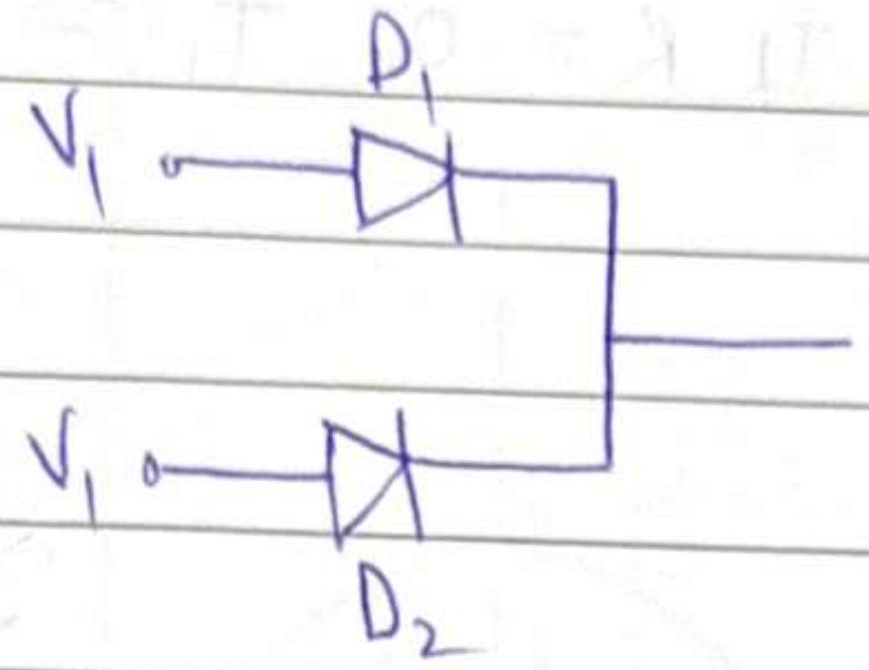
~~After $\frac{T}{2}$: $-Ri = L\frac{di}{dt}$~~

At $t = 0^+$, since D_1 has higher anode potential, it conducts

At $t = \frac{T}{2}^+$, since D_2 has " " ", it conducts

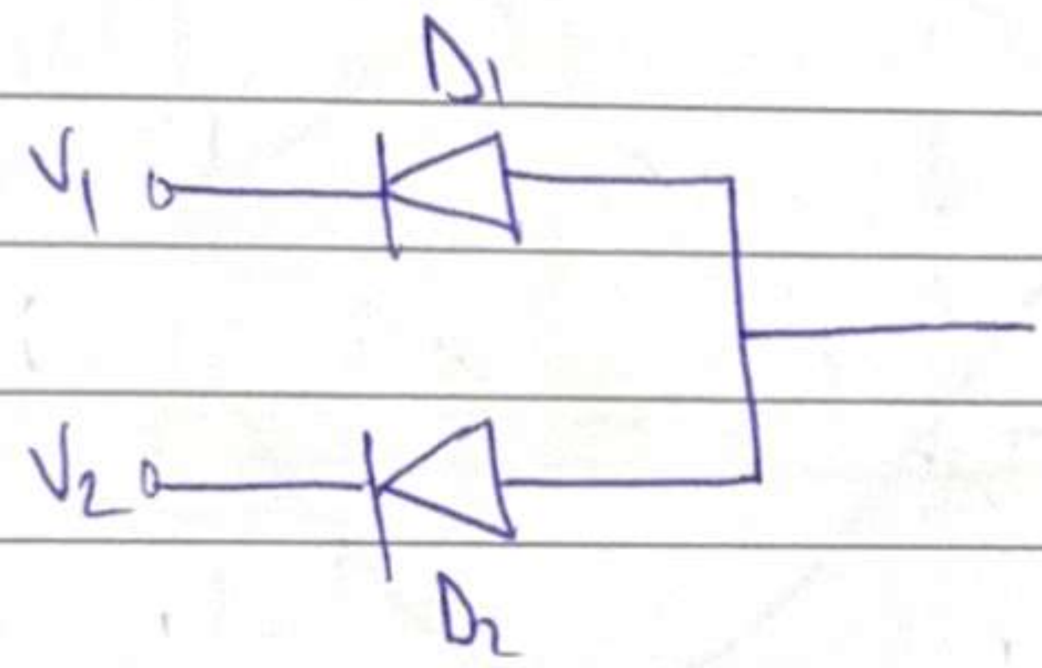
$\Rightarrow$ One of D_1, D_2 must be ON because di/dt, it forces one of the diodes to conduct

Common cathode connection of diodes



Diode with higher anode potential will conduct and other diode will be off.
(Also possible that both diodes off, but both can't be on)

Common anode connection of diodes



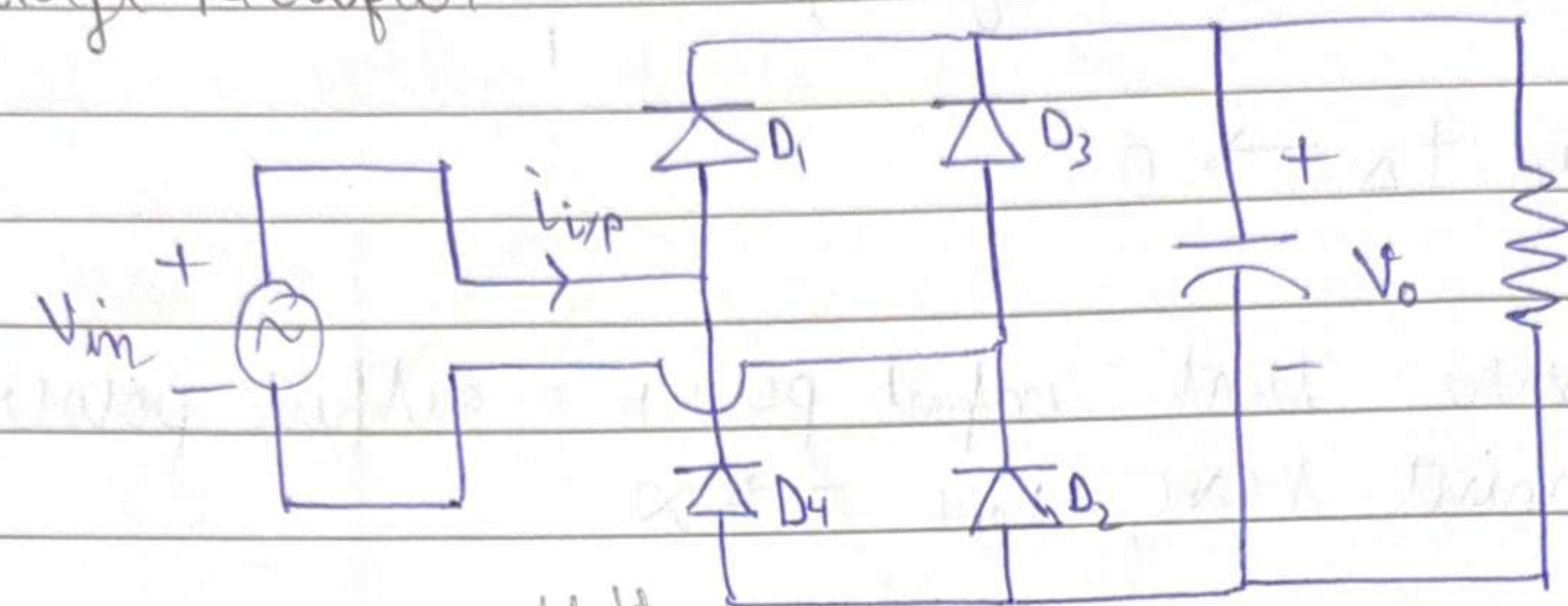
Free-wheeling
Diode has no forcing voltage, ($V_d = 0$) but current flows through it.

$\therefore$ Exponential decay of i_L from $T/2$ to T $\Rightarrow$ does not reach zero.

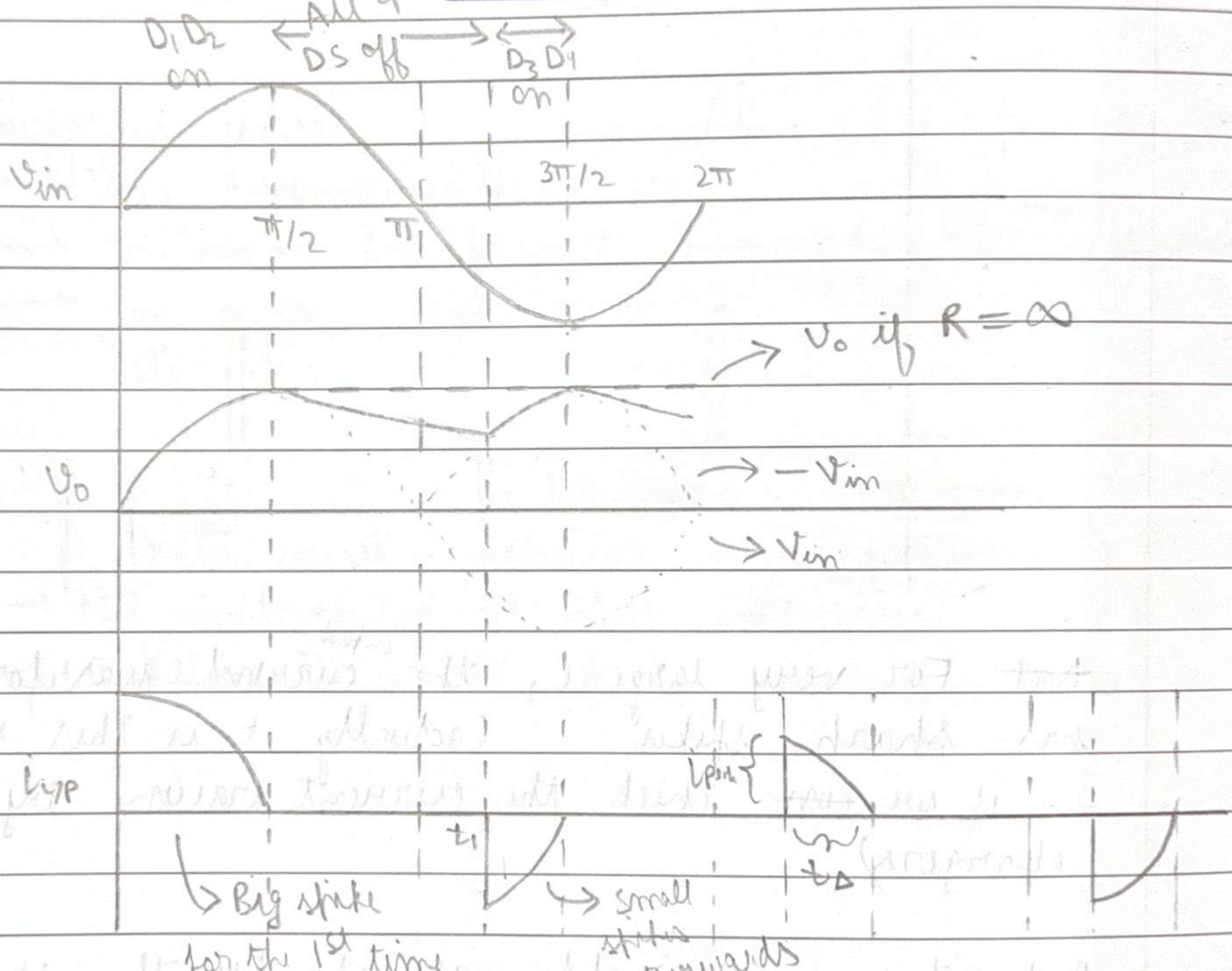
$\therefore$ At the end of cycle, $i_L \neq 0$
$\therefore$ $i_L(0) \neq 0$ (In periodic steady state)

Try to draw waveforms for RLE load ~~to~~ bridge rectifier

Bridge Rectifier



(Use common cathode and common anode rules of diodes)



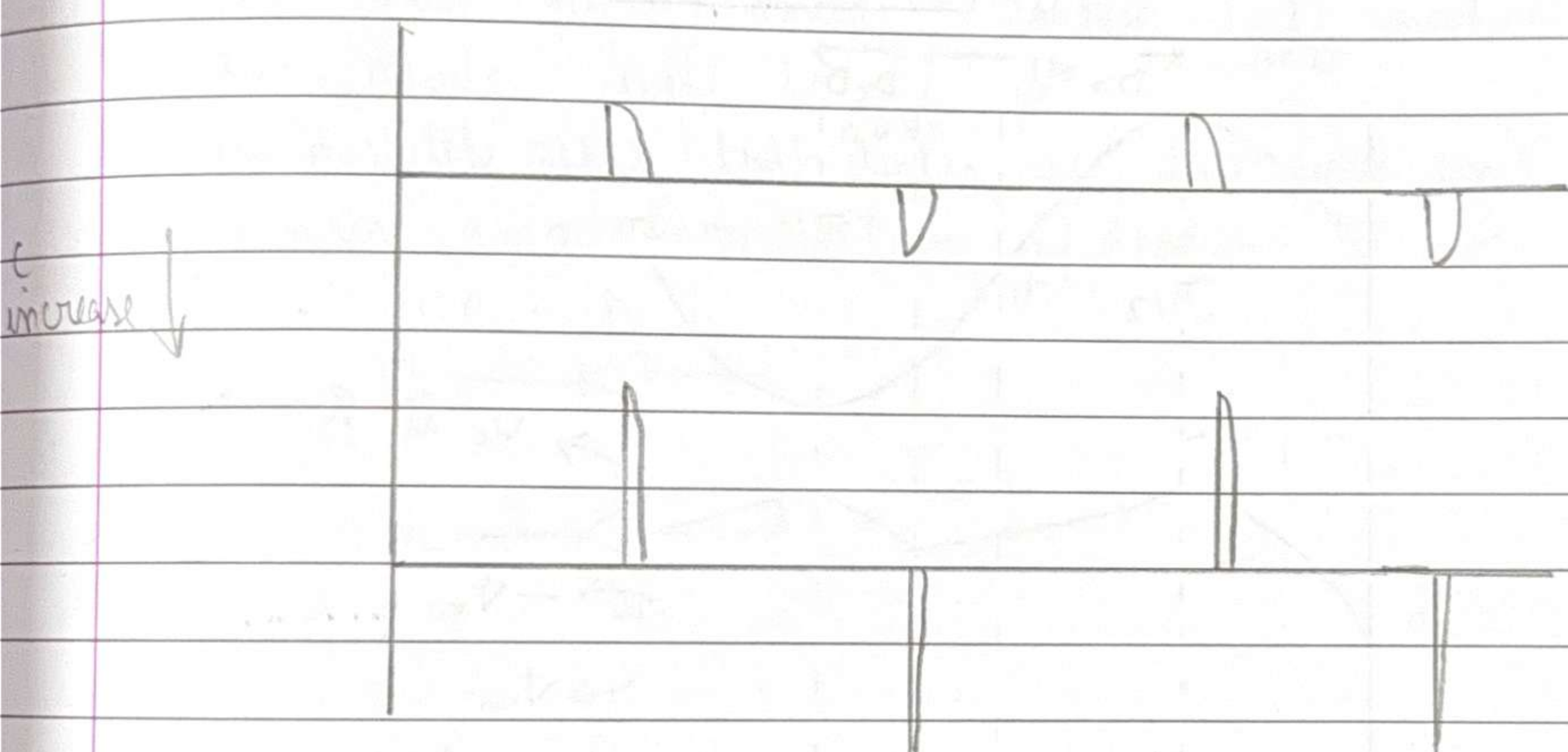
From 0 to π/2, $i_{i/p} = C\frac{dV_c}{dt}$ (ignoring current going into R branch because R is typically large)

From π/2 to t_1, $i_{i/p} = 0$

If we keep increasing C,

Clearly $t_\Delta \to 0$

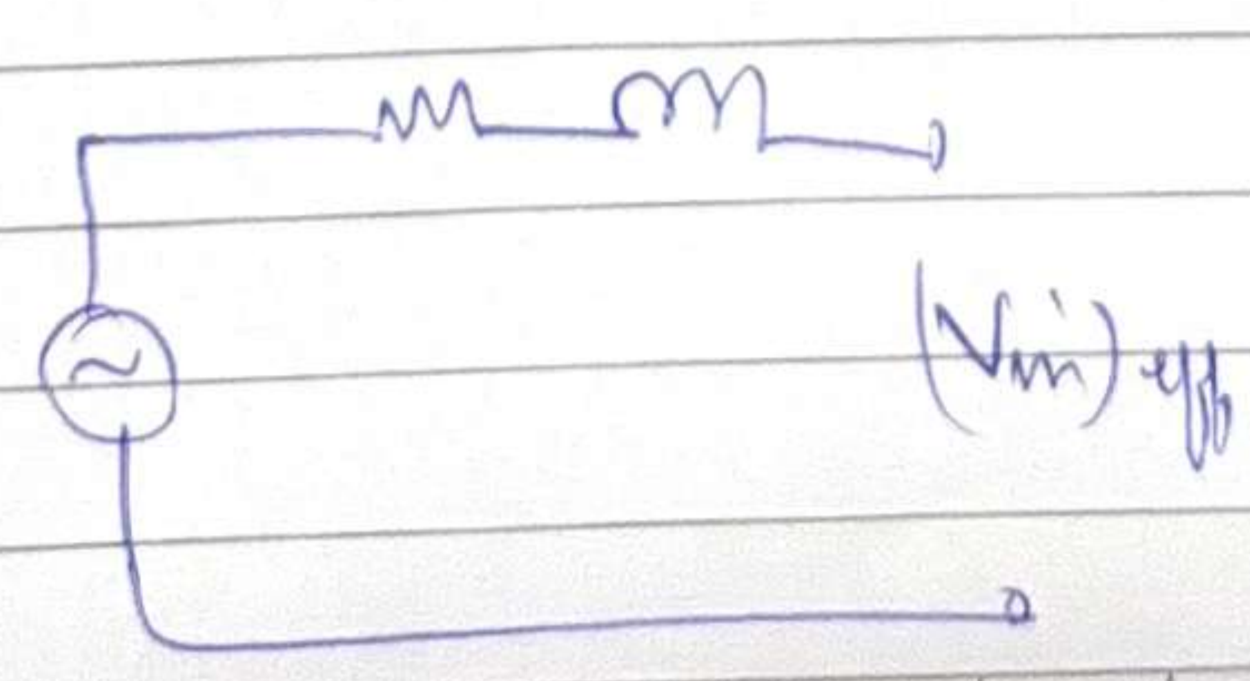
To ensure that input power = output power
We must have, $i_{peak} \to \infty$



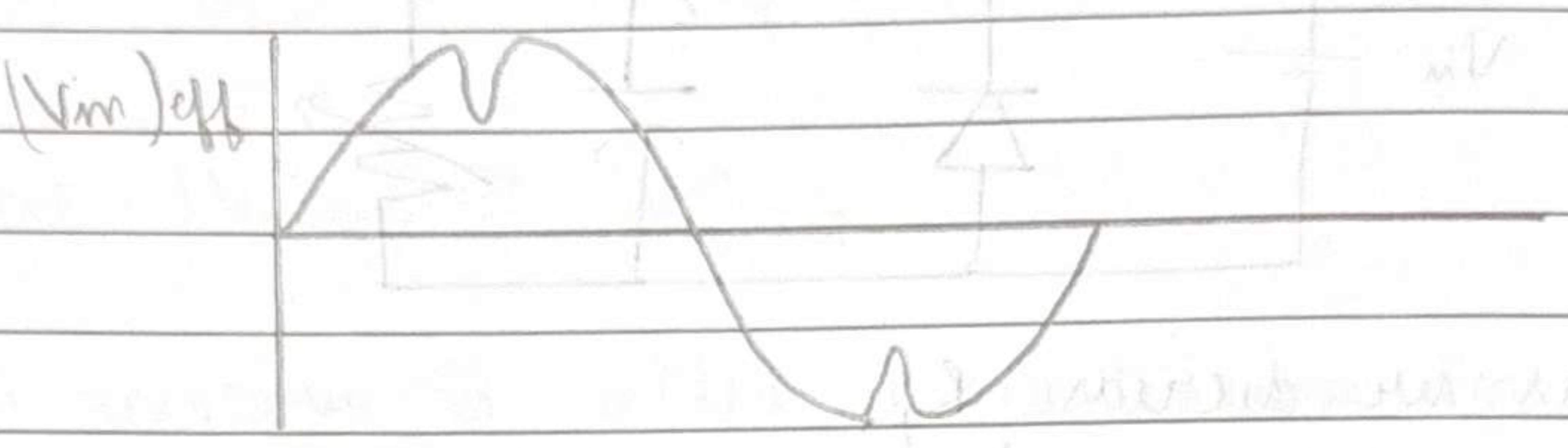
~~Not~~ For very large C, the input current waveform has sharp spikes (actually it is this way if we ~~too~~ check the current drawn by phone chargers)

→ But not very desirable, the sharper the spike, the higher will be $L\frac{di_L}{dt}$

∴ In the grid system that provides power supply at homes, there are inductances.



Hence, at the moment of the spike, there will a voltage drop in Vin



⇒ Distorted wave

→ has harmonics

→ in ~~an~~ a transformer, harmonics ⇒ more losses

→ in a fan, harmonics ⇒ humming noise

etc. etc.

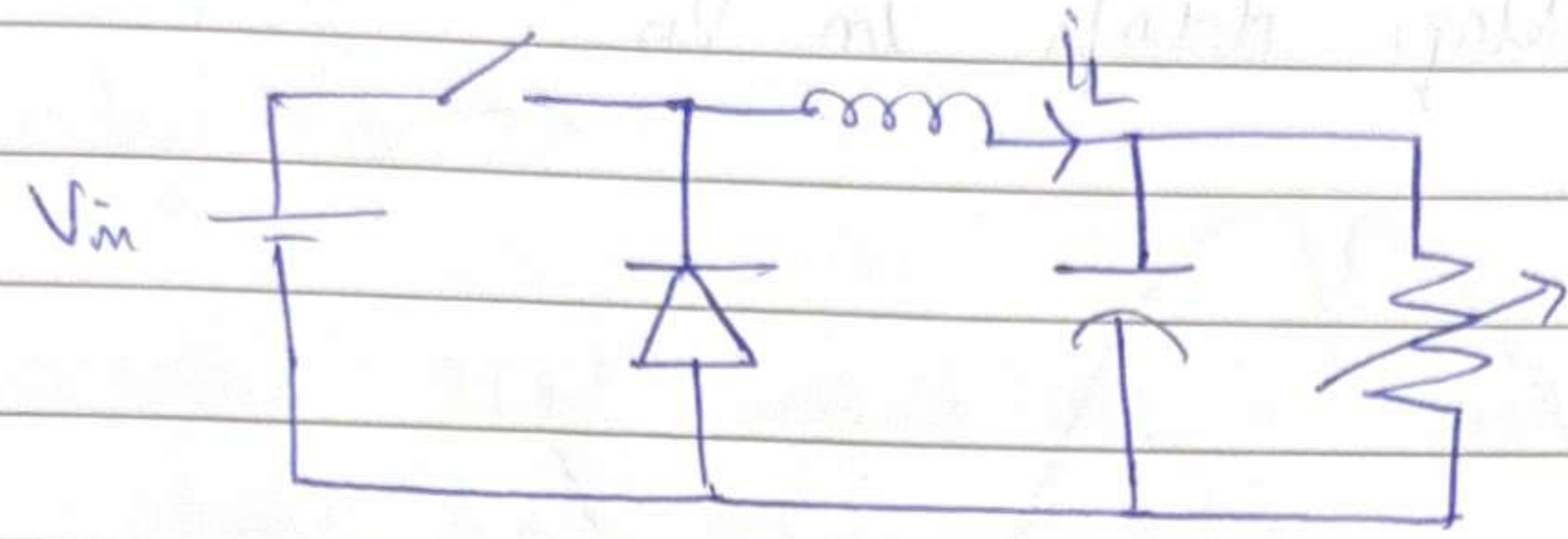
solution?

There exists a solution, but expensive

→ Not implemented in phone charger

→ Implemented in AC, fridge etc.

DC-DC Buck Converter



As we decrease R,

$V_o = D V_{in} \Rightarrow V_o$ remains constant

$i_R = \frac{V_o}{R}$ increases

$\therefore$ I_L increases

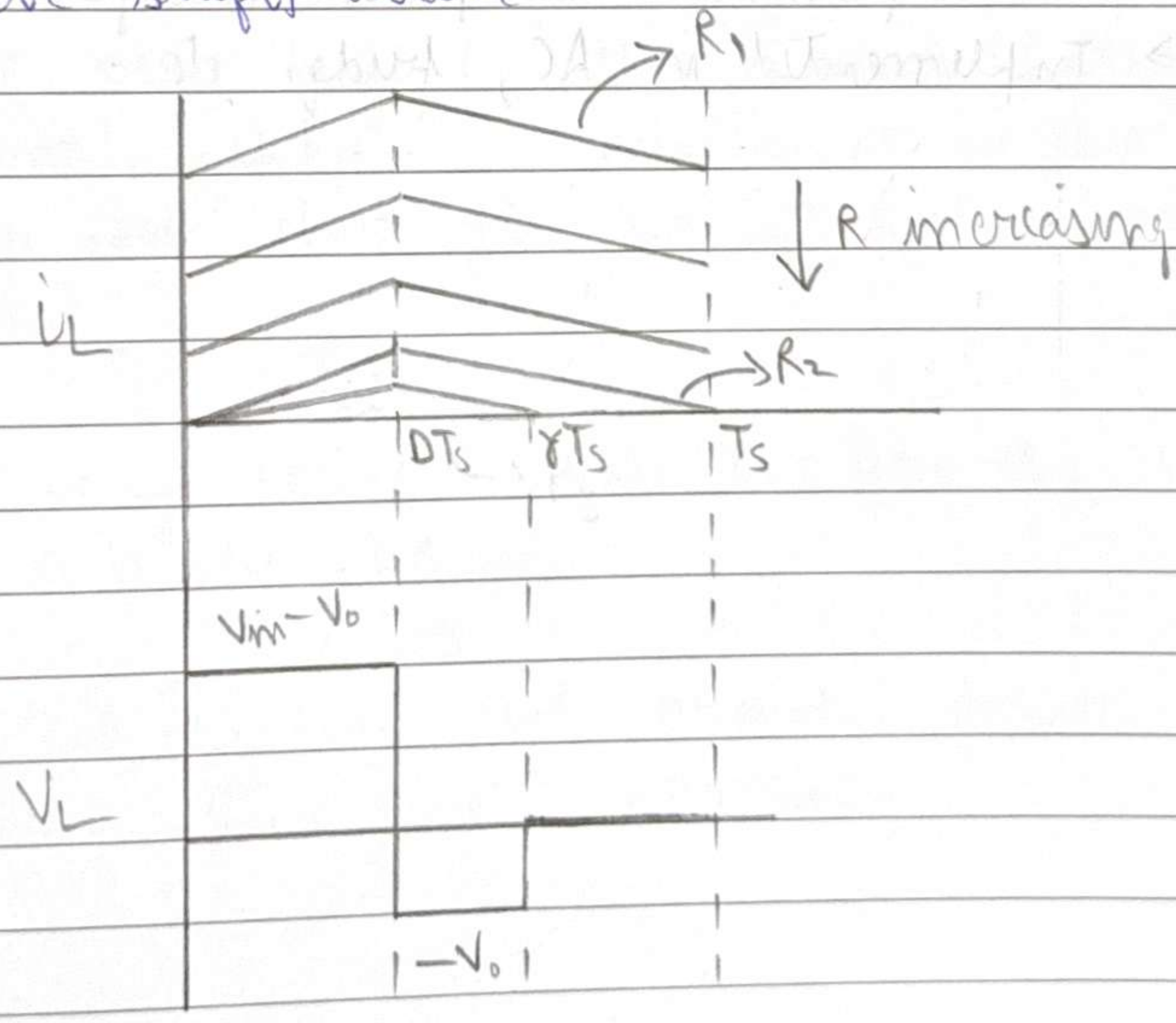
As we increase R,

V_o remains same

Slopes of i_L remain same

$I_L = \langle i_L \rangle$ decreases

$\therefore$ i_L curve shifts down



$\int_{T_s} v_L \, dt = 0$ (Volt-second law)

$\Rightarrow V_o = \frac{V_{in} D}{\gamma}, \gamma \leq 1$

Hence $(V_o)_{DCM} \geq (V_o)_{CCM}$

This scenario is called Discontinuous Conduction Mode (DCM)

Original case was Continuous Conduction Mode (CCM)

Critical case is Critical Conduction Mode

In DCM, why does i_L touch zero before T_s?
In order to reduce the ~~a~~ average $\langle i_L \rangle = I_L$, the integral of i_L $(= \int i_L dt)$ must reduce.

Hence the line must touch zero earlier.

$\gamma > D$
As we keep increasing R (i.e. decreasing load)
$\gamma \rightarrow D$.
$\therefore$ At very light loads, the circuit does not give desired load $(= DV_{in})$
Solution? $\longrightarrow$ Reduce D ! $\Rightarrow$ Closed loop control

DCM condition can happen in any kind of Power Electronic circuit.

How to find γ?

$$\langle i_L \rangle_{T_s} = \frac{V_o}{R}$$

$$\frac{1}{2} \times \frac{\gamma T_s}{T_s} \times i_L^{Peak} = \frac{D V_{in}}{\gamma R}$$

How to check for CCM or DCM?

CCM and DCM $\longrightarrow$ Defined wrt to L in any Power electronic ckt

$$I_L^{avg} < \frac{\Delta I_L}{2} \Rightarrow DCM$$

$$> \frac{\Delta I_L}{2} \Rightarrow CCM$$

HW: Find the CCM-DCM condition for Boost Conv.
D, T_s, V_{in}, I_{load} given

Larger L, means smaller ripple, means more CCM range

Larger F_s, smaller ripple, more CCM range

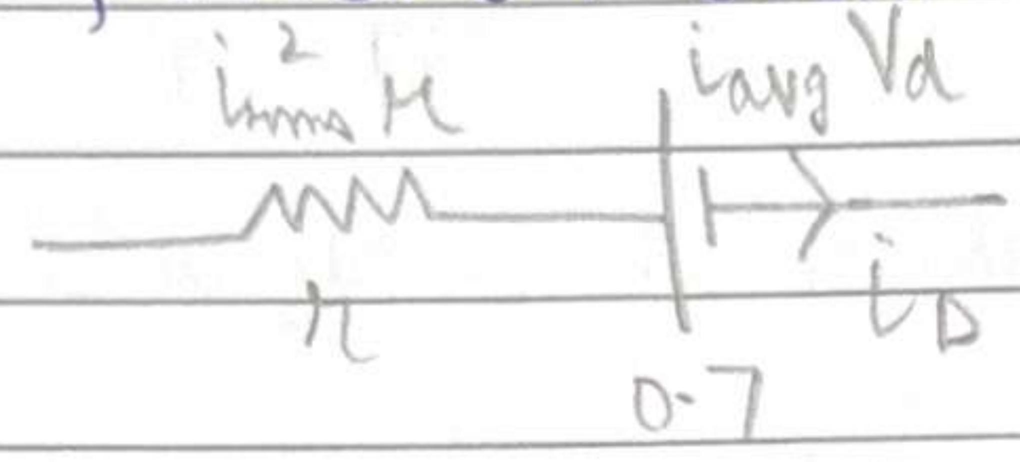
If we use small L, ~~same~~ $\xrightarrow{\text{large ripple}}$ compact chargers! - Good
need large C to filter large ripple! - Bad
Tradeoff

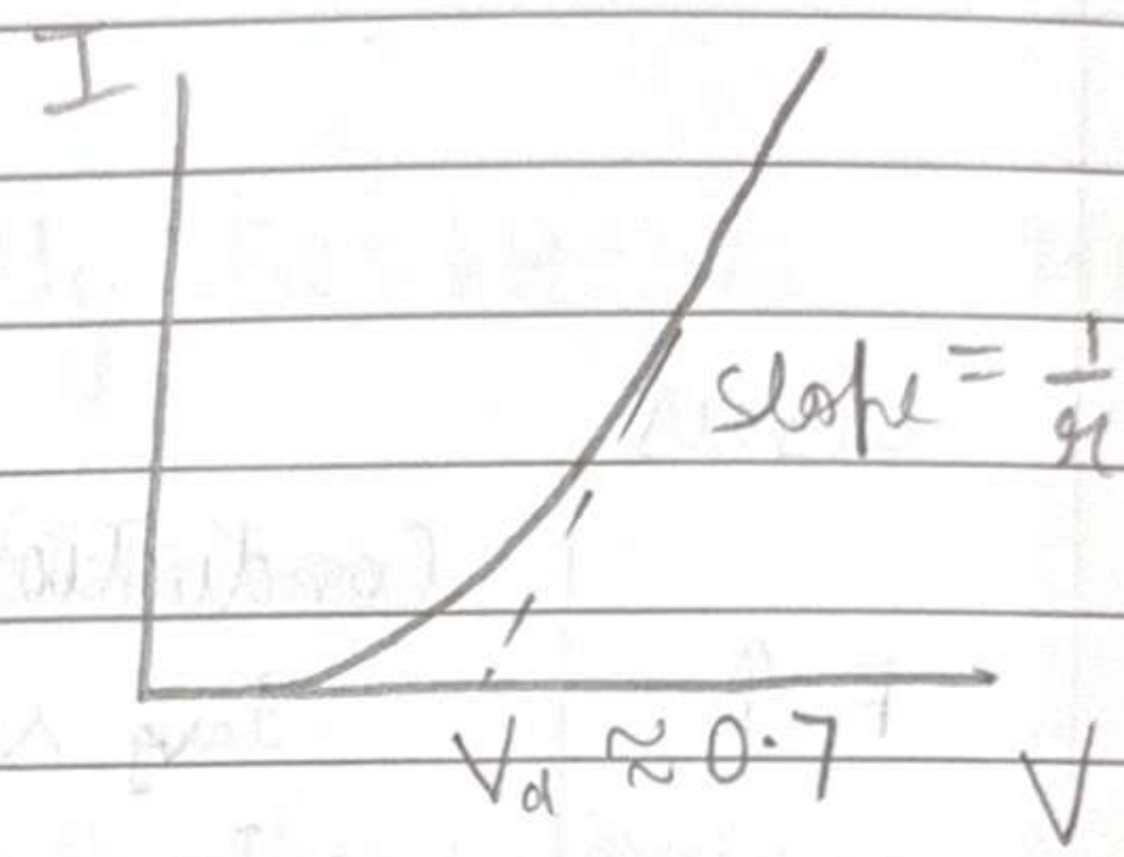
Often in commercial applications, the ~~time~~ critical line dividing CCM and DCM ~~is~~ lies in 30 - 50% range of ~~I~~ load current specification.

Power Losses

- Conduction Losses

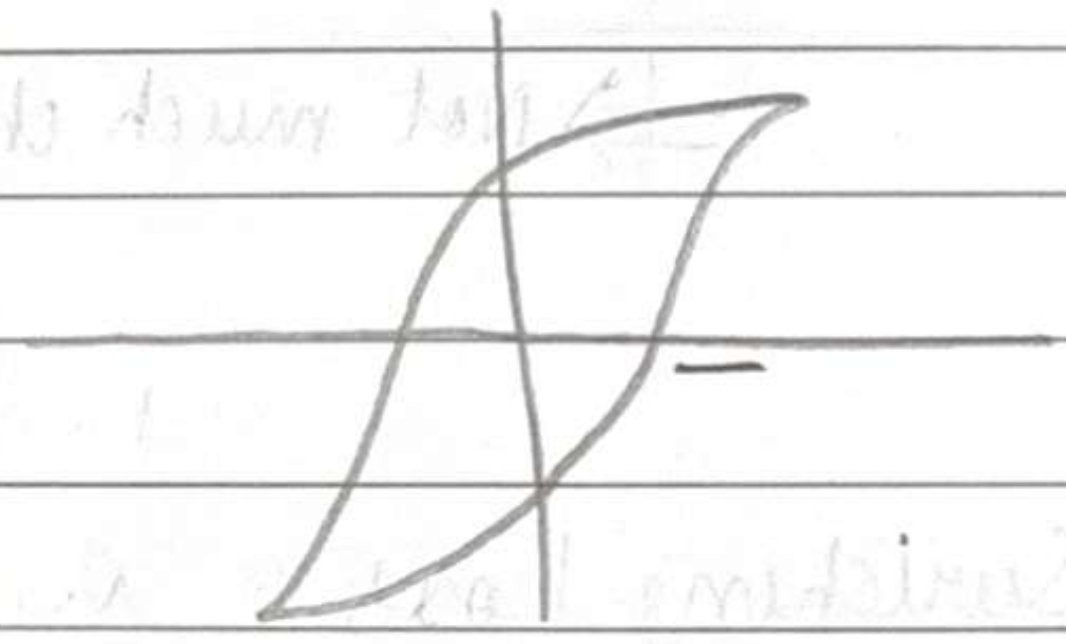
  associated with components like diode, MOSFET switch





Equivalent ckt of diode

- Hysteresis and Eddy current losses in inductor
  ($\propto B^{1.6} f$) ($\propto f^2 B^2$)



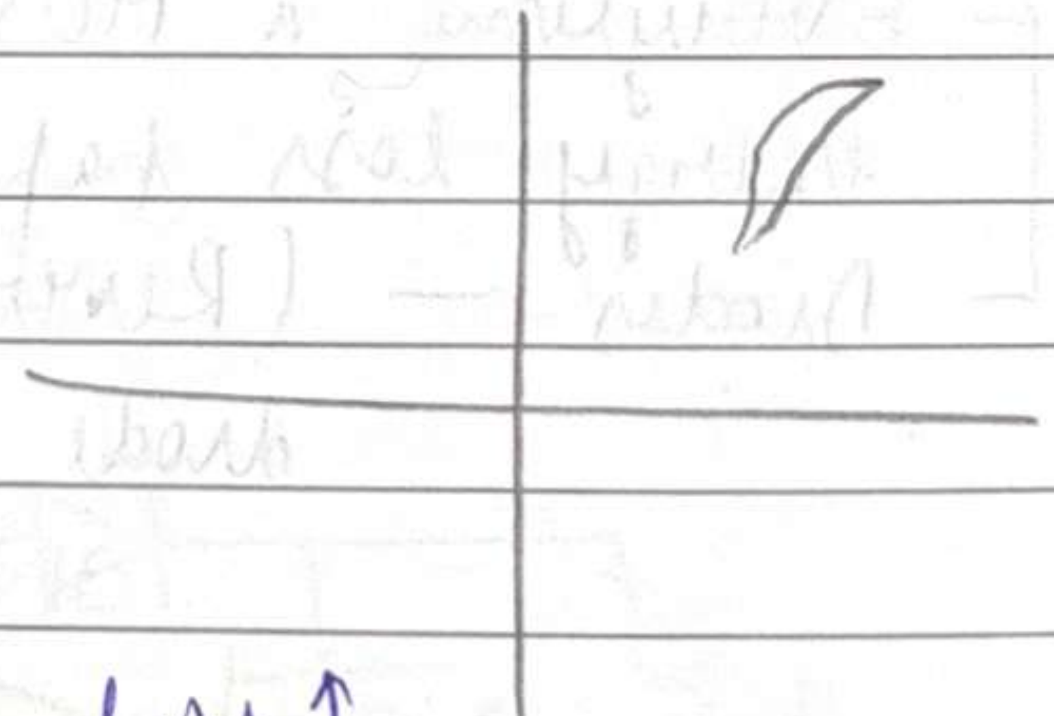
If F_s becomes 2x

Because of DC offset, generally BH curve will lie in non-linear region

$\therefore$ ΔB will not be much

$\downarrow B^{1.6}$ factor is not strong

$\uparrow f$ factor dominates



USUALLY, $F_s \uparrow \Rightarrow$ Hysteretic loss $\uparrow$

$F_s \uparrow \Rightarrow$ core loss $\uparrow$

For same average value, higher ~~rm~~ ripple $\rightarrow$ ~~higher~~ ~~lower~~ higher rms

$\therefore$ For $F_s \uparrow$, ~~ripple $\uparrow$~~ ripple $\downarrow$, $i_{rms} \downarrow$, ~~conduction loss $\uparrow$~~ conduction loss $\downarrow$

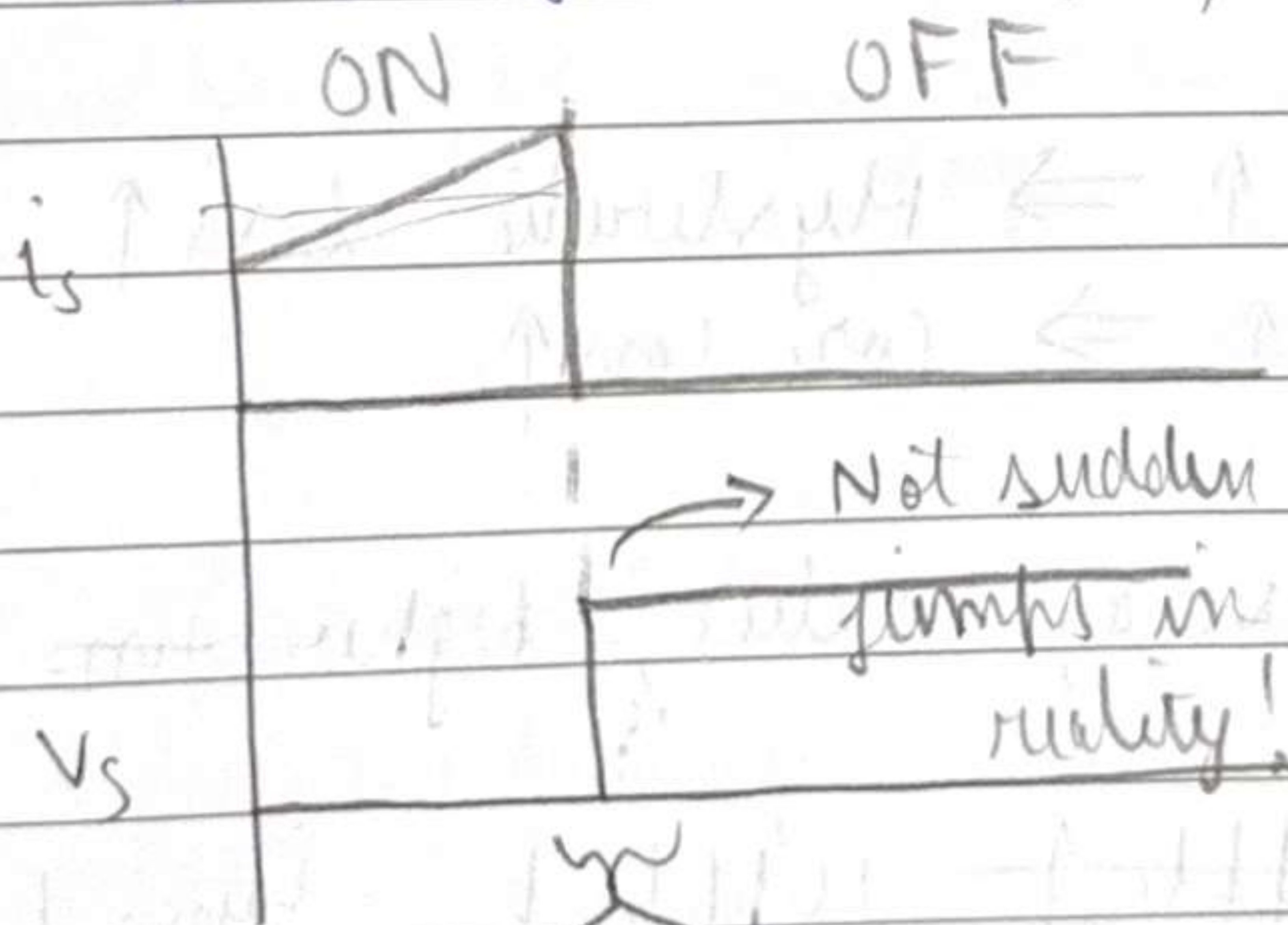
- Skin effect

  $F_s \uparrow$, skin effect $\uparrow$

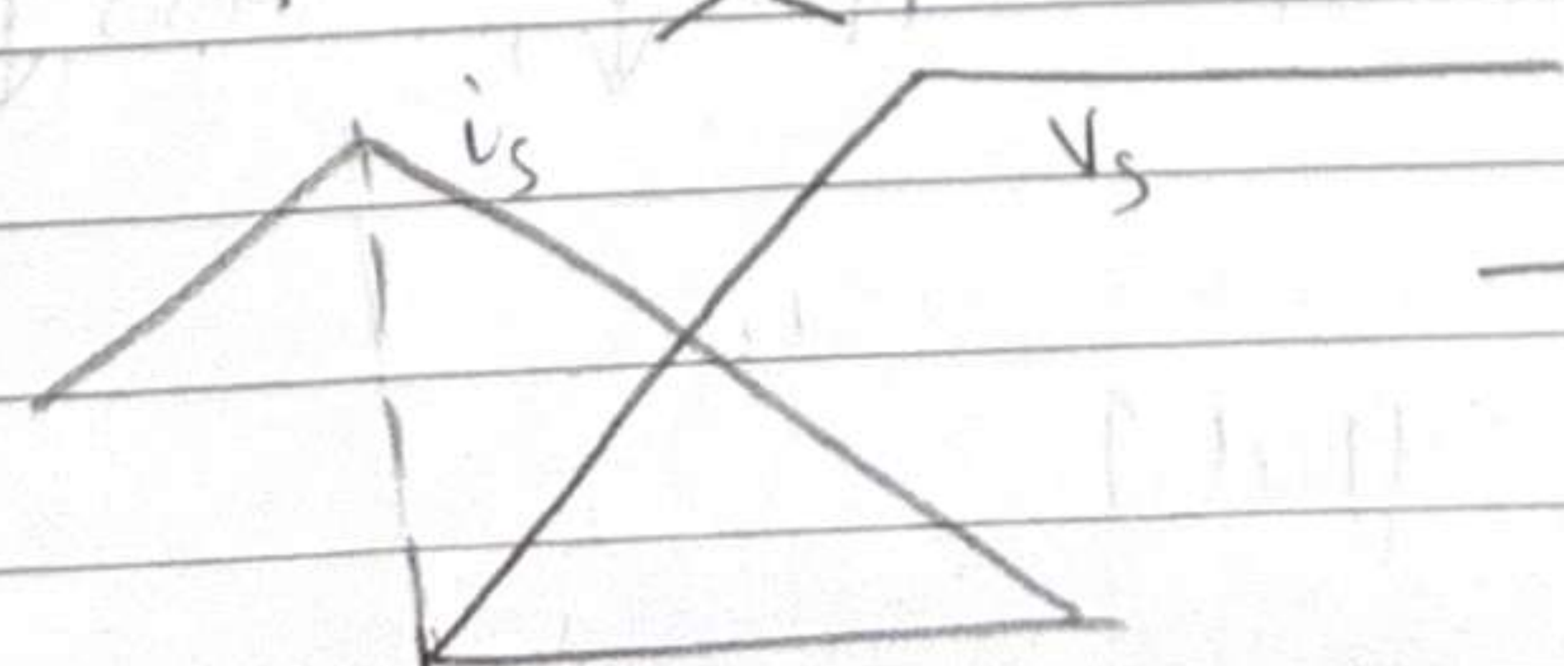
Losses

	Conduction loss	Core loss	Switching loss
$F_s \uparrow$	I_{avg} same $I_{rms} \downarrow$ R (due to skin effect) $\uparrow$	$f \downarrow$ $B \uparrow$ (non-linear relation)	Switching loss $\uparrow$
	$\Rightarrow$ Not much change	Increases	Increases

— Switching Loss
- Everytime a MOSFET is turned on / off, a small energy loss happens.
- Diodes — (Reverse Recovery) energy loss everytime diode is turned on / off.



For a boost converter



→ Overlap of high i_s and V_s
→ Power loss during switching

Generally, we see that $F_s \uparrow \Rightarrow$ losses $\uparrow$

But if we reduce F_s, we will need larger L and C.
Trade off!

~~In~~
In phone charger, F_s is typically 500 kHz to ~2 MHz

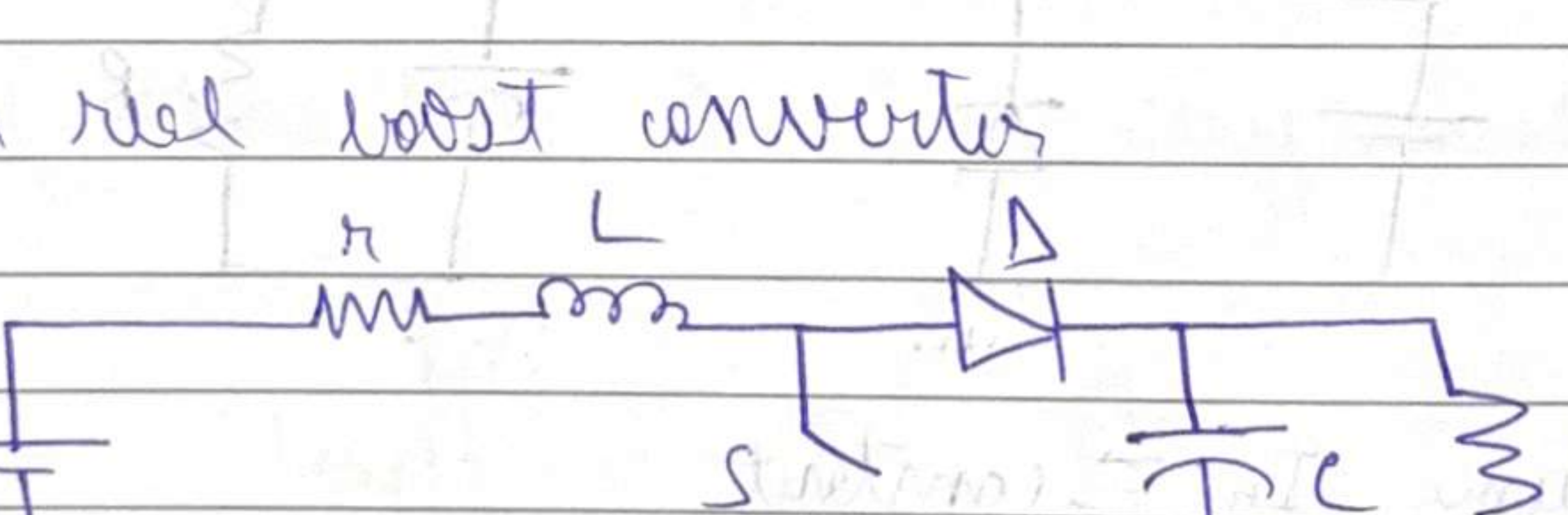
~~A~~ choice of F_s, depends on applications.

For low power applications, (like phone charger), we generally allow some amount of power loss, but want small L and C

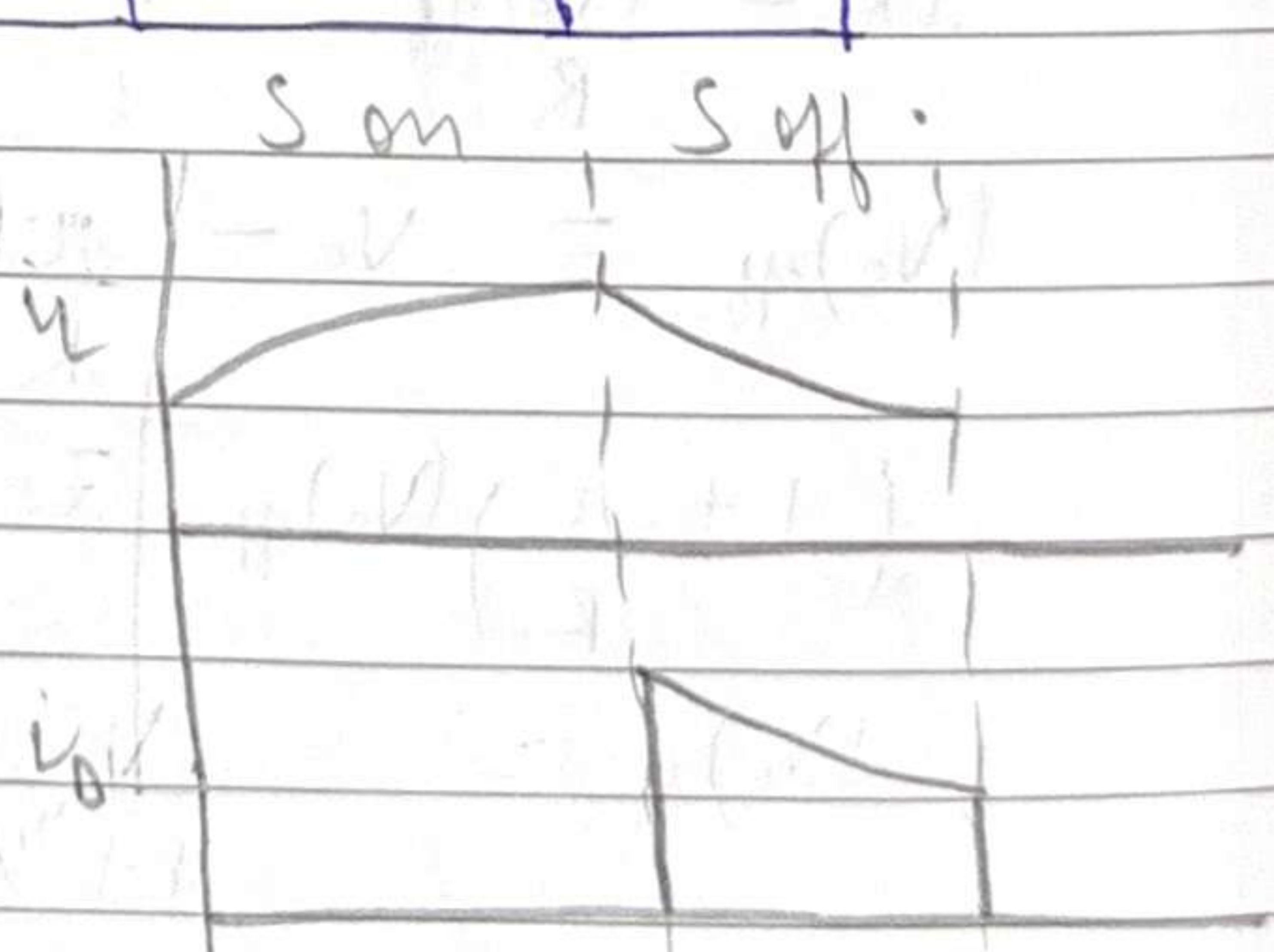
Boost Gain Limitation
In an ideal boost converter,

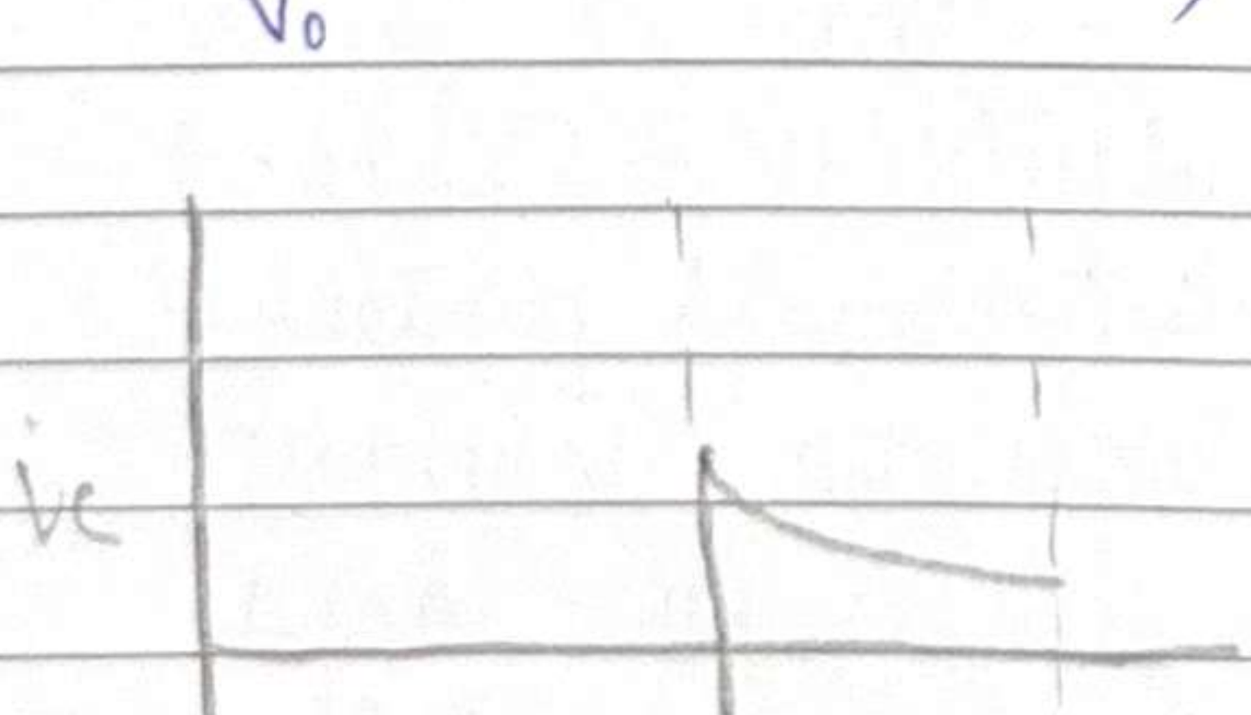
$$\frac{V_o}{V_i} = \frac{1}{1-D} \rightarrow \infty \text{ as } D \rightarrow 1$$

But in a real boost converter



$$\frac{V_{in}}{V_o} = ? \quad \text{(CCM)}$$





It comes out to be

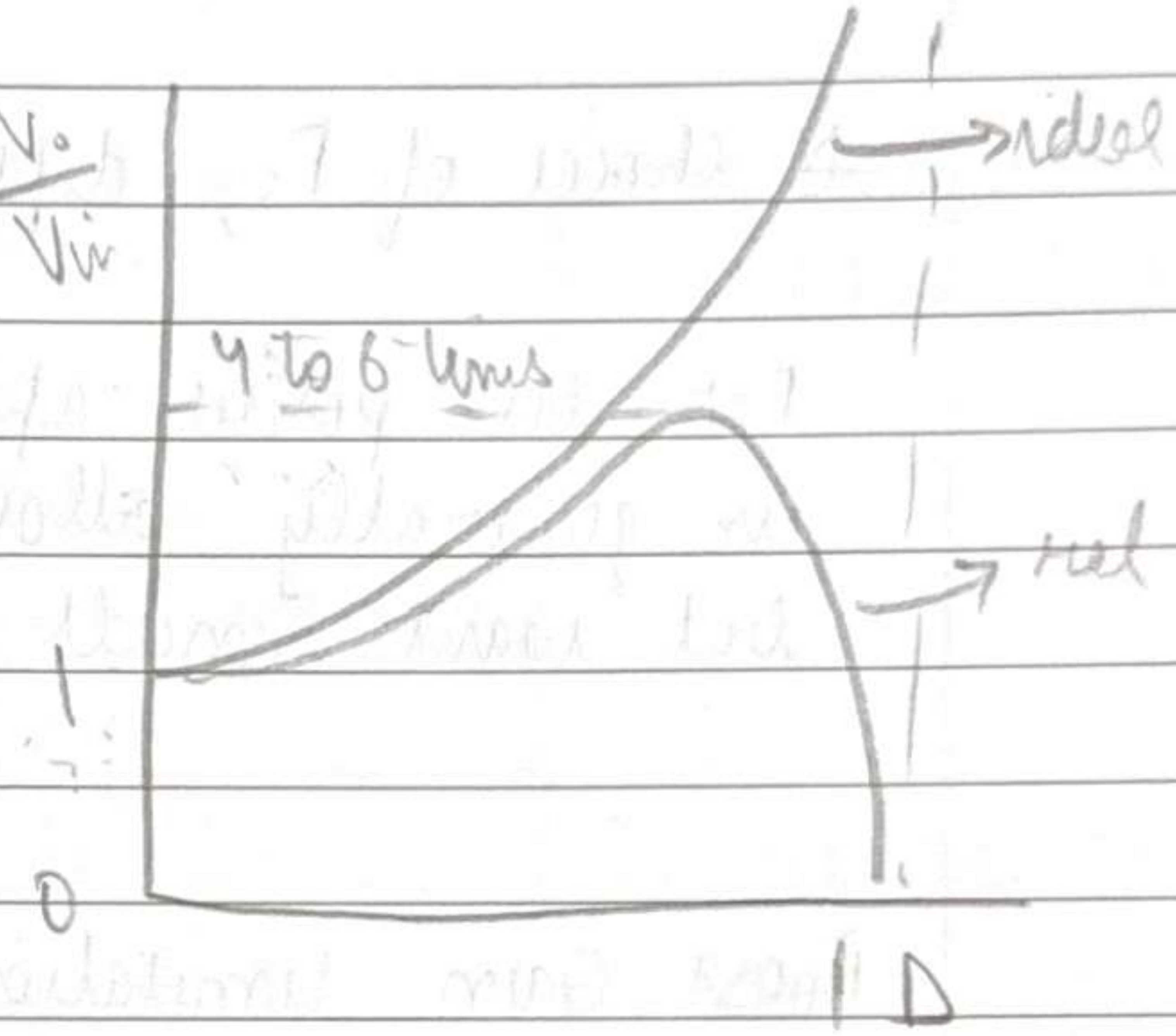
$$\frac{V_0}{V_{in}} = \frac{1}{1-D+\frac{r}{R(1-D)}} \quad (HW!!)$$

$$(V_{in})_{eff} = V_{in} - \cancel{I_L R}\ I_L r$$

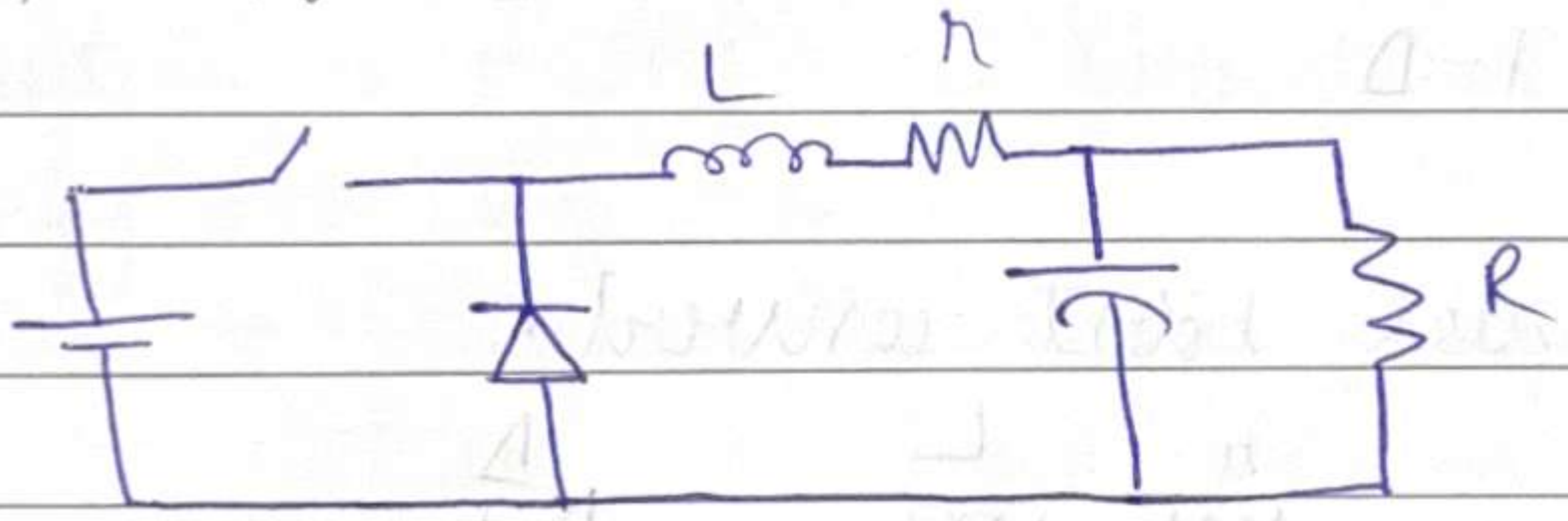
Typically $r << R$

$\cancel{I_L = V_0}$

Gain limitation!



Buck Gain Limitation



Assume I_r = constant

$$I_r = \frac{(V_0)_{eff}}{R}$$

$$(V_0)_{eff} = V_0 - \frac{r}{R}(V_0)_{eff}$$

$$\left(1+\frac{r}{R}\right)(V_0)_{eff} = V_0$$

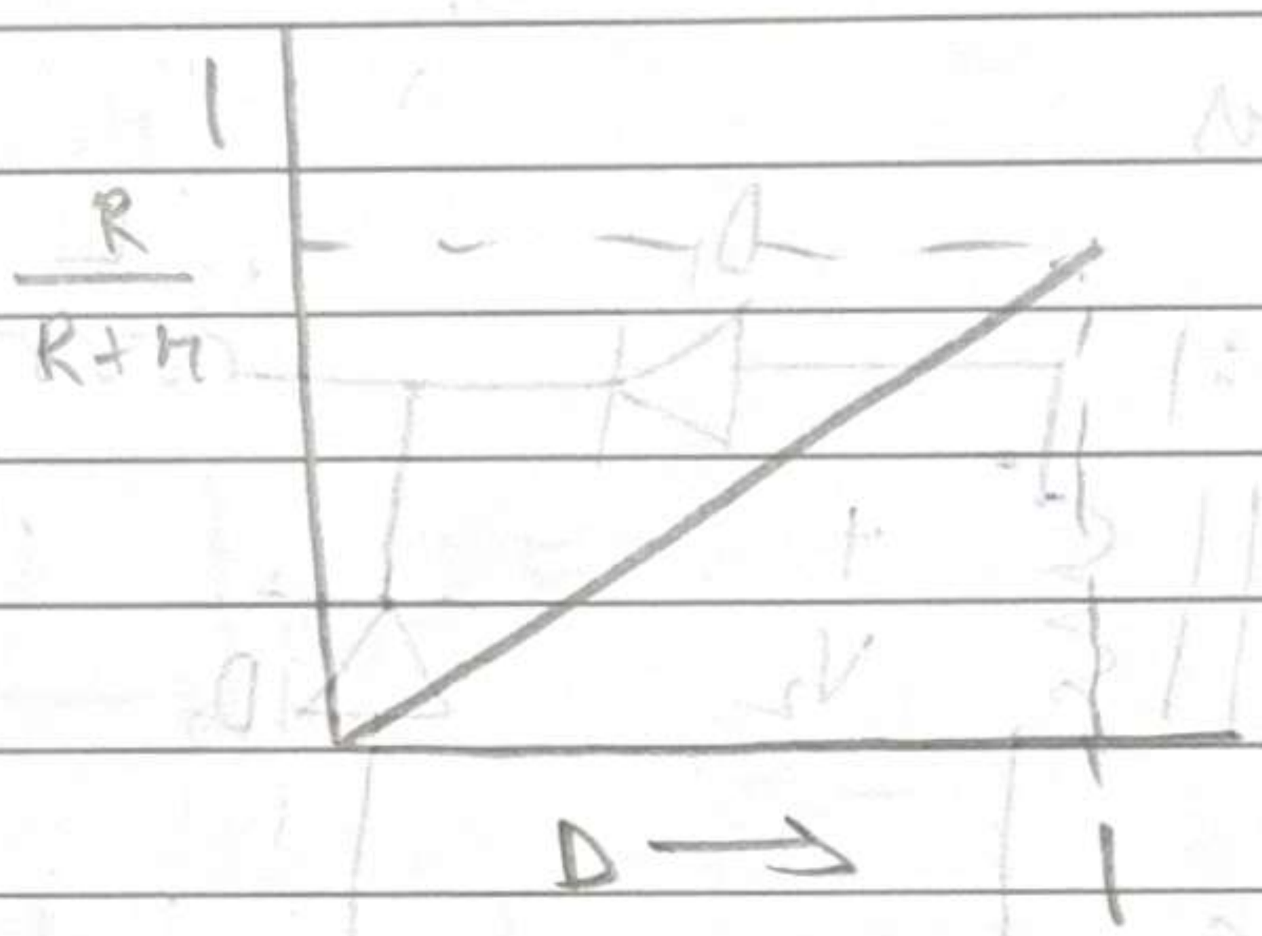
$$(V_0)_{eff} = \frac{V_0}{1+r/R}$$

$$(V_o)_{eff} = \frac{D V_{in}}{1 + r/R}$$

$$\text{Gain} = \frac{D}{1 + r/R}$$

~~to~~ We can make D as small as possible

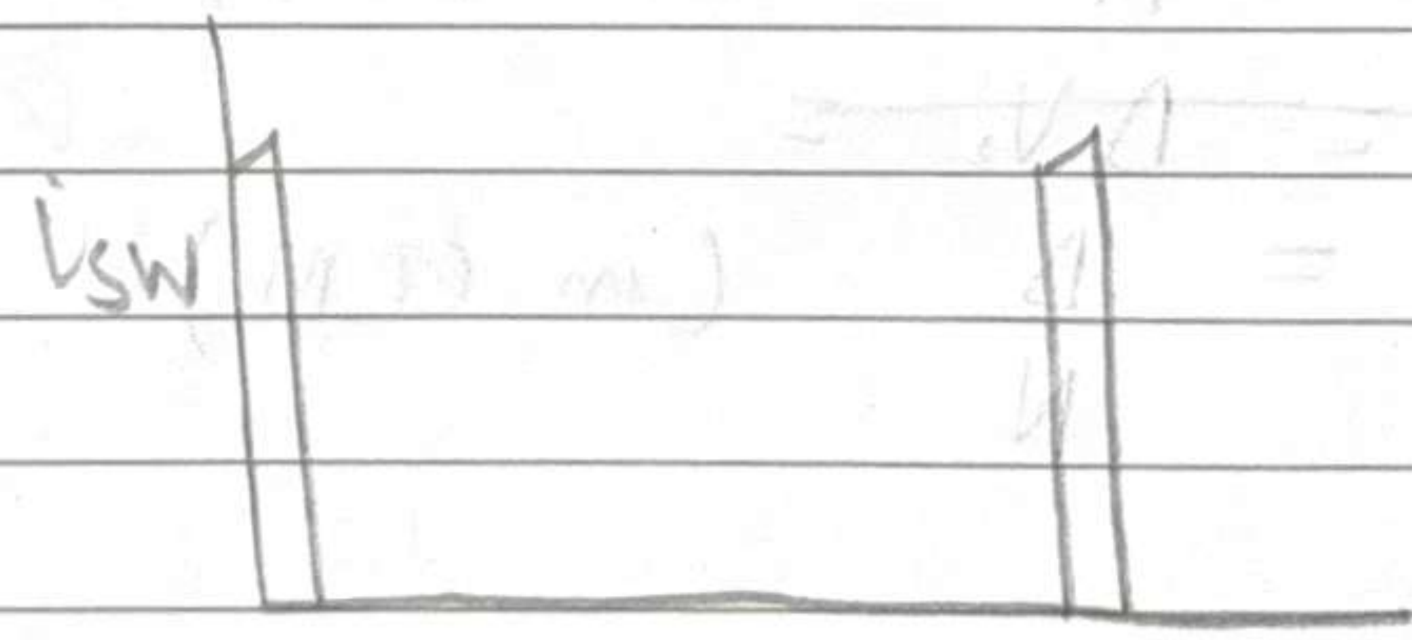
$\therefore$ No limitation on attenuation



But even then, practically we don't use buck converter for significant attenuation.

Why ?

Significant attenuation $\Rightarrow$ ~~Smo~~ Very small D



$(i_s)_{avg}$ is small

$(i_s)_{rms}$ is relatively small

$\therefore$ Switching / Conduction loss in switch $\uparrow\uparrow$

Increases required rating of the switch

Also efficiency $\downarrow\downarrow$ due to loss $\uparrow\uparrow$

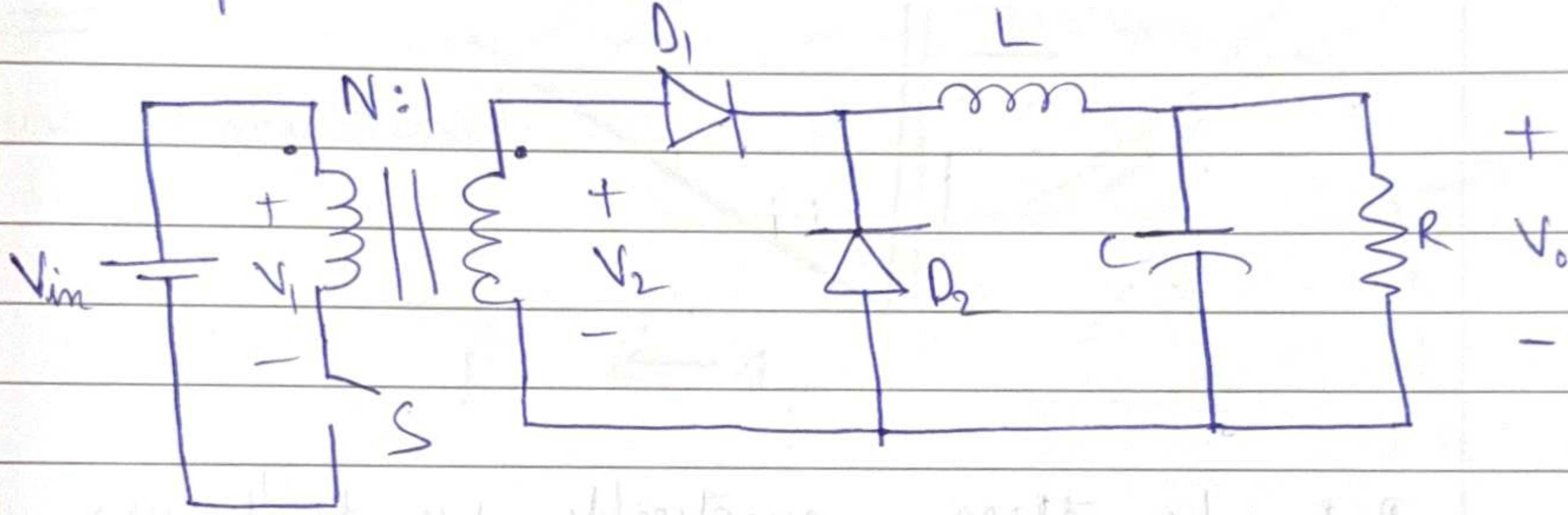
Basically in $\lim D \to 0$, buck convertor acts like a linear regulator

Buck, boost etc are non-isolated converters

∴ Can't use buck / boost for large attenuation / gain practically

Another class of converter types —

Isolated DC/DC converters
↓
Use transformers



Assume ideal transformer for now

~~$\frac{V_2}{V_1} = \frac{1}{N}$~~

~~$V_o = D V_2 =$~~

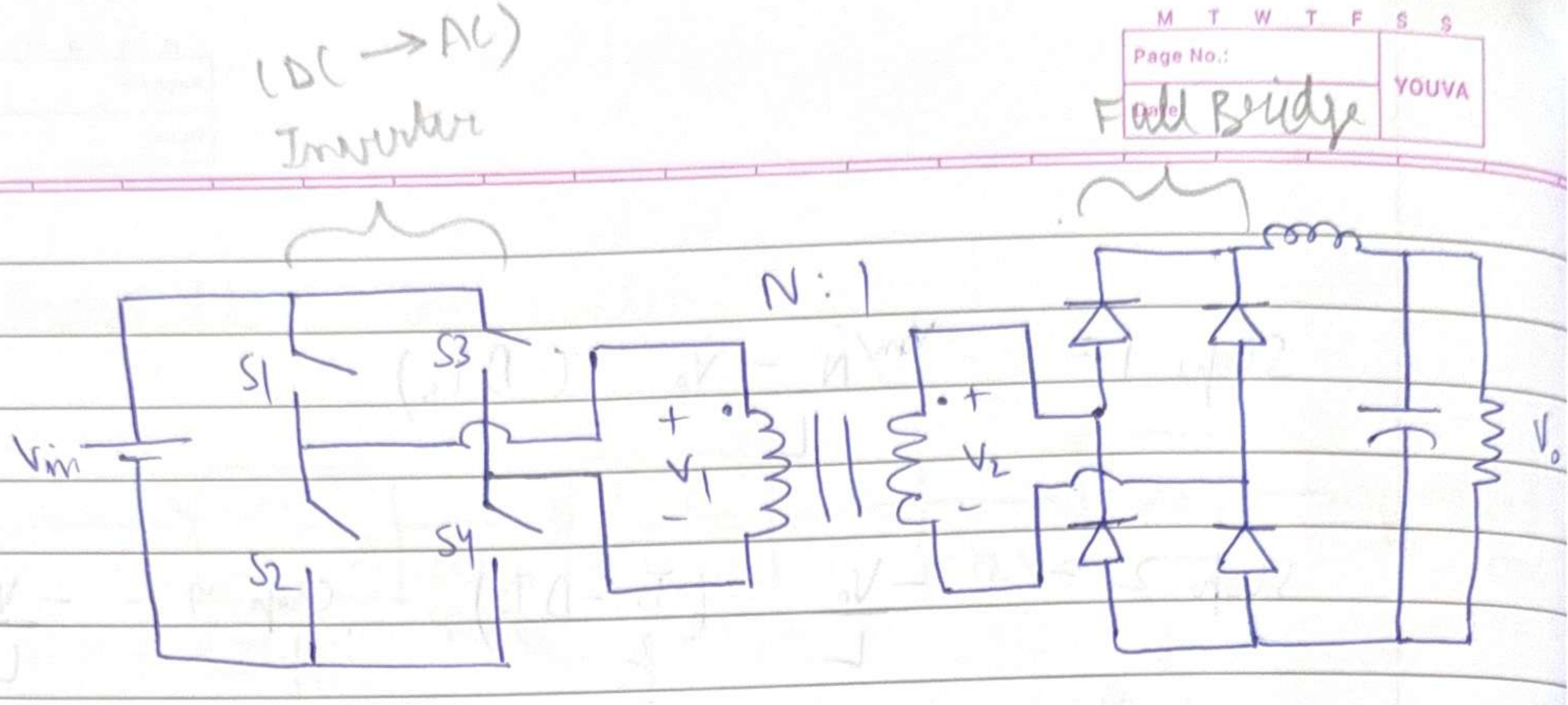
$$\frac{V_o}{V_{in}} = \frac{D}{N} \quad \text{(in CCM)}$$

This solves the problem we had.

If we want $\frac{V_o}{V_{in}} = \frac{1}{100}$

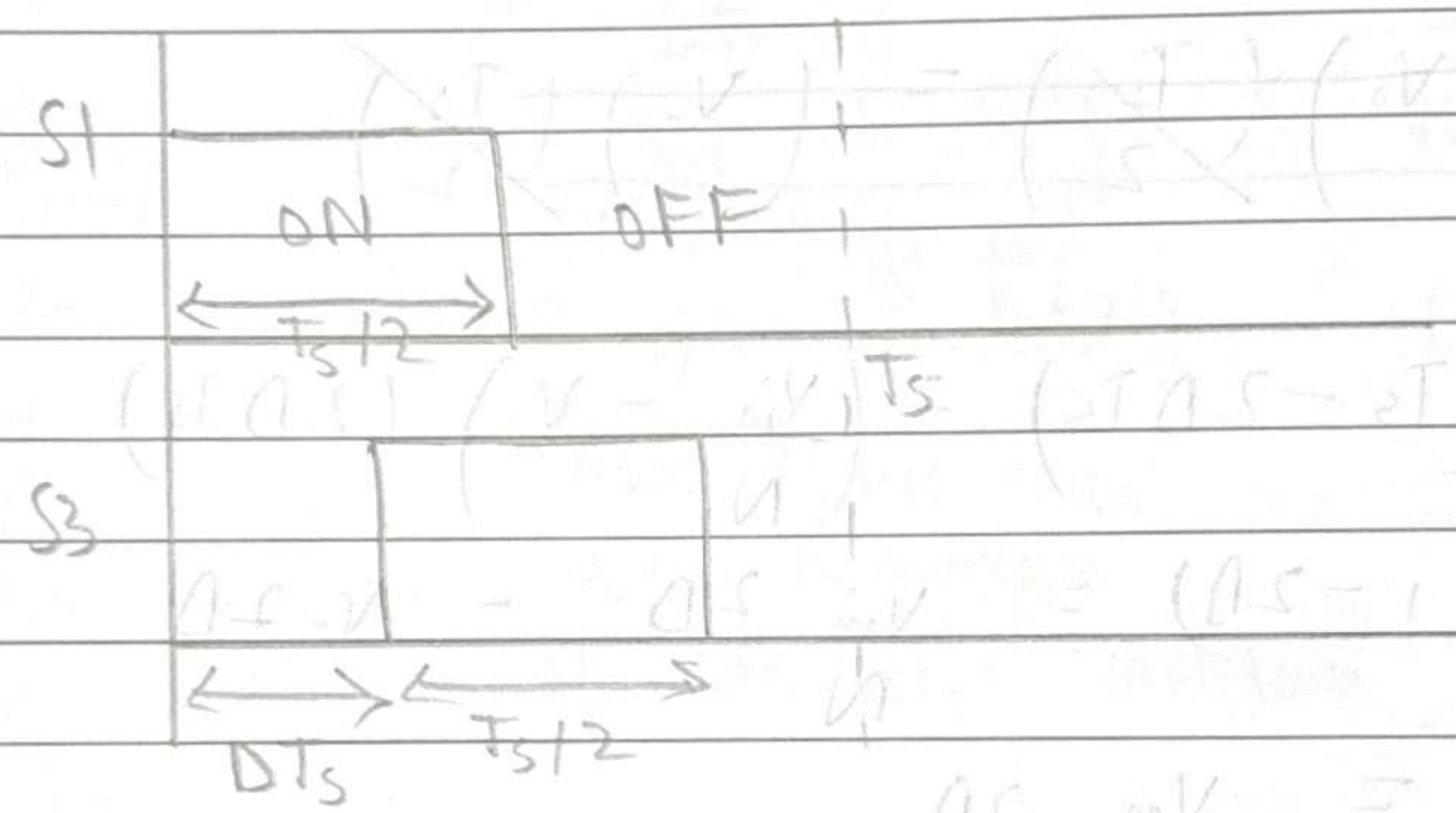
we don't need to set very small D, can set high N !!

This is called DC/DC Forward converter



Take $S1 = \overline{S2}$

$S3 = \overline{S4}$



$\frac{V_o}{V_i} = ?$ (CCM)

When $S1 = 1,\ S3 = 0$

$V_1 = V_{in}$

$V_2 = V_{in}/N$

$S1 = 1,\ S3 = 1$

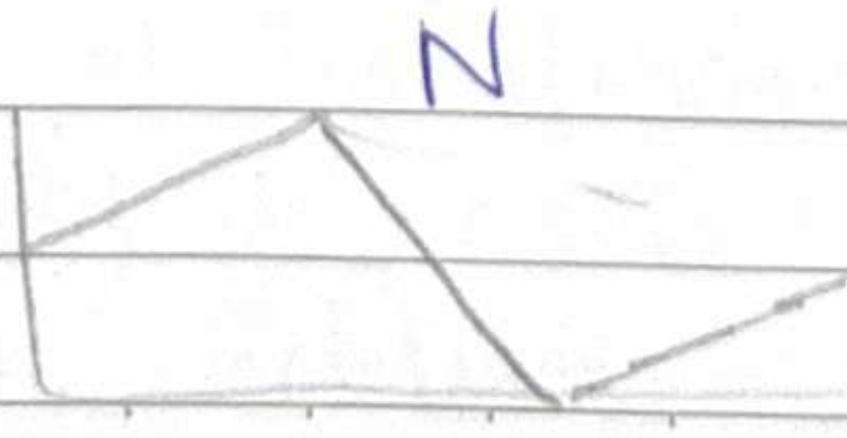
$V_1 = 0 \Rightarrow V_2 = 0$

$S1 = 0,\ S3 = 1$

$V_2 = \frac{-V_{in}}{N}$

$S1 = S3 = 0$

$V_2 = 0$



$$\text{Slope 1} = \frac{V_{in}/N - V_o}{L} \,(DT_s)$$

$$\text{Slope 2} = \frac{-V_o}{L}\left(\frac{T_s}{2} - DT_s\right) \qquad \text{Slope 4} = \frac{-V_o}{L}\left(\frac{T_s}{2} - DT_s\right)$$

$$\text{Slope 3} = \frac{V_{in}/N - V_o}{L}\,(DT_s)$$

$$\cancel{\left(\frac{V_{in}}{N} - V_o\right)\left(\frac{T_s}{2}\right) = \left(V_o\right)\left(\frac{T_s}{2}\right)}$$

$$V_o\left(T_s - 2DT_s\right) = \left(\frac{V_{in}}{N} - V_o\right)(2DT_s)$$

$$V_o(1-2D) = \frac{V_{in}}{N}2D - V_o 2D$$

$$V_o = V_{in}\frac{2D}{N}$$

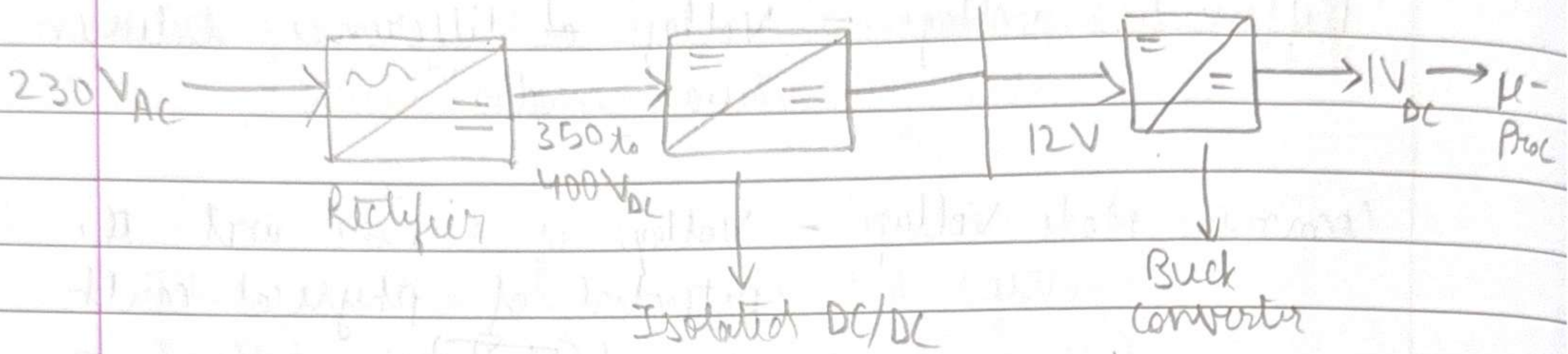
$$\frac{V_o}{V_{in}} = \frac{2D}{N}$$

This is called Phase-shifted Full-Bridge Converter

Recap - ~~Step~~ Buck & Boost not practically feasible for large step-up or step-down

- Still very useful for smaller step up or step down.

Example - Data Centers



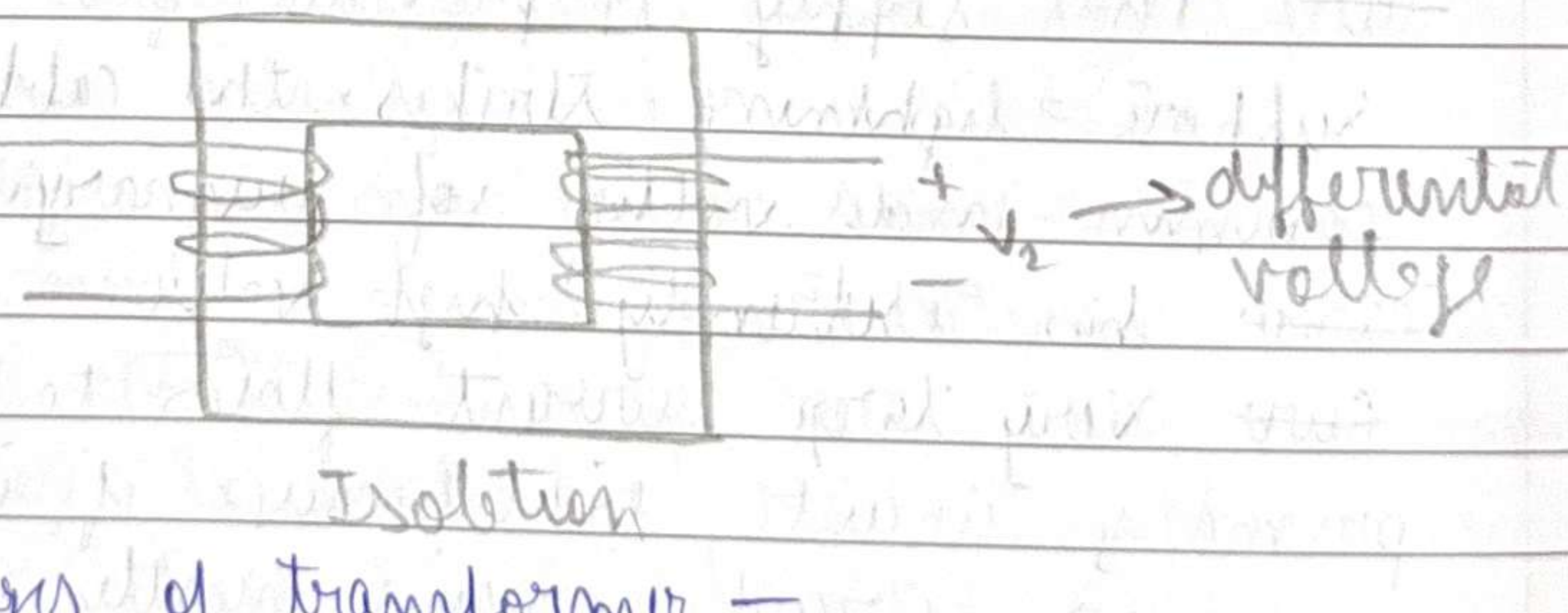
Sits on a separate shelf (i.e. sufficient space available)
Also there are some advantages of electrical 'isolation'.

Sits on the chip itself, very close to μ-proc (i.e. space limitation)

Electrical isolation

There is no direct electrical connection between left and right sides of the circuit.

E.g., a transformer has electrical isolation between primary and secondary coils.



Isolation

2 advantages of transformer —

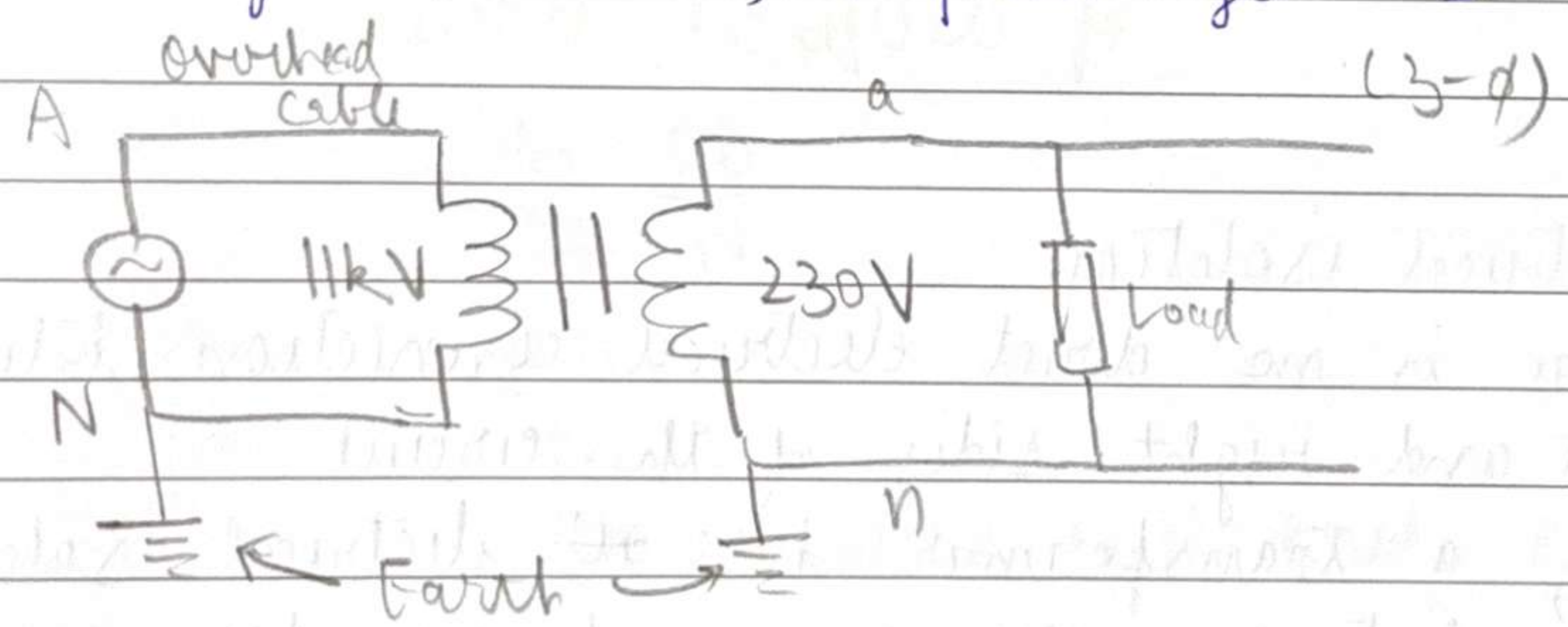
- Large step up / down possible
- Provides isolation

Why do we want isolation?

Differential voltage – Voltage ~~at~~ difference between two nodes

Common Mode Voltage – Voltage of a node wrt the potential of physical earth (धरती)

In the power supply at our homes, there is a 3-ϕ transformer.
There are 4 wires (3 phases and 1 neutral).
The neutral is usually 'earthed', i.e. connected to physical earth, to avoid floating voltage of neutral.
To ensure that if I ~~A~~ touch neutral while standing on earth, I ~~q~~ don't get a shock.



~~Sut~~ The supply is provided from overhead cable
Suppose lightning strikes the cables, so suddenly common-mode voltage of primary coils momentarily ~~carry~~ has extremely high value
~~Curr~~ Very large current flows to the earth in primary circuit, but because of isolation, secondary circuit remains mostly unaffected.

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Isolation — protects the components on the secondary coil from large currents, due to ~~from~~ sudden large common-mode voltage fluctuations on primary side.

There are two types of isolated systems —
- Earthed secondary
- Unearthed secondary

} Primary is earthed

There is some advantage of both types —

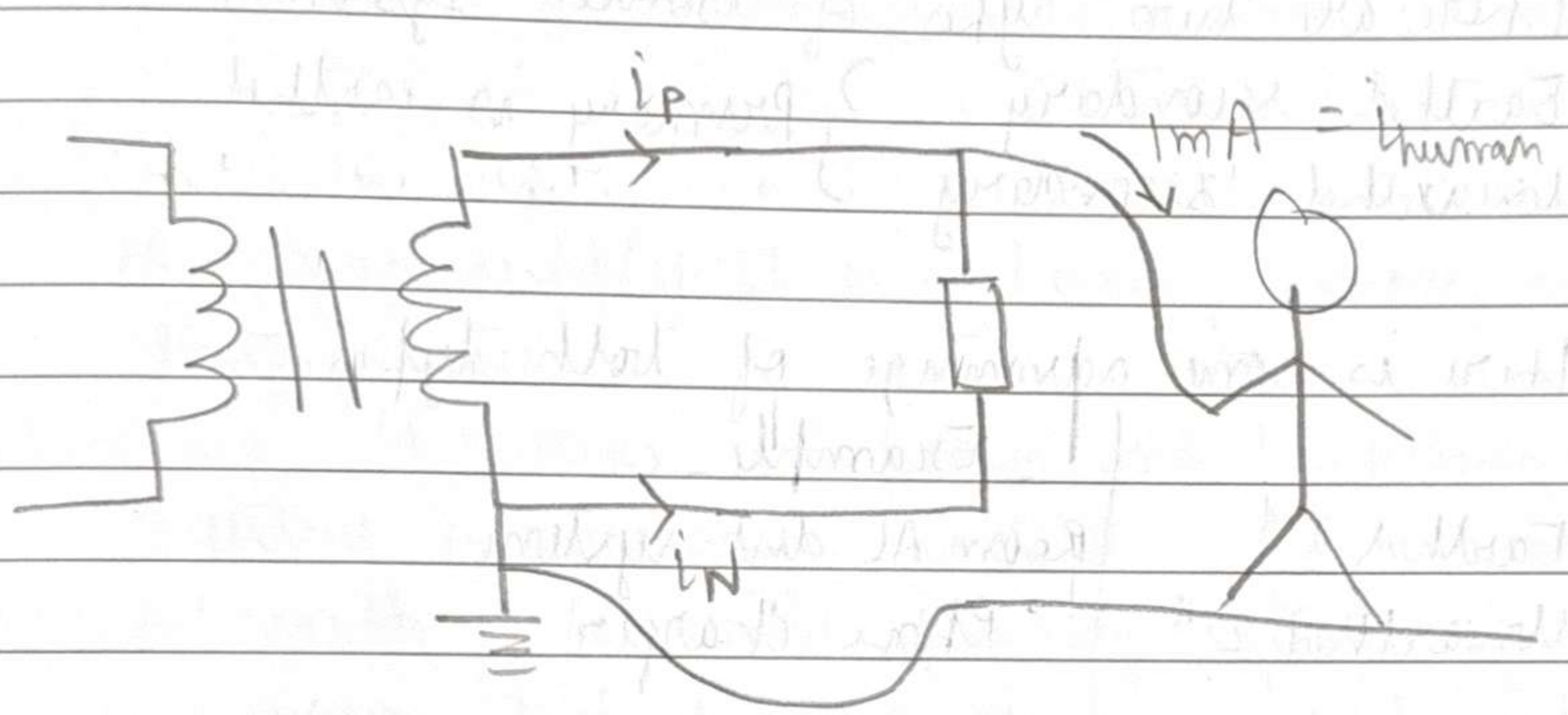
	Example
Earthed 2^{nd}	Room AC dist. system
Unearthed 2^{nd}	Phone charger

In an earthed system, upon touching the live wire, I may get a shock because of voltage difference b/w live wire and earth.

In non-earthed system, ~~the~~ common ~~v~~ mode voltage of live wire is not defined wrt धरती, ~~(air may act as parasitic connection which~~ ~~which~~ there is no path for circuit to be completed, so even if I touch the live, current won't flow through me (ideally).
In reality, air may act as some sort of parasitic resistive connection between live wire and धरती and may cause some common mode voltage to develop at live. In that, we may get a shock, but still safer option.

Shock protection in Earthed system

MCB, Fuses — X don't protect humans.
- only protect the appliances
- will ~~hurt~~ help avoid fires but not direct shock.



$i_p + i_n = i_{human}$

Components like ELCB / RCCB check the value of $i_p + i_n$, and trip (i.e. ~~tu~~ break the ckt) when $i_p + i_n$ exceeds some threshold, based on safety standard of humans.

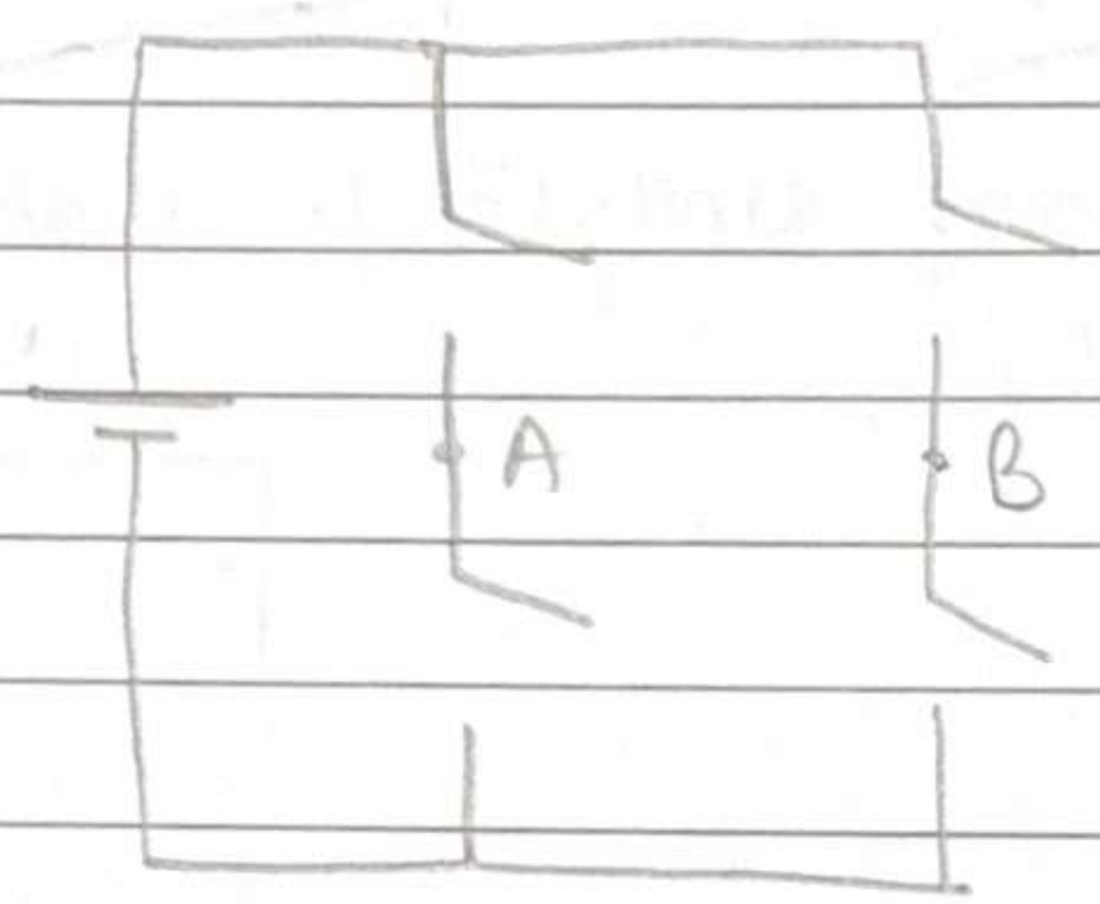
Advantage of earthed system — common-mode voltage is well defined, so we know how strong we need to build insulations and other protection measures.

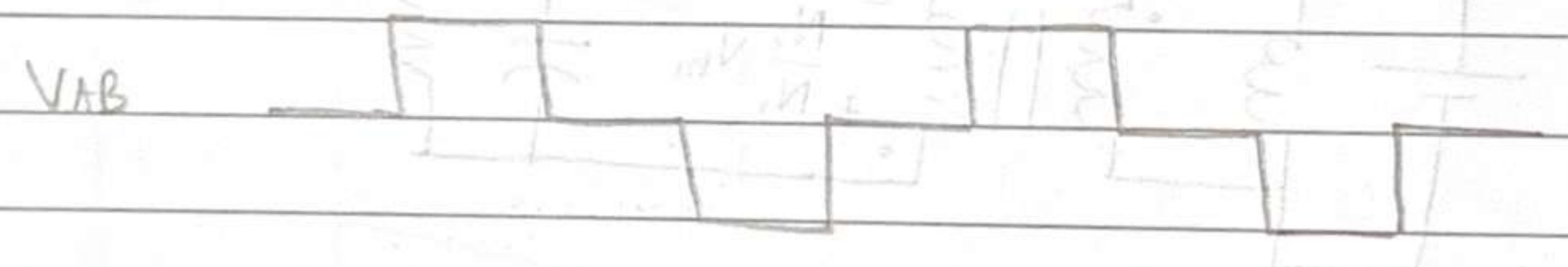
For 100 V to 80 V,
can use Buck, Forward

Buck → higher η, lower cost, smaller size

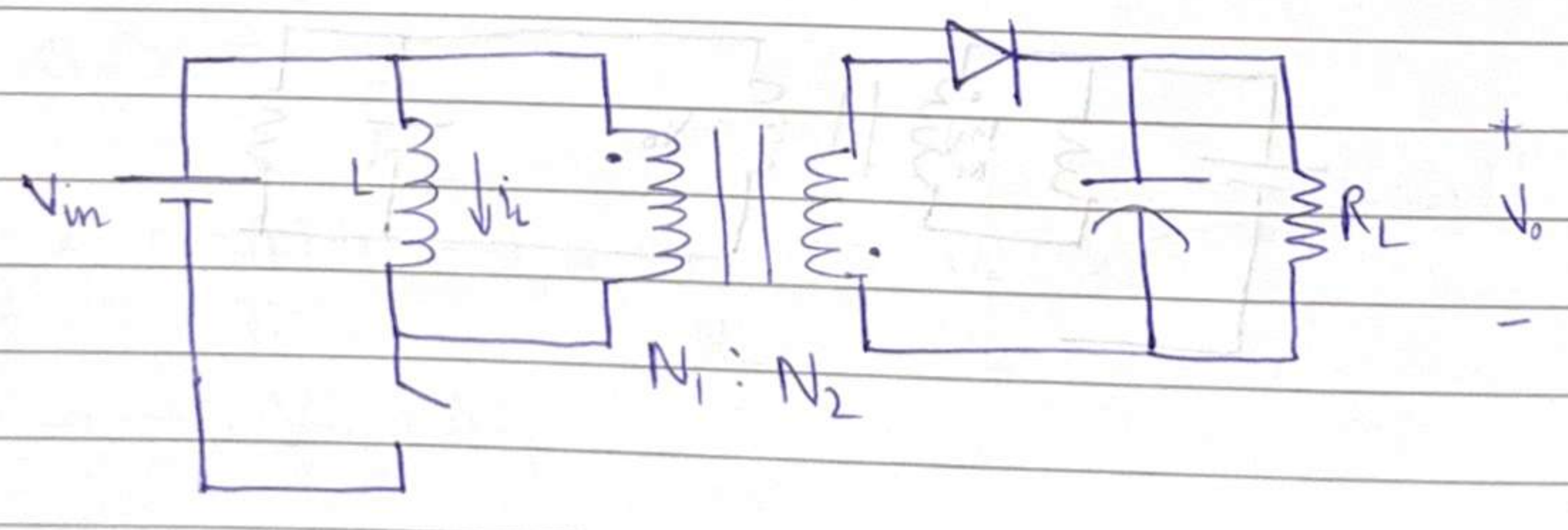
Forward → isolation, Large gain

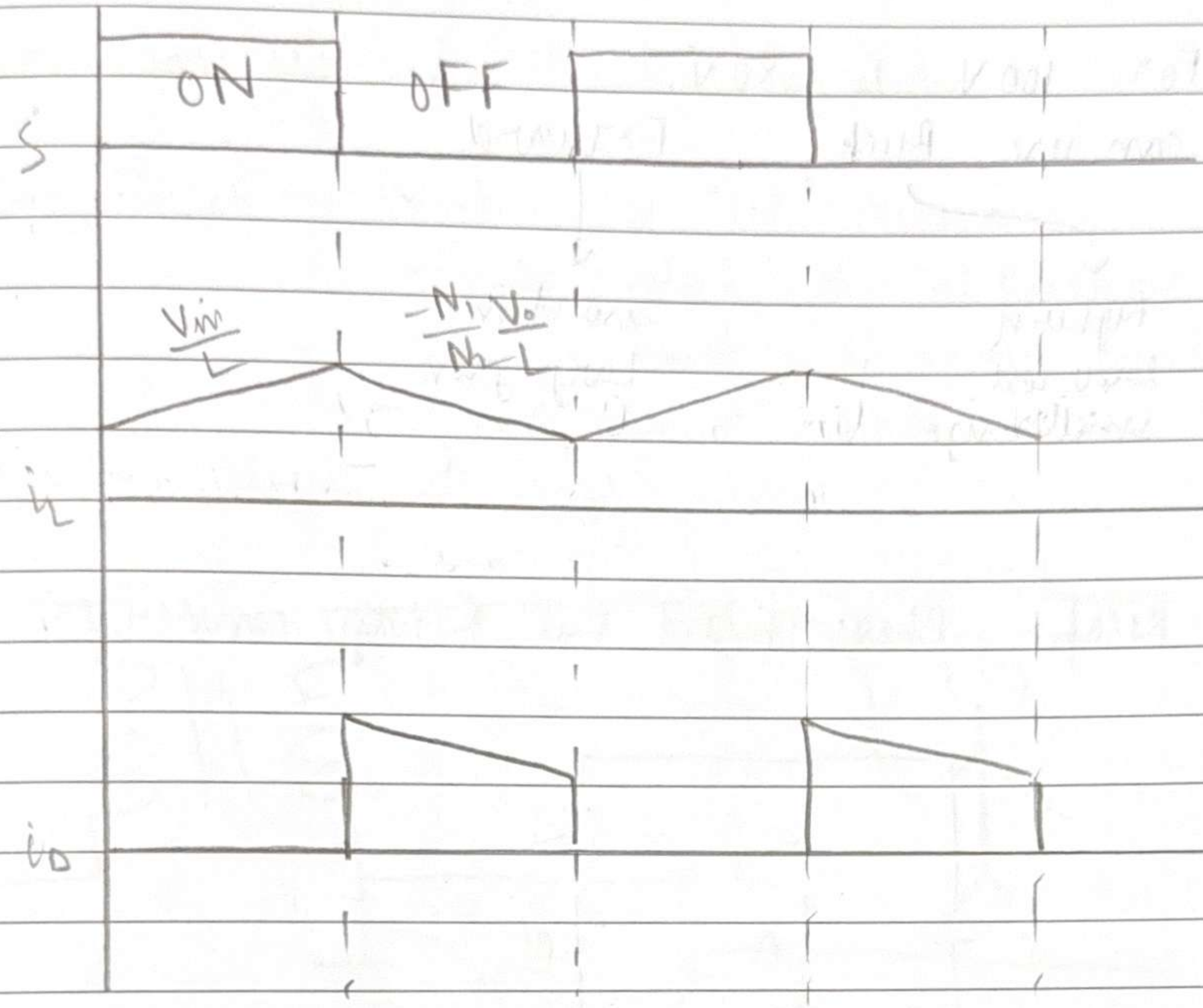
Recap - Phase Shifted Full Bridge converter



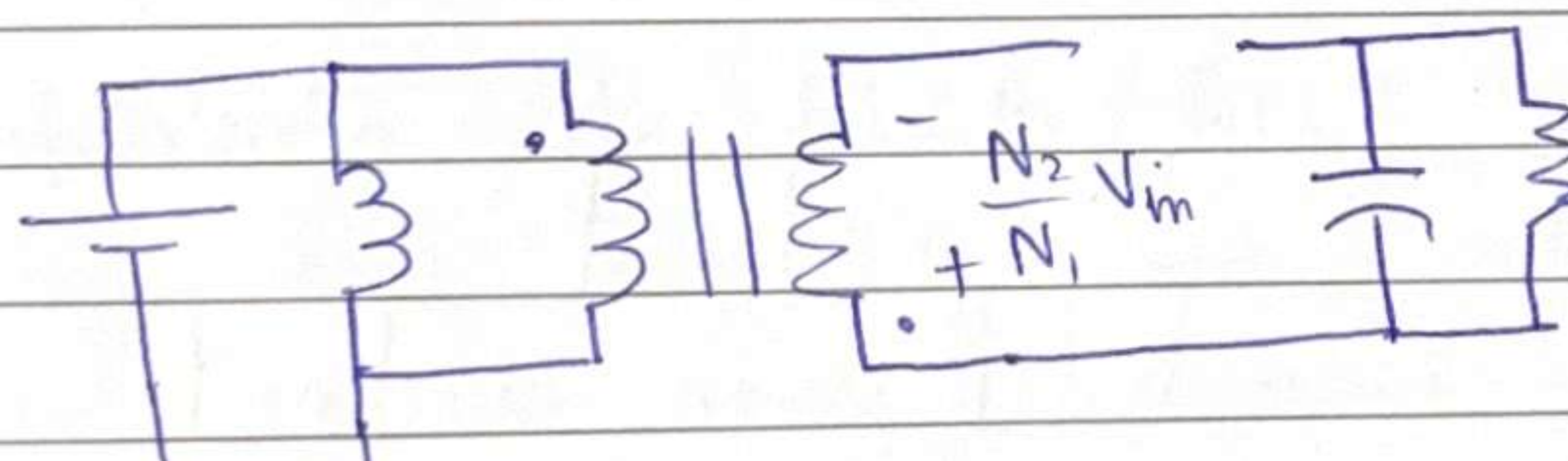


"bipolar"

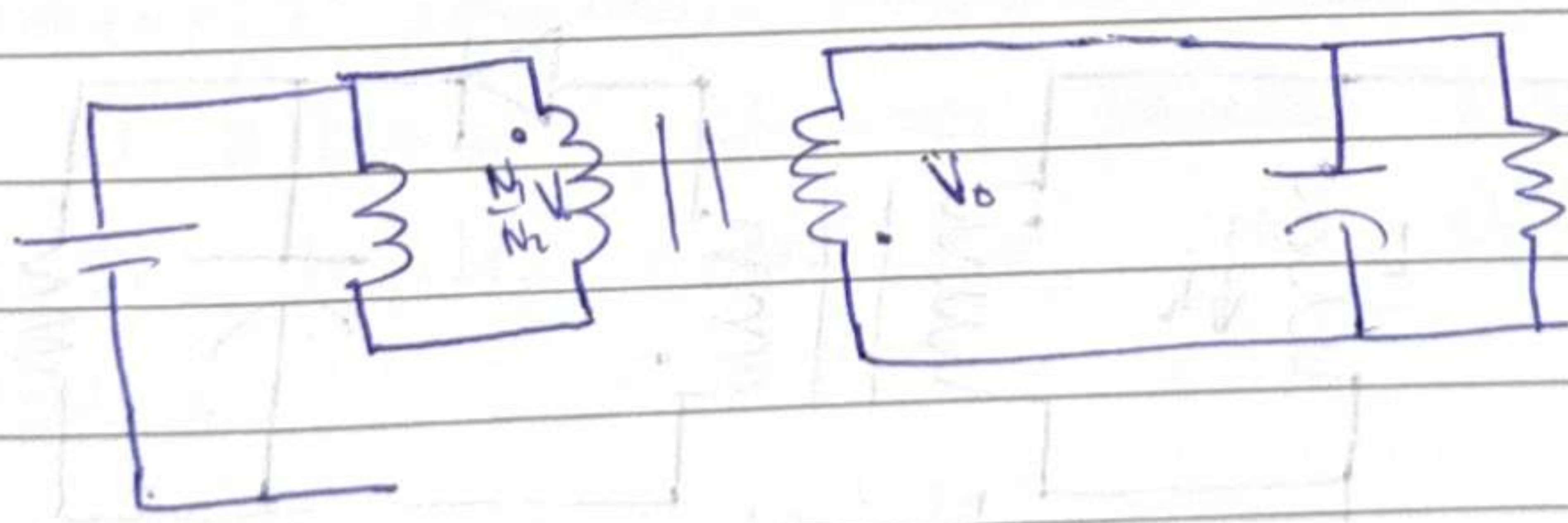




When S = ON,



When S = OFF,



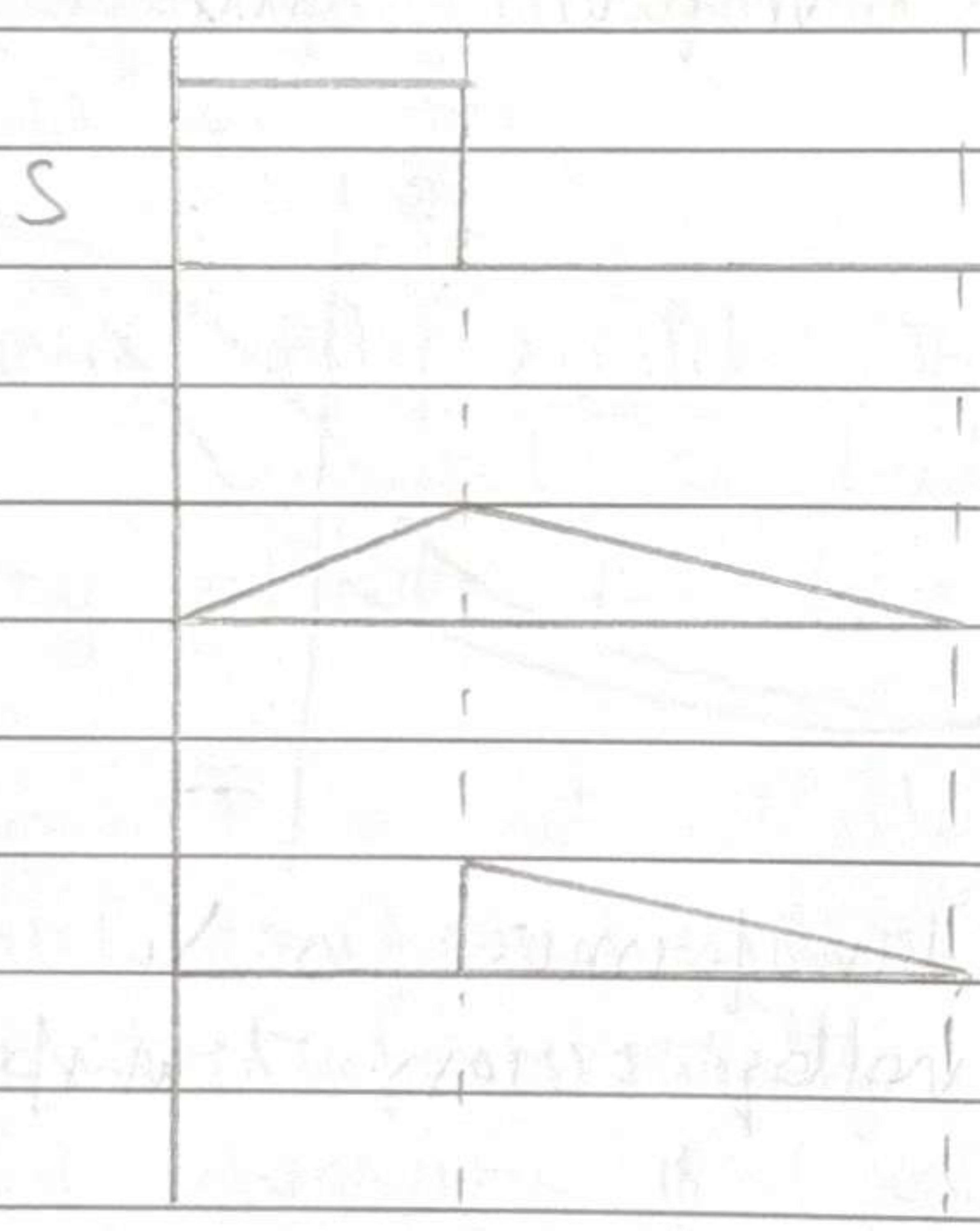
$$V_{in} D = \frac{N_1}{N_2} V_o (1-D)$$

$$\frac{V_o}{V_{in}} = \frac{N_2}{N_1} \cdot \frac{D}{1-D}$$

HW: solve in DCM mode
- Determine the condition for CCM/DCM
- Determine expression for V_o/V_{in}

This circuit is called Flyback converter

DCM Mode of Flyback Converter (In critical conduction mode)



$$\langle i_D \rangle = \frac{V_o}{R}$$

$$\frac{1}{2} \times \frac{(1-D)T_s}{T_s} \times i_{Lpeak} \times \frac{N_1}{N_2} = \frac{V_o}{R}$$ → Wrong!!

$$i_{peak} = \frac{V_{in}}{L} \times DT_s$$

$$\therefore \frac{D(1-D)}{2} \frac{V_{in}}{LF_s} \frac{N_1}{N_2} = \frac{V_o}{R}$$

$\rightarrow$ current

$$\frac{D(1-D)}{2}\,\frac{V_{in}}{LF_s}\,\frac{N_1}{N_2} \le \frac{V_o}{R} \quad \text{(condition for CCM)}$$

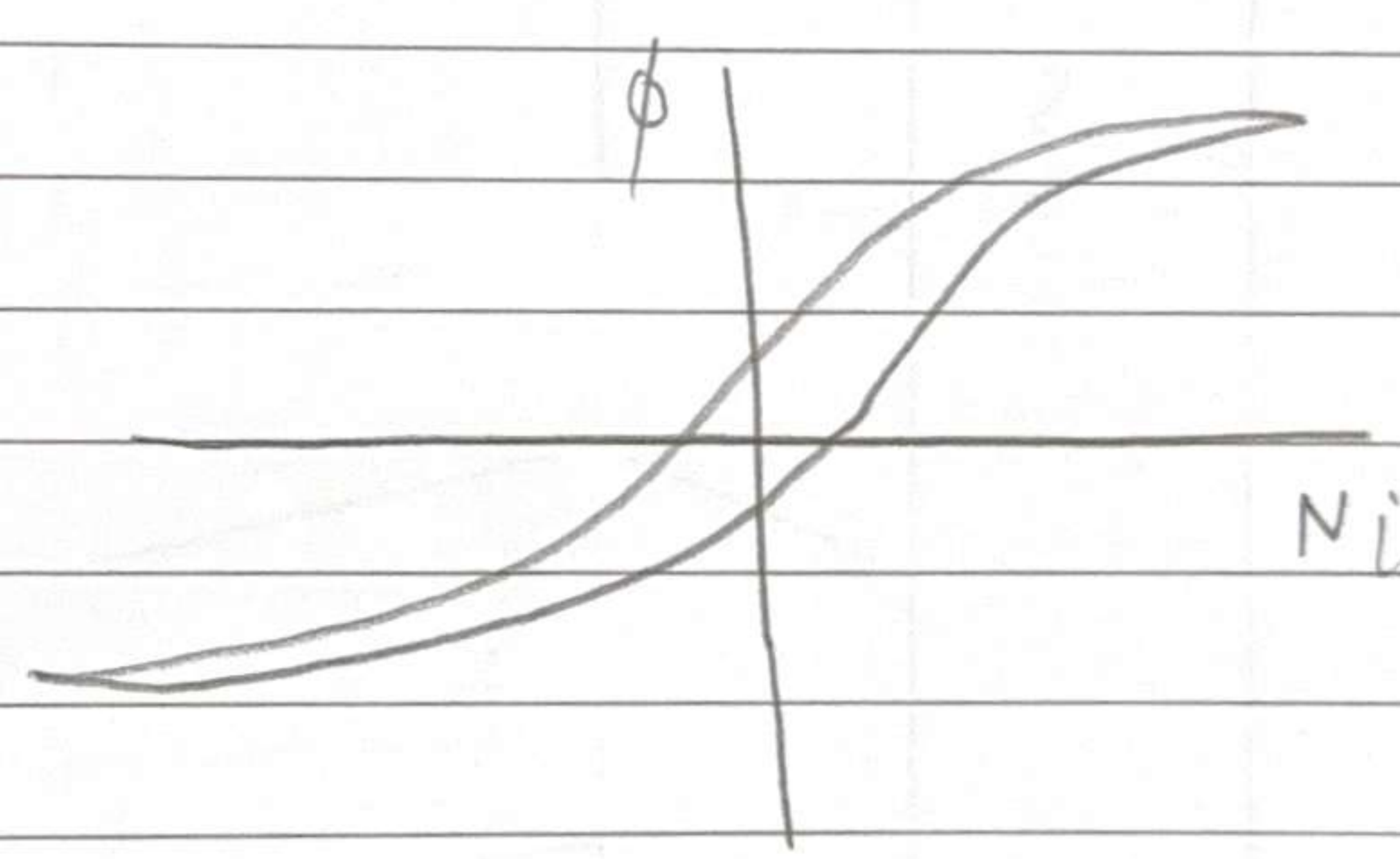
wrt $V_o = \dfrac{N_2}{N_1}\,\dfrac{D}{1-D}$

Increasing L, F_s takes the circuit farther from DCM condition.

CCM/DCM is defined wrt to inductor current.
If it touches zero then it is DCM,
If not then CCM.

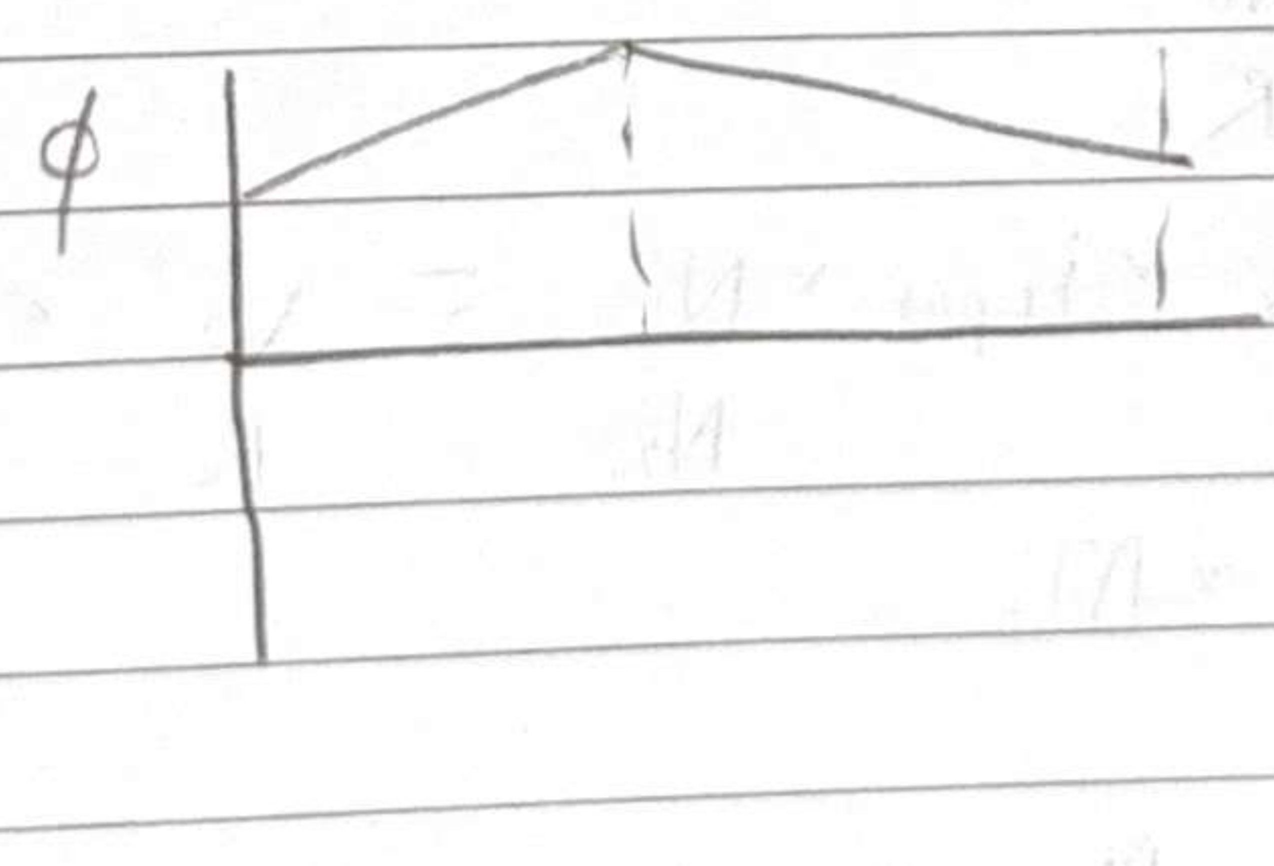
For the flyback, will the transformer saturate?

$$e = N\frac{d\phi}{dt}$$



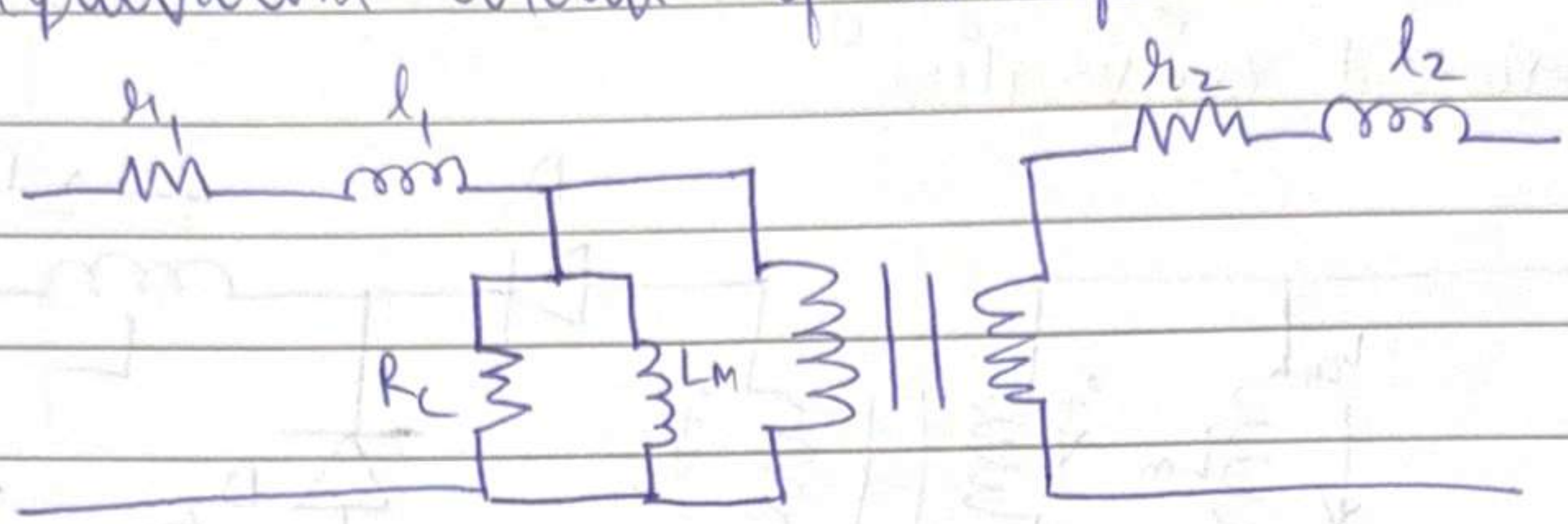
Voltage across primary of transformer is V_L
$\langle V_L \rangle = 0 \Rightarrow$ no avg dc voltage across transformer

$$\frac{d\phi}{dt} = \frac{V_L}{N} \propto i_L$$



For non-ideal transformer

Recall equivalent circuit of transformer



Note that in this equivalent ckt, L_m is already in same place as the L in flyback circuit.

Hence, practically we don't need to separately add an L in the flyback circuit, the internal magnetizing inductance of the transformer does the job.

Hence, only switch, transformer, Diode, capacitor required to build circuit.

∴ Very popular for low power, low cost applications.

Since L is not actually added in the circuit as a "real" component, there is no i_L,
So to experimentally decide CCM / DCM mode, we measure ϕ ($\propto i_L$)

Non-ideal Transformer for PSFB

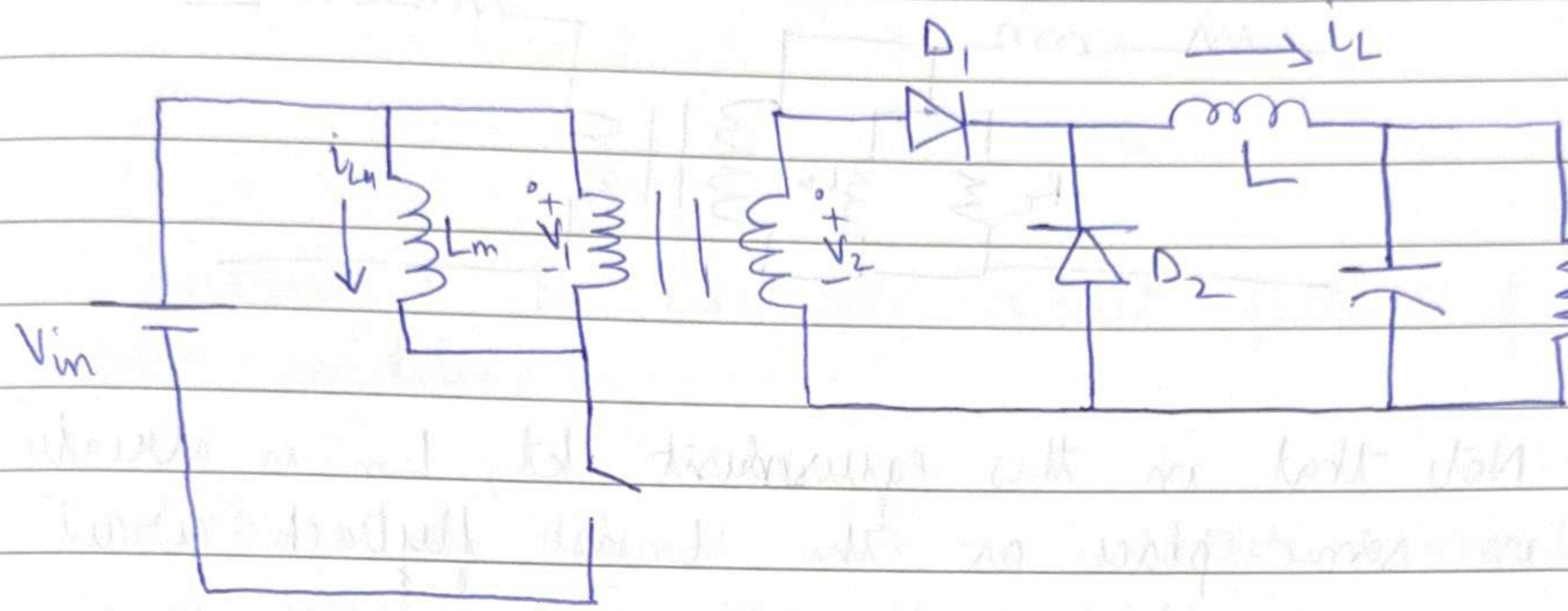
Here, $\langle e_1 \rangle = \langle V_{AB} \rangle = 0$ hence no problem (no saturation)

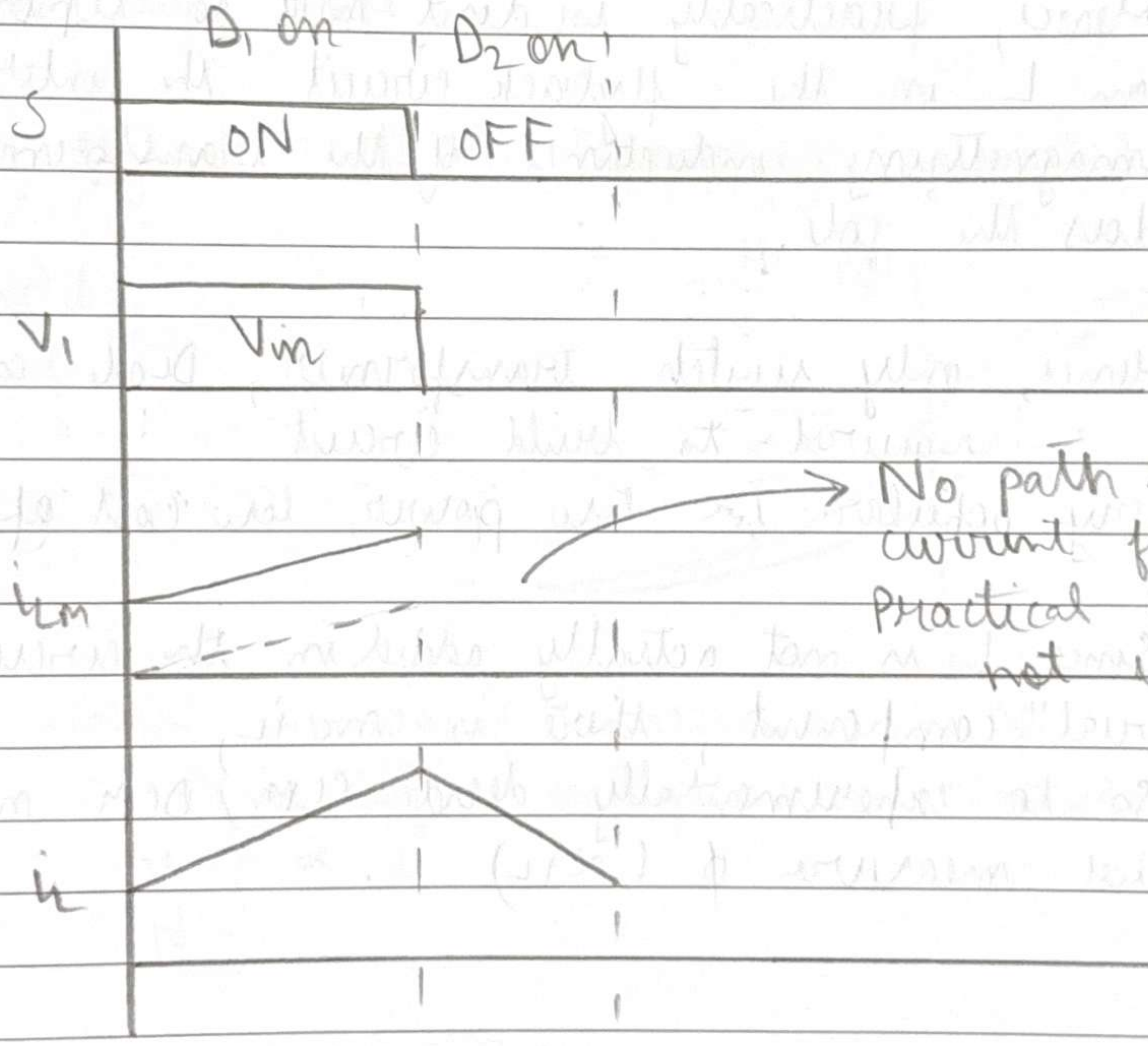
In case $\langle e_1 \rangle \neq 0$ then $e = N\dfrac{d\phi}{dt}$

ϕ will keep increasing and ultimately saturate.

Hence, we are safe as long as $\langle e_1 \rangle = 0$

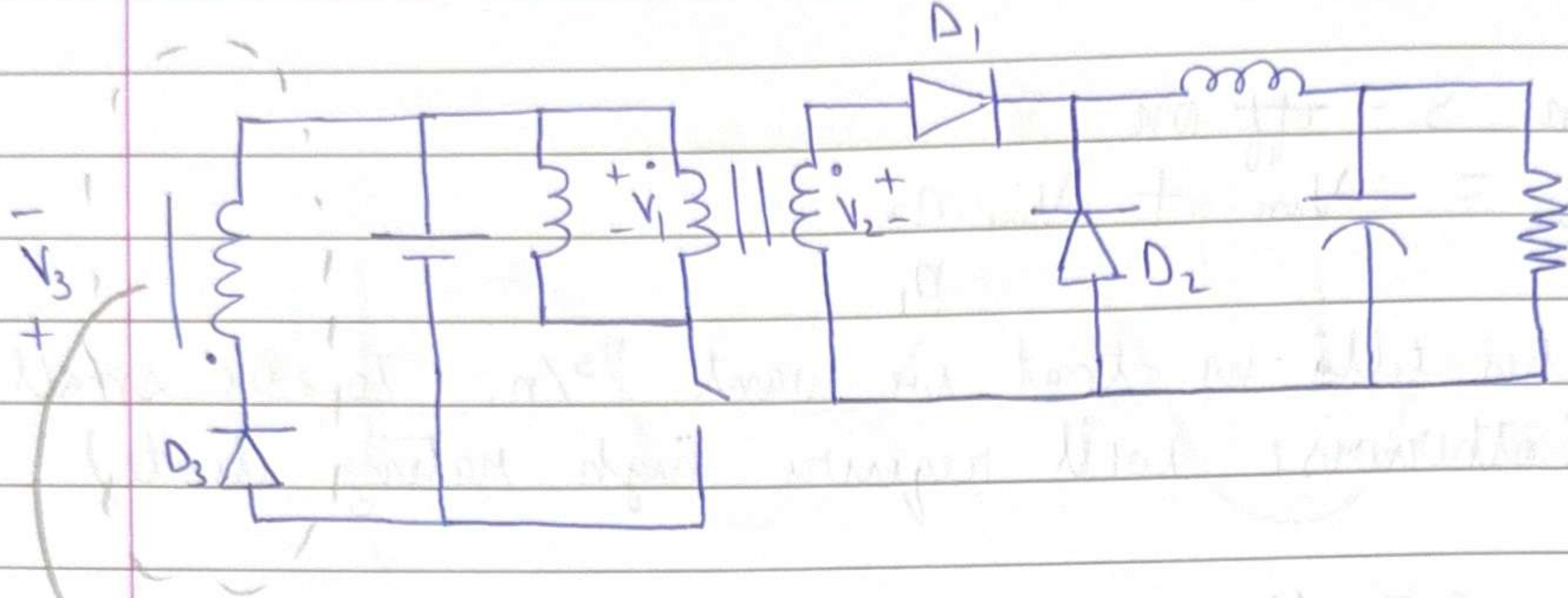
Effect of magnitizing inductance (L_m) / saturation on Forward converter



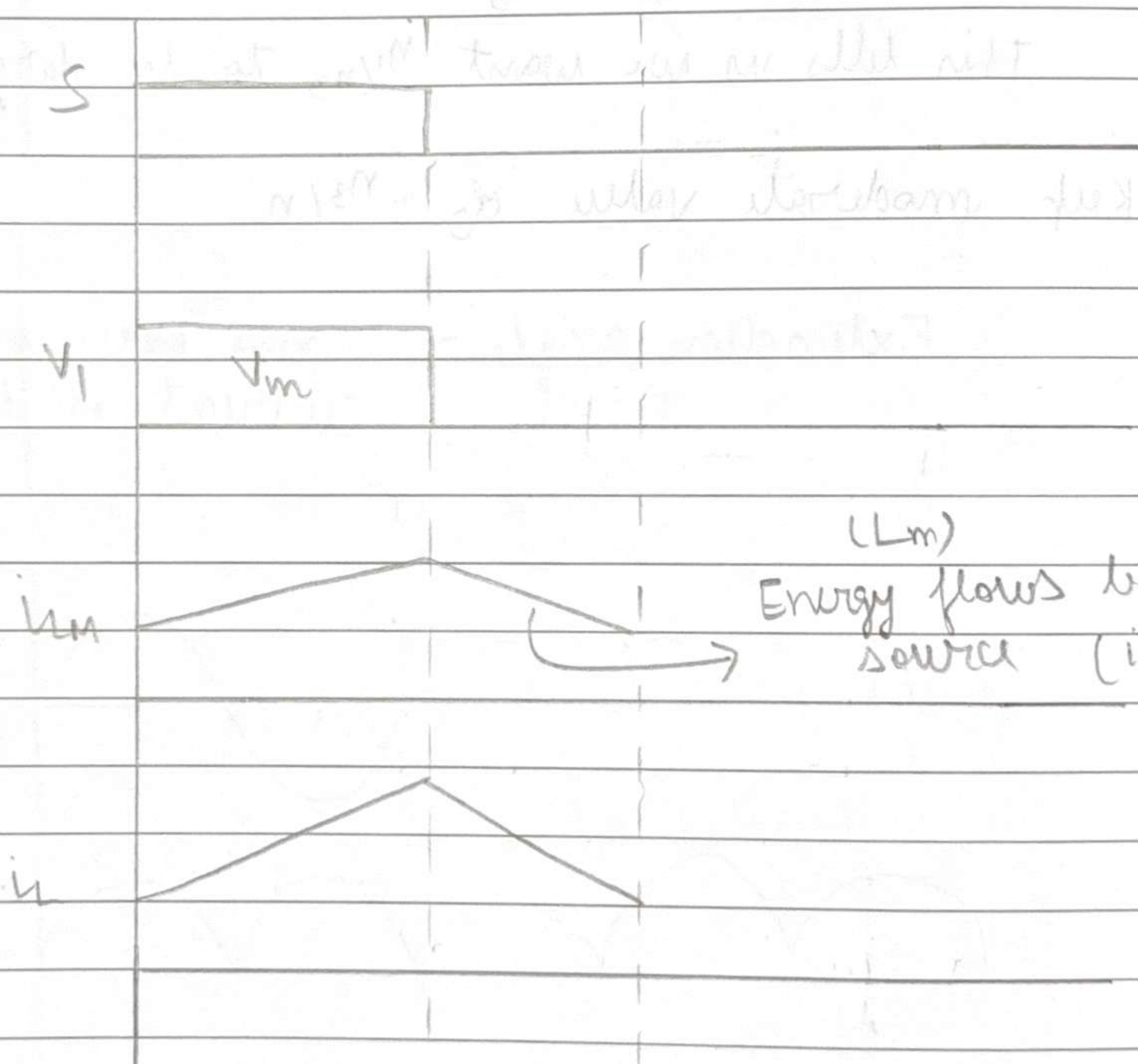


No path for current flow, ⟹ Practical ckt will not work

Hence to make the circuit work in practice, we need to add an additional circuitry



→ Tertiary coil of transformer



(Lm)
Energy flows back to source ($i_{LM}\downarrow$)

D_3 off $\quad$ D_3 on

↳ Current through secondary is zero (since D_1 off)

To ensure power conservation, current flows through tertiary.

When $S = \cancel{off}$ on

$$V_{D_3} = V_{in} + V_{in}\frac{n_3}{n_1}$$

This tells us that we want n_3/n_1 to be small (otherwise will require high rating diode)

When $S =$ off,

$$V_{S_1} = V_{in} + V_{in}\frac{n_1}{n_3}$$

This tells us we want n_1/n_3 to be large

$\therefore$ Keep moderate value of n_3/n_1

Extinction angle — range over which current is decreasing

Fourier series

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$
(periodic)

- We use Fourier series to simplify non-sinusoidal AC circuit analysis.
- We also use it to define quality of sin I or V

Ideally, we want V and I to be sin, because
① — Network components cause $\int (\,)dt$ and $\frac{d}{dt}(\,)$
sine will remain sine. Not true for other shapes.
② — Supply to induction motor needs to be sine

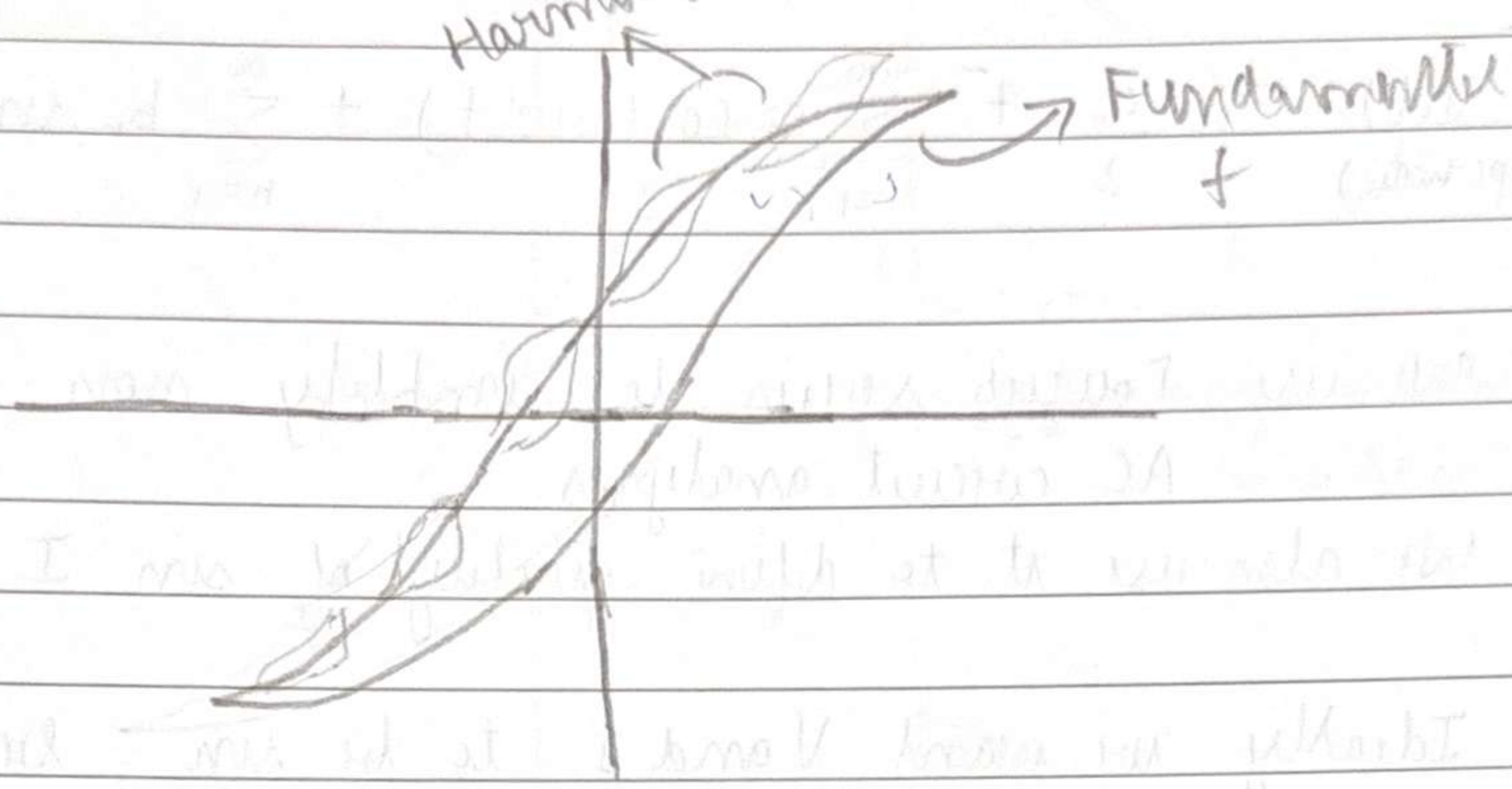
In induction motor, pure sine current causes a rotating magnetic field at synchronous speed.
If we supply square wave current to induction motor, the rotating mag. field will have many harmonic frequency components.
The rotor will effectively 'latch' onto one of the ~~to~~ frequency components. The other ~~fra~~ freq. components of rotating mag. field will aid rotor's rotation ~~and~~ some time and oppose it some time ⇒ vibrations and additional losses
⇒ Humming noise
Globally motors are one of the primary consumers of electrical energy

~~③~~ —

PTO

$I = I_1 \cos(\omega t) + I_n \cos(n\omega t)$

③ — Higher harmonics in transformers increase losses

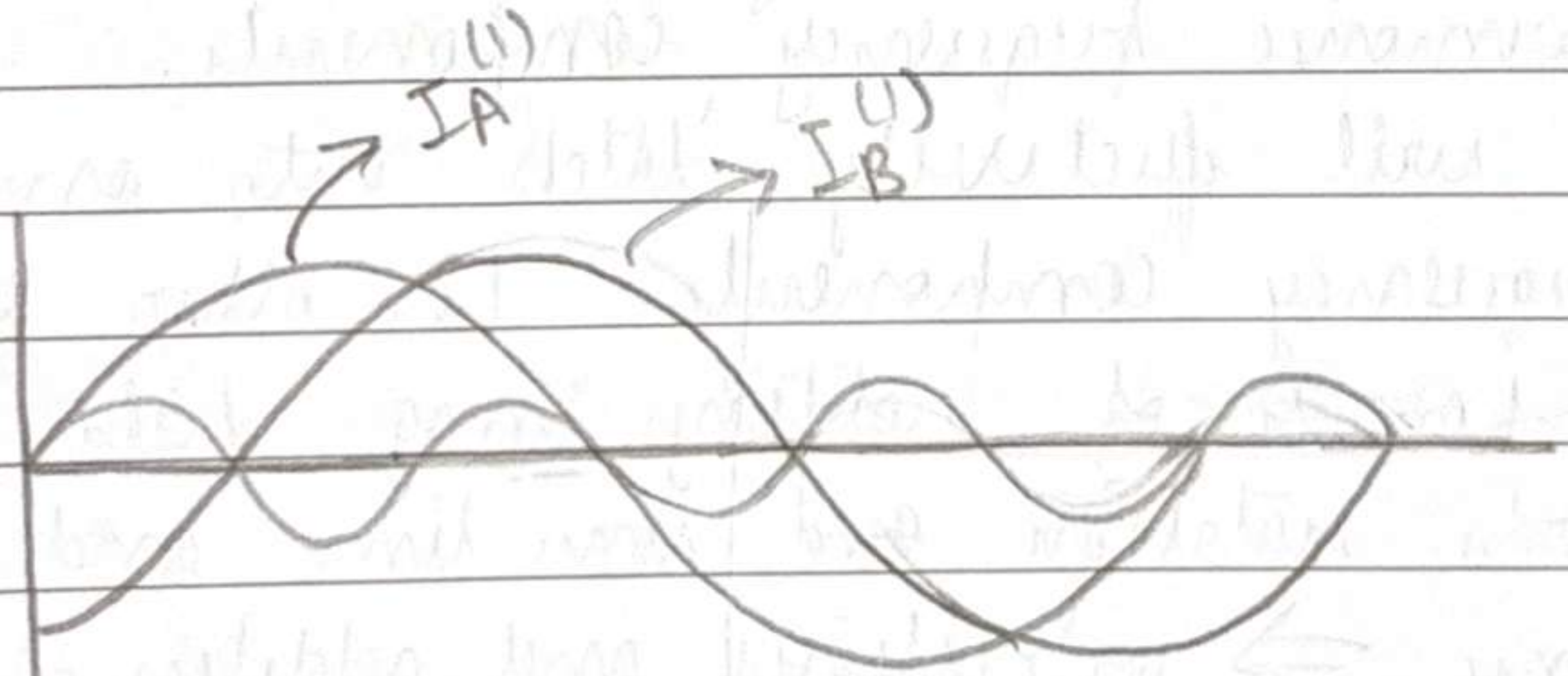


Harmonics cause smaller hysteresis ~~loss~~ loops along the larger fundamental frequency loop.
$\Rightarrow$ increased hysteresis loss.

Harmonics $\Rightarrow$ increased core loss

Skin effect $\propto$ frequency
$\therefore$ Harmonics $\Rightarrow$ increased skin effect loss

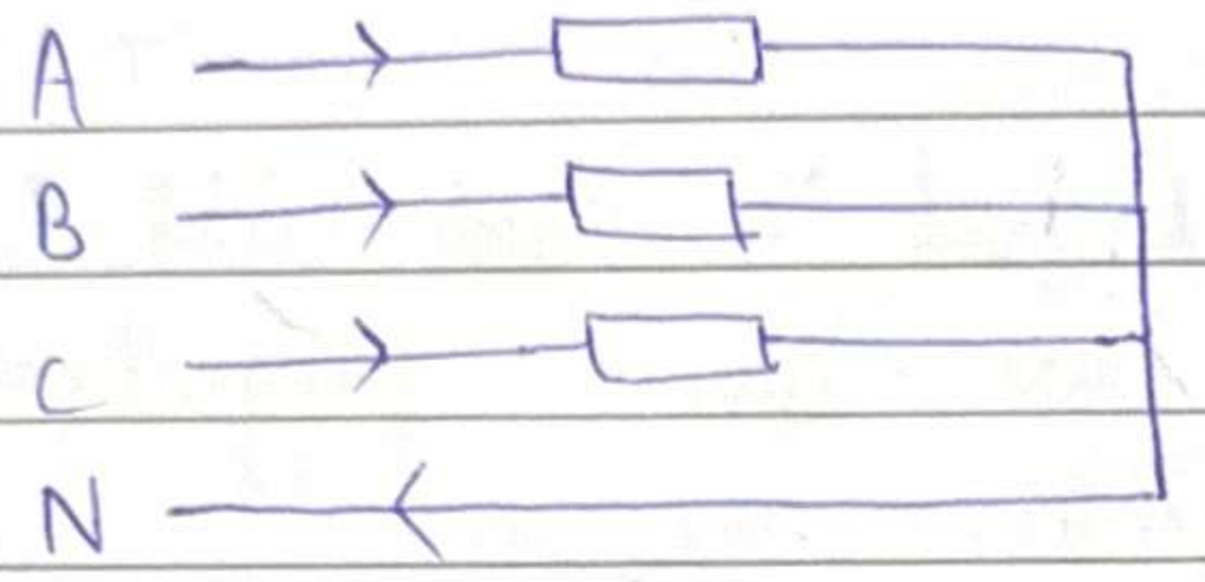
④ —



$I_A^{(1)} = A\cos(\omega t)$
$I_B^{(1)} = A\cos(\omega t - 120^\circ)$
$= A\cos\left[\omega\left(t - \frac{T}{3}\right)\right]$

Now, $I_A^{(3)} = A^{(3)}\cos(3\omega t)$
$I_B^{(3)} = A^{(3)}\cos(3\omega t)$ same !!

In 3-ϕ system, higher frequency components, even in a balanced load, don't sum to zero.



(Balanced load)

We saw that $I_A^{(1)} + I_B^{(1)} + I_C^{(1)} = 0$

But $I_A^{(3)} = I_B^{(3)} = I_C^{(3)}$

$\therefore \quad I_A^{(3)} + I_B^{(3)} + I_C^{(3)}$

hence due to harmonics, neutral will have to carry current $\Rightarrow$ increased cost because of thick wire requirement

So even if voltage supply is pure sin,
- For linear loads (R, L, C) balanced, I will be sin, so $I_N = 0$
- But for non-linear loads (~~ca~~ include diodes, switches) like diode rectifier etc., I will not be sin, it will have harmonic components, so $I_N \neq 0$.

N will have to carry large 3rd harmonic current. (~~At~~ Also, happens for 3rd, 9th, 15th and so on harmonics)

5- High freq. components in power lines can inject high frequencies into nearby communication lines.

BLDC has rectifier and inverter
(has diodes) ⇒ non-linear loads

IM has only RLC ⇒ linear loads

Usually there are 'standards' to limit harmonics drawn / injected by the loads connected in power system

Linear loads ⇒ input V and output I will have same frequency

Non-linear loads ⇒ May have different frequency (presence of harmonics)

Linear loads (R, L, C) ⇒ If V_s is pure sin with f_o
I drawn is also pure sin with f_o

Non linear loads (D, Swi + RLC) ⇒ V_s is pure sin with f_o
I drawn could be of multiple freq

BLDC ⇒ Non-linear
IM ⇒ Linear

Is lift a linear or non-linear load?
Lifts have IM.
Even then, that IM has variable frequency supply, which causes speed of lift be continuously adjusted during as/descent
That control circuit includes rectifiers ⇒ non-linear

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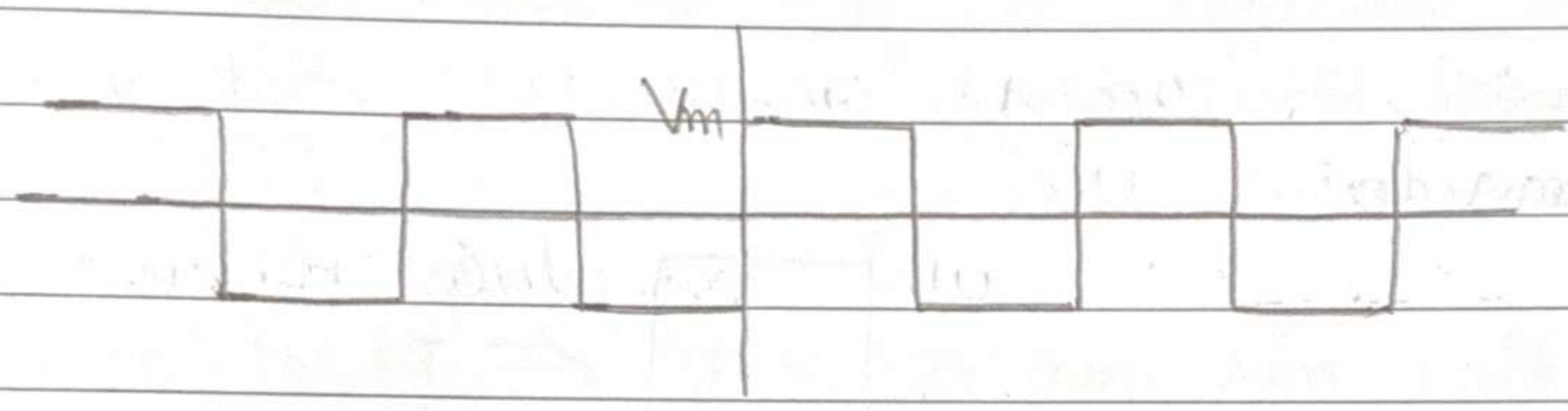
For $f(t)$ with half-wave symmetry,
if; $f(t) = -f(t - \frac{T}{2})$,

even harmonics are absent

Consider again the square wave



RMS values of -

$V_0 = 0$

$V_1 = \frac{2\sqrt{2}\, V_m}{\pi}$

$V_2 = 0$

$V_3 = \frac{2\sqrt{2}\, V_m}{3\pi}$

$V_4 = 0$

$V_5 = \frac{2\sqrt{2}\, V_m}{5\pi}$

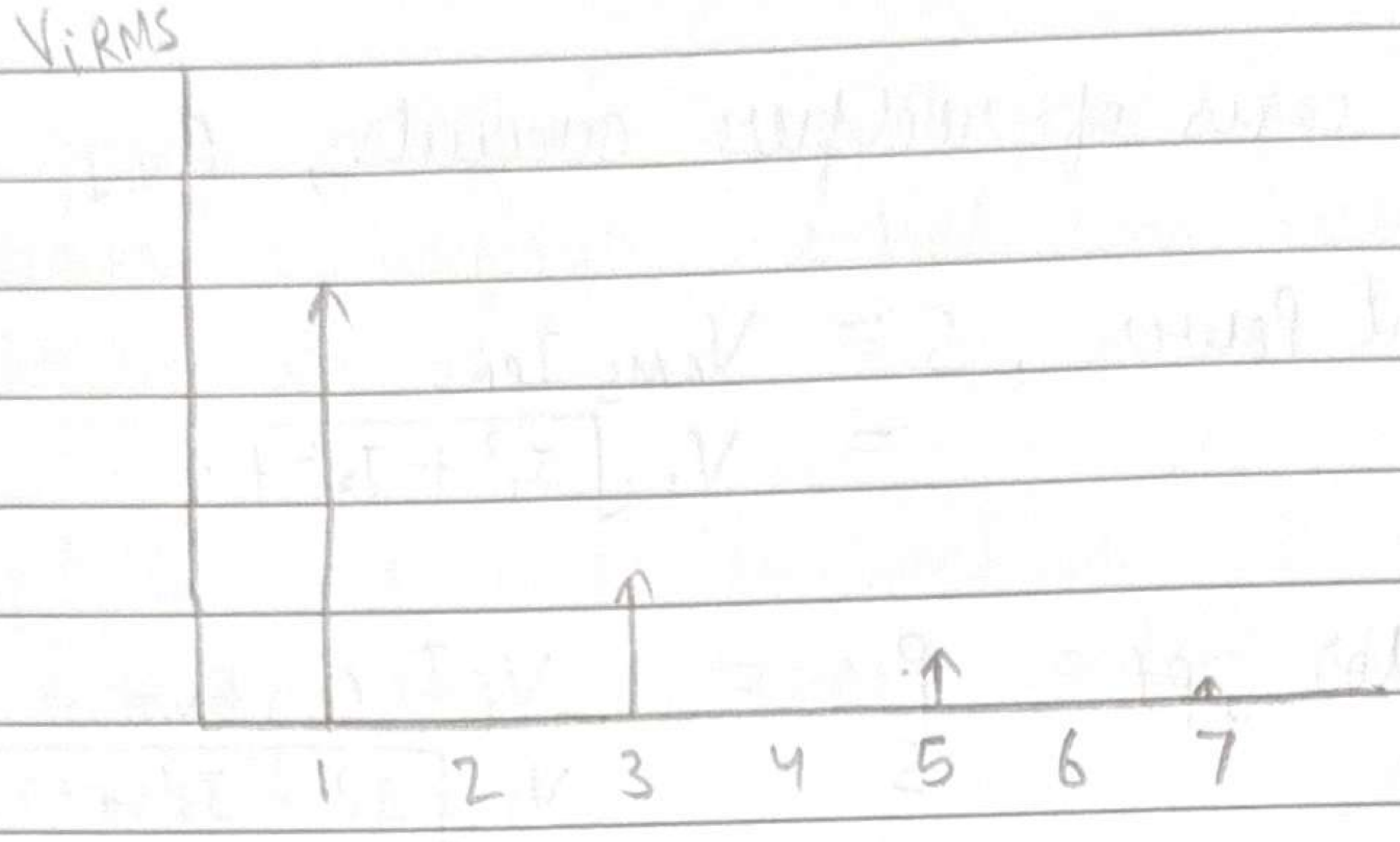
and so on

$(V)_{RMS} = V_m \cancel{/\sqrt{2}}$ (Here)

$V_{RMS} = \sqrt{V_1^{RMS^2} + V_2^{RMS^2} + V_3^{RMS^2} + \dots}$

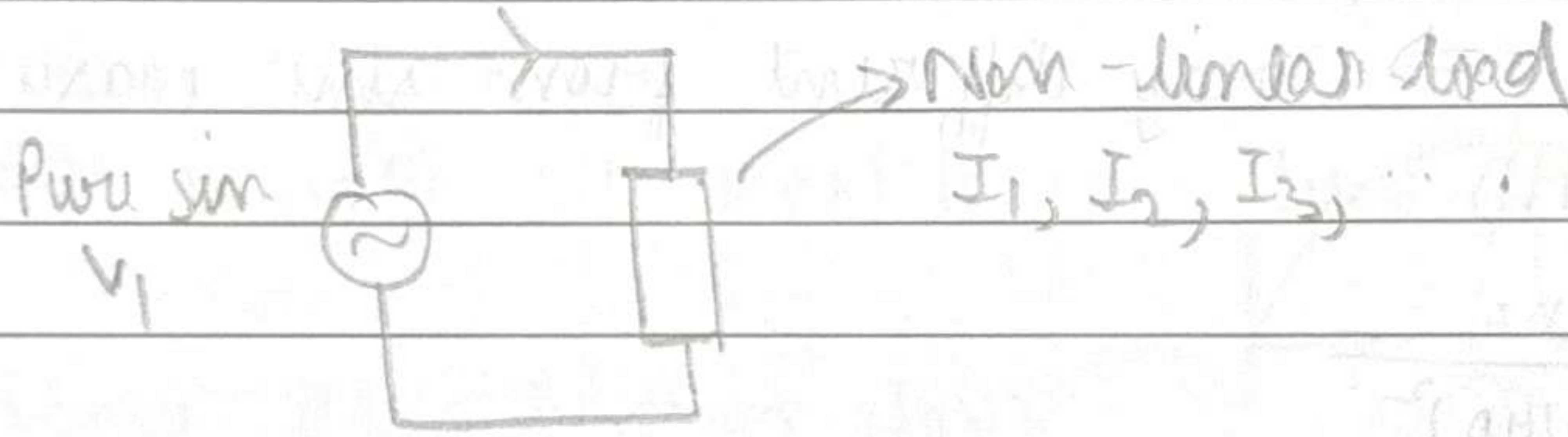
HW: Derive this

$$\frac{\frac{\pi^2}{6}-1}{\pi^2/6} = 1-\frac{6}{\pi^2}$$



Most power electronic circuits have half-wave symmetric currents

Exception — Half-wave diode rectifier etc.

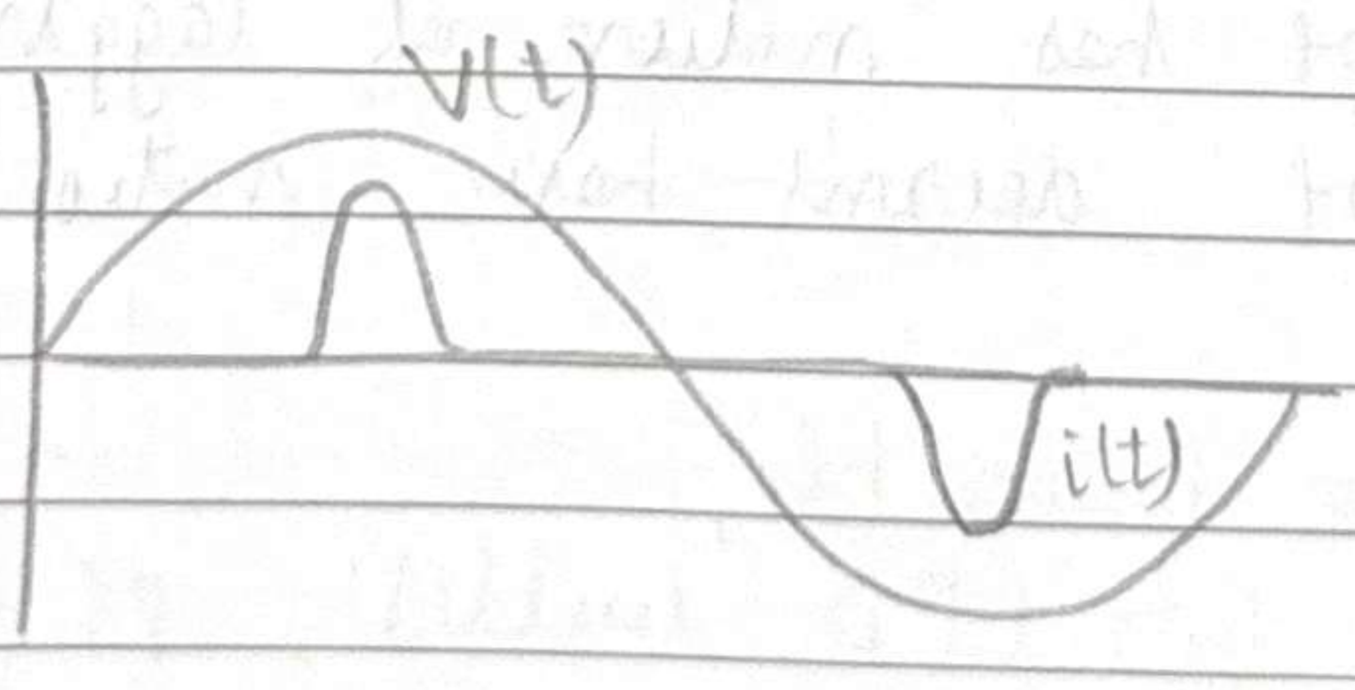


$$P = \frac{1}{T}\int_0^T v(t)\, i(t)\, dt$$

$$P = V_1 I_1 \cos\theta_{V_1 I_1}$$

(all other current components give zero integral)

Hence real power is only delivered by fundamental component.



In most cases of rectifier circuits, $\theta_{V_1 I_1}$ is close to 0.

Apparent Power, $S = V_{RMS} I_{RMS}$
$= V_1 \sqrt{I_1^2 + I_2^2 + \cdots}$

Power Factor, $pf = \frac{P_1}{S} = \frac{V_1 I_1 \cos\theta_{V_1 I_1}}{V_1 \sqrt{I_1^2 + I_2^2 + \cdots}}$

If harmonics are absent, $I_{RMS} = I_1$
$pf = \cos\theta_{V_1 I_1}$!!

Total Harmonic Distortion, $THD = \frac{\sqrt{I_2^2 + I_3^2 + \cdots}}{I_1}$ $(\times 100\%)$

THD = 0 ⇒ pure sine wave
Large THD ⇒ very different from sine wave.
THD could be defined for i or V

$\therefore$ $pf = \frac{\cos\theta_{V_1 I_1}}{\sqrt{1 + (i_{THD})^2}}$

$$pf = \underbrace{\cos\theta_{V_1 I_1}}_{\text{Displacement pf}} \times \underbrace{\frac{1}{\sqrt{1 + (i_{THD})^2}}}_{\text{Distortion pf}}$$

Usually V_{THD} is very low ($\approx$ 2 to 5%)
i_{THD} in diode rectifier could be 70 to 100 %

Displacement pf has notion of lagging / leading
Distortion pf doesn't have notion of leading/ lagging

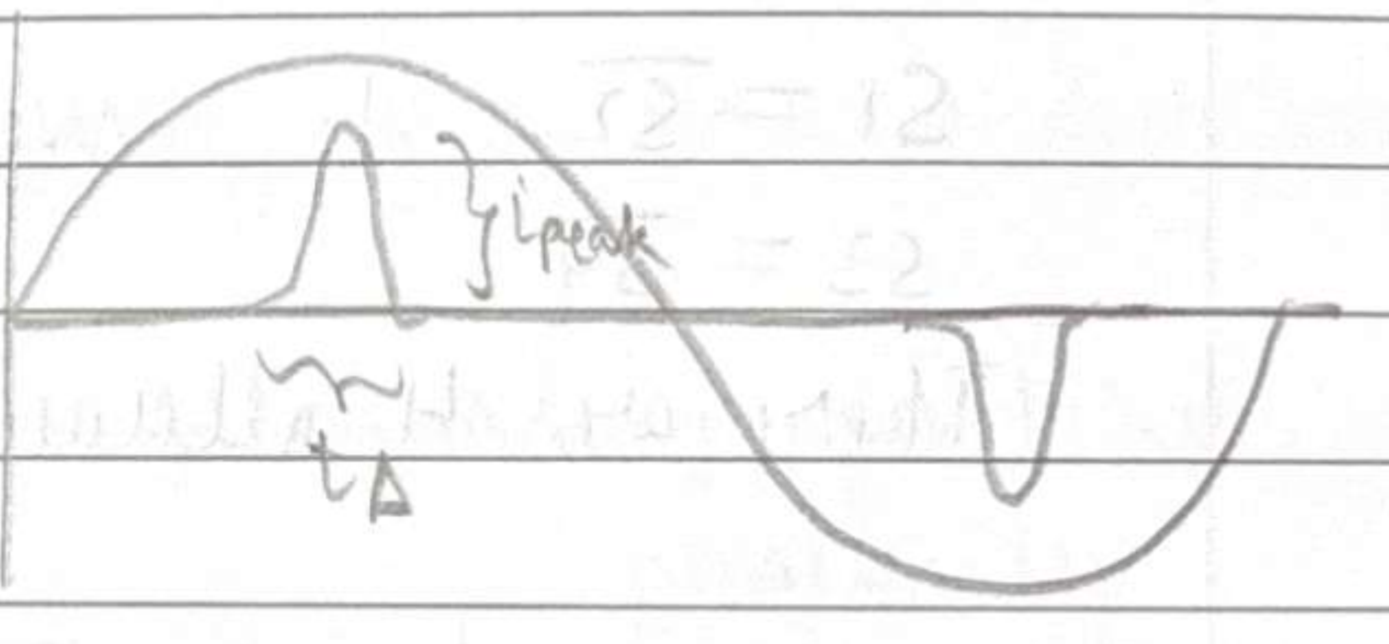
Practical ~~Bulb~~ Example,
Typically for LED bulbs, pf = 0.9
which gives $i_{THD} \approx 50\%$

Q) A full bridge rectifier circuit has pf = 0.8. As ~~always~~ a designer, what can you do to improve its pf?

Option 1) We used to add a C to improve pf earlier. Won't work here because adding C improves displacement pf. Here we have distortion pf.

Option 2) Add some filter on AC side so that current drawn from input does not have large high frequency components
⇒ smaller THD

Option 3) Reduce C,
⇒ $t_\Delta \uparrow$
$i_{peak} \downarrow$
Hence $i(t)$ becomes closer to a pure sin
⇒ pf improves



BUT, reducing C ⇒ Ripple increases
Tradeoff!!